Econ148Proj03_final

May 4, 2026

1 Imports and Initialization

Code cell purpose: Cell below is extremely important! It imports the relevant packages to be utilized throughout the entire notebook, also choosing desired display settings. Cell must be ran prior to any other cells are to ensure proper reproducibility. This matters because every later section depends on the same libraries and display choices being loaded before any analysis is run.

[1]:
```
## Initializing the notebook
%pip install numpy pandas matplotlib pytrends statsmodels scikit-learn joblib␣
 ↪lightgbm

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import pytrends

import statsmodels.api as sm
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tsa.stattools import adfuller

from sklearn.metrics import mean_squared_error
from IPython.display import display

from pytrends.request import TrendReq
from pytrends.exceptions import TooManyRequestsError

import os
import time
import itertools
import warnings
from pathlib import Path
from joblib import Parallel, delayed

warnings.filterwarnings("ignore")
pd.set_option("display.max_columns", 100)
pd.set_option("display.float_format", lambda x: f"{x:,.2f}")
```

```
Requirement already satisfied: numpy in /srv/conda/lib/python3.11/site-packages
(2.2.6)
Requirement already satisfied: pandas in /srv/conda/lib/python3.11/site-packages
(2.2.3)
Requirement already satisfied: matplotlib in /srv/conda/lib/python3.11/site-
packages (3.10.8)
Collecting pytrends
  Using cached pytrends-4.9.2-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: statsmodels in /srv/conda/lib/python3.11/site-
packages (0.14.4)
Requirement already satisfied: scikit-learn in /srv/conda/lib/python3.11/site-
packages (1.6.0)
Requirement already satisfied: joblib in /srv/conda/lib/python3.11/site-packages
(1.5.3)
Collecting lightgbm
  Using cached lightgbm-4.6.0-py3-none-manylinux_2_28_x86_64.whl.metadata (17
kB)
Requirement already satisfied: python-dateutil>=2.8.2 in
/srv/conda/lib/python3.11/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /srv/conda/lib/python3.11/site-
packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /srv/conda/lib/python3.11/site-
packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in
/srv/conda/lib/python3.11/site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /srv/conda/lib/python3.11/site-
packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/srv/conda/lib/python3.11/site-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/srv/conda/lib/python3.11/site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in
/srv/conda/lib/python3.11/site-packages (from matplotlib) (24.2)
Requirement already satisfied: pillow>=8 in /srv/conda/lib/python3.11/site-
packages (from matplotlib) (11.1.0)
Requirement already satisfied: pyparsing>=3 in /srv/conda/lib/python3.11/site-
packages (from matplotlib) (3.3.2)
Requirement already satisfied: requests>=2.0 in /srv/conda/lib/python3.11/site-
packages (from pytrends) (2.32.3)
Collecting lxml (from pytrends)
  Using cached
lxml-6.1.0-cp311-cp311-manylinux_2_26_x86_64.manylinux_2_28_x86_64.whl.metadata
(4.0 kB)
Requirement already satisfied: scipy!=1.9.2,>=1.8 in
/srv/conda/lib/python3.11/site-packages (from statsmodels) (1.14.1)
Requirement already satisfied: patsy>=0.5.6 in /srv/conda/lib/python3.11/site-
packages (from statsmodels) (1.0.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
```

```
/srv/conda/lib/python3.11/site-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /srv/conda/lib/python3.11/site-
packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/srv/conda/lib/python3.11/site-packages (from requests>=2.0->pytrends) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /srv/conda/lib/python3.11/site-
packages (from requests>=2.0->pytrends) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/srv/conda/lib/python3.11/site-packages (from requests>=2.0->pytrends) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in
/srv/conda/lib/python3.11/site-packages (from requests>=2.0->pytrends)
(2026.2.25)
Using cached pytrends-4.9.2-py3-none-any.whl (15 kB)
Using cached lightgbm-4.6.0-py3-none-manylinux_2_28_x86_64.whl (3.6 MB)
Using cached
lxml-6.1.0-cp311-cp311-manylinux_2_26_x86_64.manylinux_2_28_x86_64.whl (5.2 MB)
Installing collected packages: lxml, lightgbm, pytrends
                        3/3
[pytrends]1/3 [lightgbm]
Successfully installed lightgbm-4.6.0 lxml-6.1.0 pytrends-4.9.2
Note: you may need to restart the kernel to use updated packages.
```

2 Data Loading, Cleaning, and Exploratory Data Analysis

Our code covers work for Track A, Project 3. In this section, we prepare the weekly FRED initial unemployment insurance claims data. For the final submission, the notebook expects a folder titled `Data` in the same directory as this notebook. Within that folder should be the four required FRED CSV files labeled `CAICLAIMS.csv`, `HIICLAIMS.csv`, `INICLAIMS.csv`, and `NYICLAIMS.csv`. Generally, the raw FRED CSV files that we utilize here are already quite clean, however, we still wanted to make sure that they contained good time series for the process of forecasting. Mainly, this section performs work such as checking for missing values, duplicates, date parsing, negative values, etc. Additionally, work is done for summary statistics, annual patterns, and other relevant EDA steps.

2.1 1. Setting up the Datasets and Loading Helper Functions

Our project, as aforementioned, utilizes four state-level FRED initial claims series. In the cell below, all data is loaded through the local `Data` folder, and there are also a couple of brief helper functions created which are beneficial for the general flow of code to come later.

Code cell purpose: This cell, being early in our setup, is basic in purpose. Specifically, it works to define things such as our list of states, labels for the states, the local data path that is being used, and a couple brief helper functions which are utilized for both loading/cleaning our FRED CSVs. This matters because these shared names and helpers keep the rest of the notebook reproducible, avoiding repeated file-path and column-cleaning logic.

[2]:
```
## Datasets of interest, naming them, etc.
datasets = ["CAICLAIMS", "NYICLAIMS", "INICLAIMS", "HIICLAIMS"]

state_names = {
    "CAICLAIMS": "California",
    "NYICLAIMS": "New York",
    "INICLAIMS": "Indiana",
    "HIICLAIMS": "Hawaii"
}

## Relevant local path for data
data_dir = Path("Data")

## Function to check for relevant path
def claims_path(code):
    path = data_dir / f"{code}.csv"
    if not path.exists():
        raise FileNotFoundError(f"You are missing {path}. Please ensure to␣
 ↪include this CSV in the Data folder.")
    return path

## Function for date column presence
def date_column(raw):
    options = [col for col in raw.columns if "date" in col.lower()]
```

```
    return options[0] if options else raw.columns[0]

## Function for claims column checking
def claims_column(raw, code, date_col):
    if code in raw.columns:
        return code
    options = [col for col in raw.columns if col != date_col]
    if not options:
        raise ValueError(f"There were no claims column found for {code}.")
    return options[0]

## Function to return base claims in dataframe
def base_claims_frame(raw, code):
    date_col = date_column(raw)
    value_col = claims_column(raw, code, date_col)
    return pd.DataFrame({
        "Date": pd.to_datetime(raw[date_col], errors="coerce"),
        "Claims_Raw": pd.to_numeric(raw[value_col], errors="coerce"),
        "StateCode": code,
        "State": state_names[code]
    })

## Function to have cleaned claims dataframe
def clean_claims_frame(df):
    df = df.sort_values("Date").drop_duplicates(subset=["Date"], keep="first").
 ↪reset_index(drop=True)
    df["Claims_Was_Missing"] = df["Claims_Raw"].isna()
    df["Claims"] = df["Claims_Raw"].interpolate(method="linear",␣
 ↪limit_direction="both")
    return df

## Function that takes raw values and cleans them using prior function
def load_and_clean_claims(code):
    raw = pd.read_csv(claims_path(code))
    raw.columns = raw.columns.astype(str).str.strip()
    return clean_claims_frame(base_claims_frame(raw, code))
```

One important thing to note about the above `clean_claims_frame` function. Specifically, with our current structure, we have that the raw FRED CSV values end up being preserved in the `Claims_Raw` column. However, as some observations are found to be missing in the original claims column, i.e. Hawaii, a flag is generated called `Claims_Was_Missing` to indicate problem rows. Additionally, for the purpose of good modeling, a cleaned variable titled `Claims` is generated in which the aforementioned problem values are filled-in through the use of linear interpolation within each state's time series. Interpolated values are NOT the same as original reported claims values, instead being a manual pre-processing step that we take such that we can actually run time-series models on complete weekly data.

2.2 2. Loading All of Our Relevant State Files

This cell works to generate a single cleaned dataframe for each state, afterwards storing them in an object titled `claims_dfs`. We also have another object, `combined_claims`, which is a stacked dataframe that is useful for relevant EDA work to produce tables and plots of interest.

Code cell purpose: The cell below simply loads the four relevant state files, stacks them such that there is only one dataframe for analysis, and then generates date-based variables which are useful for EDA work. This matters because the later EDA and modeling sections all assume the same cleaned, stacked data structure with consistent date fields.

[3]:
```
## Generating the object
claims_dfs = {}

## Working to loop through the datasets, making claims_dfs dataset
for code in datasets:
    df = load_and_clean_claims(code)
    claims_dfs[code] = df
    globals()[code] = df

## Generated the combined_claims dataset
combined_claims = pd.concat(claims_dfs.values(), ignore_index=True)
combined_claims["Year"] = combined_claims["Date"].dt.year
combined_claims["Month"] = combined_claims["Date"].dt.month
combined_claims["Week"] = combined_claims["Date"].dt.isocalendar().week.
 ↪astype(int)

print("Loaded datasets:")
for code in datasets:
    print(f"{code}: {claims_dfs[code].shape[0]:,} rows")

display(combined_claims.head())
```

```
Loaded datasets:
CAICLAIMS: 2,097 rows
NYICLAIMS: 2,096 rows
INICLAIMS: 2,096 rows
HIICLAIMS: 2,141 rows
```

	Date	Claims_Raw	StateCode	State	Claims_Was_Missing	Claims	\
0	1986-02-08	75,972.00	CAICLAIMS	California	False	75,972.00	
1	1986-02-15	54,595.00	CAICLAIMS	California	False	54,595.00	
2	1986-02-22	68,513.00	CAICLAIMS	California	False	68,513.00	
3	1986-03-01	65,424.00	CAICLAIMS	California	False	65,424.00	
4	1986-03-08	66,436.00	CAICLAIMS	California	False	66,436.00	

	Year	Month	Week
0	1986	2	6
1	1986	2	7

```
2  1986       2      8
3  1986       3      9
4  1986       3     10
```

2.3 3. Checking General Data Quality

Prior to extensive modeling, we check to ensure that each of the four time series has usable structure. Specifically, we look to ensure that there is only one observation each week, valid dates, no duplicates, numeric and non-negative claims, etc. Additionally, work is done to record raw missing claims values, i.e. in the case of Hawaii where there are indeed some missing observations.

Code cell purpose: Cell below works to generate a basic data-quality table. Utilized for verification purposes, mainly in regard to dates, missing values, duplicates, negative values, etc. This matters because confirming these basics early helps ensure later forecast errors reflect model performance rather than data-format problems.

[4]:
```
## Generating the object
quality_rows = []

## Looping through each CSV such that we have our cleaned dataset
for code in datasets:
    df = claims_dfs[code].copy()
    date_diffs = df["Date"].diff().dropna().dt.days

    quality_rows.append({
        "dataset": code,
        "state": state_names[code],
        "rows": len(df),
        "start_date": df["Date"].min().date(),
        "end_date": df["Date"].max().date(),
        "missing_dates": int(df["Date"].isna().sum()),
        "missing_claims_raw": int(df["Claims_Raw"].isna().sum()),
        "missing_claims_after_clean": int(df["Claims"].isna().sum()),
        "interpolated_claims": int(df["Claims_Was_Missing"].sum()),
        "duplicate_dates_after_clean": int(df["Date"].duplicated().sum()),
        "negative_claims": int((df["Claims"] < 0).sum()),
        "nonweekly_gaps": int((date_diffs != 7).sum()),
        "min_gap_days": date_diffs.min() if len(date_diffs) > 0 else np.nan,
        "max_gap_days": date_diffs.max() if len(date_diffs) > 0 else np.nan
    })

## Displaying results after converting to dataframe, seeing how much is␣
 ↪missing, negative, duplicated, etc.
quality_summary = pd.DataFrame(quality_rows)
display(quality_summary)
```

```
     dataset       state  rows  start_date    end_date  missing_dates  \
0  CAICLAIMS  California  2097  1986-02-08  2026-04-11              0
1  NYICLAIMS    New York  2096  1986-02-15  2026-04-11              0
2  INICLAIMS     Indiana  2096  1986-02-15  2026-04-11              0
3  HIICLAIMS      Hawaii  2141  1985-04-06  2026-04-11              0

   missing_claims_raw  missing_claims_after_clean  interpolated_claims  \
```

```
0                   5                         0                   5
1                   0                         0                   0
2                   0                         0                   0
3                  48                         0                  48

   duplicate_dates_after_clean  negative_claims  nonweekly_gaps  min_gap_days  \
0                            0               0               0             7
1                            0               0               0             7
2                            0               0               0             7
3                            0               0               0             7

   max_gap_days
0             7
1             7
2             7
3             7
```

2.4 4. Inspecting the Missing Values Present

Below, we have a table which showcases the original rows that contain missing values for the claims prior to interpolation/estimation. Important to have this such that both us (and others working through the notebook) are able to understand which specific rows are odd observations.

Code cell purpose: Cell below provides transparency in our cleaning process, mainly by listing out the rows which contained missing claims prior to any interpolation. This matters because the notebook uses interpolated modeling values later, so we need to clearly show which observations were originally missing.

[5]:
```
## Defining our missing_rows object accordingly
missing_rows = (
    combined_claims
    .loc[combined_claims["Claims_Was_Missing"], ["State", "StateCode", "Date",␣
 ↪"Claims_Raw", "Claims"]]
    .sort_values(["State", "Date"])
)

## Getting the actual amount
print(f"Total originally missing claims values: {len(missing_rows):,}")

## Listing out all rows missing raw claims
if len(missing_rows) > 0:
    display(missing_rows.head(20))
else:
    print("There were no raw claims values which were missing found here.")
```

```
Total originally missing claims values: 53
```

	State	StateCode	Date	Claims_Raw	Claims
58	California	CAICLAIMS	1987-03-21	NaN	54,067.00
73	California	CAICLAIMS	1987-07-04	NaN	47,279.60
74	California	CAICLAIMS	1987-07-11	NaN	47,188.20
75	California	CAICLAIMS	1987-07-18	NaN	47,096.80
76	California	CAICLAIMS	1987-07-25	NaN	47,005.40
6290	Hawaii	HIICLAIMS	1985-04-13	NaN	1,365.96
6291	Hawaii	HIICLAIMS	1985-04-20	NaN	1,361.91
6292	Hawaii	HIICLAIMS	1985-04-27	NaN	1,357.87
6293	Hawaii	HIICLAIMS	1985-05-04	NaN	1,353.82
6294	Hawaii	HIICLAIMS	1985-05-11	NaN	1,349.78
6295	Hawaii	HIICLAIMS	1985-05-18	NaN	1,345.73
6296	Hawaii	HIICLAIMS	1985-05-25	NaN	1,341.69
6297	Hawaii	HIICLAIMS	1985-06-01	NaN	1,337.64
6298	Hawaii	HIICLAIMS	1985-06-08	NaN	1,333.60
6299	Hawaii	HIICLAIMS	1985-06-15	NaN	1,329.56
6300	Hawaii	HIICLAIMS	1985-06-22	NaN	1,325.51
6301	Hawaii	HIICLAIMS	1985-06-29	NaN	1,321.47
6302	Hawaii	HIICLAIMS	1985-07-06	NaN	1,317.42

6303	Hawaii	HIICLAIMS	1985-07-13	NaN	1,313.38
6304	Hawaii	HIICLAIMS	1985-07-20	NaN	1,309.33

2.5 5. Details Regarding Duplicates and Gaps in Dates

This cell works to display any cases in which we have state-date duplicates or non-weekly date gaps, both of which are major issues for our procedures. Desire no problem rows after cleaning process.

Code cell purpose: Cell below directly prints the checks for duplicate date and non-weekly gaps in dates. This matters because weekly forecasting models require an evenly spaced time index with one observation per state-week.

[6]:
```
## Looping through each of our four states and ensuring that there are no␣
 ↪problem rows after cleaning
for code in datasets:
    df = claims_dfs[code].copy()

    ## Checking the duplicated dates after cleaning (want NONE)
    duplicated_dates = df.loc[df["Date"].duplicated(keep=False), ["Date",␣
 ↪"Claims"]]

    ## Checking the date gaps other than exactly 7 days (want NONE)
    gap_check = df[["Date"]].copy()
    gap_check["gap_days"] = gap_check["Date"].diff().dt.days
    nonweekly_gaps = gap_check.loc[gap_check["gap_days"].notna() &␣
 ↪(gap_check["gap_days"] != 7)]

    print(f"\n{code} - {state_names[code]}")
    print(f"Duplicated dates after cleaning: {len(duplicated_dates):,}")
    print(f"Non-weekly gaps after cleaning: {len(nonweekly_gaps):,}")

    if len(nonweekly_gaps) > 0:
        display(nonweekly_gaps.head(10))
```

```
CAICLAIMS - California
Duplicated dates after cleaning: 0
Non-weekly gaps after cleaning: 0

NYICLAIMS - New York
Duplicated dates after cleaning: 0
Non-weekly gaps after cleaning: 0

INICLAIMS - Indiana
Duplicated dates after cleaning: 0
Non-weekly gaps after cleaning: 0

HIICLAIMS - Hawaii
Duplicated dates after cleaning: 0
Non-weekly gaps after cleaning: 0
```

2.6 6. Previewing Rows and Exploring Basic Shape of Dataset

Here, we go through and complete a quick row-level sanity check for each of the four states. What makes this useful is that, through such checks, we confirm that date range and claims values are logical and look reasonable for use later on, saving us trouble when attempting to run more complex summaries.

Code cell purpose: Below, the cell simply previews the initial and last rows for each state to make sure that our loaded series appear logical. This matters because a simple first-and-last-row check can reveal date parsing or ordering issues before deeper summary statistics are produced.

[7]:
```
## Loading the first and last couple rows for each state's claims
for code in datasets:
    print(f"\n{code} - {state_names[code]}")
    display(claims_dfs[code].head(5))
    display(claims_dfs[code].tail(5))
```

```
CAICLAIMS - California
```

	Date	Claims_Raw	StateCode	State	Claims_Was_Missing	Claims
0	1986-02-08	75,972.00	CAICLAIMS	California	False	75,972.00
1	1986-02-15	54,595.00	CAICLAIMS	California	False	54,595.00
2	1986-02-22	68,513.00	CAICLAIMS	California	False	68,513.00
3	1986-03-01	65,424.00	CAICLAIMS	California	False	65,424.00
4	1986-03-08	66,436.00	CAICLAIMS	California	False	66,436.00

	Date	Claims_Raw	StateCode	State	Claims_Was_Missing	\
2092	2026-03-14	38,273.00	CAICLAIMS	California	False	
2093	2026-03-21	37,819.00	CAICLAIMS	California	False	
2094	2026-03-28	38,010.00	CAICLAIMS	California	False	
2095	2026-04-04	40,140.00	CAICLAIMS	California	False	
2096	2026-04-11	40,874.00	CAICLAIMS	California	False	

	Claims
2092	38,273.00
2093	37,819.00
2094	38,010.00
2095	40,140.00
2096	40,874.00

```
NYICLAIMS - New York
```

	Date	Claims_Raw	StateCode	State	Claims_Was_Missing	Claims
0	1986-02-15	20647	NYICLAIMS	New York	False	20647
1	1986-02-22	21499	NYICLAIMS	New York	False	21499
2	1986-03-01	21738	NYICLAIMS	New York	False	21738
3	1986-03-08	22049	NYICLAIMS	New York	False	22049
4	1986-03-15	20144	NYICLAIMS	New York	False	20144

```
           Date  Claims_Raw  StateCode     State  Claims_Was_Missing  Claims
2091 2026-03-14       13862  NYICLAIMS  New York               False   13862
2092 2026-03-21       13699  NYICLAIMS  New York               False   13699
2093 2026-03-28       14935  NYICLAIMS  New York               False   14935
2094 2026-04-04       13343  NYICLAIMS  New York               False   13343
2095 2026-04-11       21488  NYICLAIMS  New York               False   21488
```

INICLAIMS - Indiana

```
        Date  Claims_Raw  StateCode    State  Claims_Was_Missing  Claims
0 1986-02-15        8089  INICLAIMS  Indiana               False    8089
1 1986-02-22        7304  INICLAIMS  Indiana               False    7304
2 1986-03-01        7746  INICLAIMS  Indiana               False    7746
3 1986-03-08        7724  INICLAIMS  Indiana               False    7724
4 1986-03-15        6279  INICLAIMS  Indiana               False    6279
```

```
           Date  Claims_Raw  StateCode    State  Claims_Was_Missing  Claims
2091 2026-03-14        2608  INICLAIMS  Indiana               False    2608
2092 2026-03-21        2418  INICLAIMS  Indiana               False    2418
2093 2026-03-28        2279  INICLAIMS  Indiana               False    2279
2094 2026-04-04        2727  INICLAIMS  Indiana               False    2727
2095 2026-04-11        3629  INICLAIMS  Indiana               False    3629
```

HIICLAIMS - Hawaii

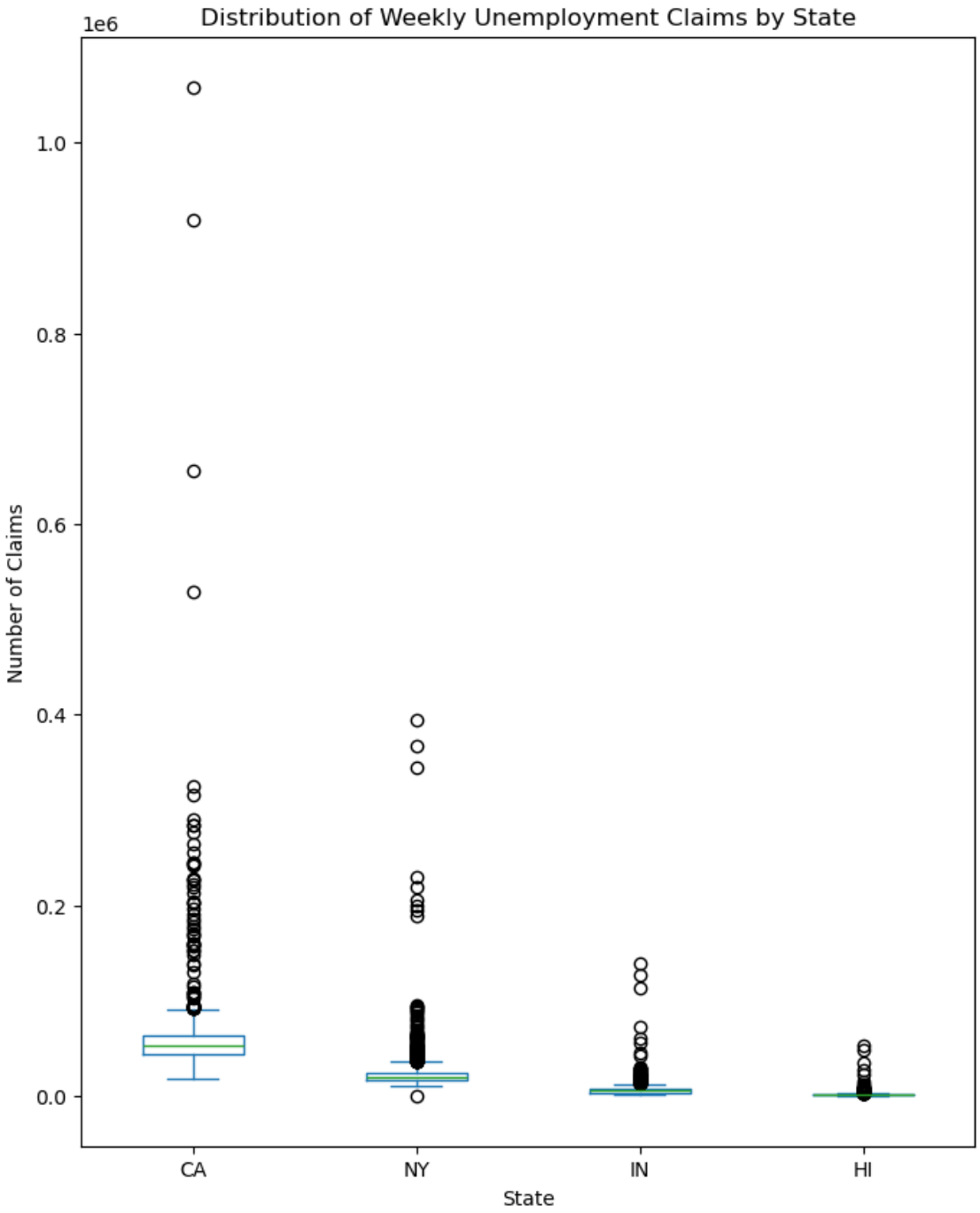
```
        Date  Claims_Raw  StateCode   State  Claims_Was_Missing   Claims
0 1985-04-06    1,370.00  HIICLAIMS  Hawaii               False 1,370.00
1 1985-04-13         NaN  HIICLAIMS  Hawaii                True 1,365.96
2 1985-04-20         NaN  HIICLAIMS  Hawaii                True 1,361.91
3 1985-04-27         NaN  HIICLAIMS  Hawaii                True 1,357.87
4 1985-05-04         NaN  HIICLAIMS  Hawaii                True 1,353.82
```

```
           Date  Claims_Raw  StateCode   State  Claims_Was_Missing   Claims
2136 2026-03-14    1,025.00  HIICLAIMS  Hawaii               False 1,025.00
2137 2026-03-21    1,597.00  HIICLAIMS  Hawaii               False 1,597.00
2138 2026-03-28    1,548.00  HIICLAIMS  Hawaii               False 1,548.00
2139 2026-04-04    1,126.00  HIICLAIMS  Hawaii               False 1,126.00
2140 2026-04-11    1,072.00  HIICLAIMS  Hawaii               False 1,072.00
```

Code cell purpose: Looking at some distributions. This matters because claims levels differ sharply by state, so distribution plots help us understand scale, skew, and outliers before modeling.

[8]:
```
## Setting up a basic dataframe object to work with
df_all = pd.concat([claims_dfs[code]["Claims"] for code in datasets],axis=1)
df_all.columns = ["CA", "NY", "IN", "HI"]

## Boxplot for distribution of weekly unemployment claims per state
df_all.plot(kind="box", figsize=(8,10))
plt.title("Distribution of Weekly Unemployment Claims by State")
```

```
plt.ylabel("Number of Claims")
plt.xlabel("State")
plt.show()
```

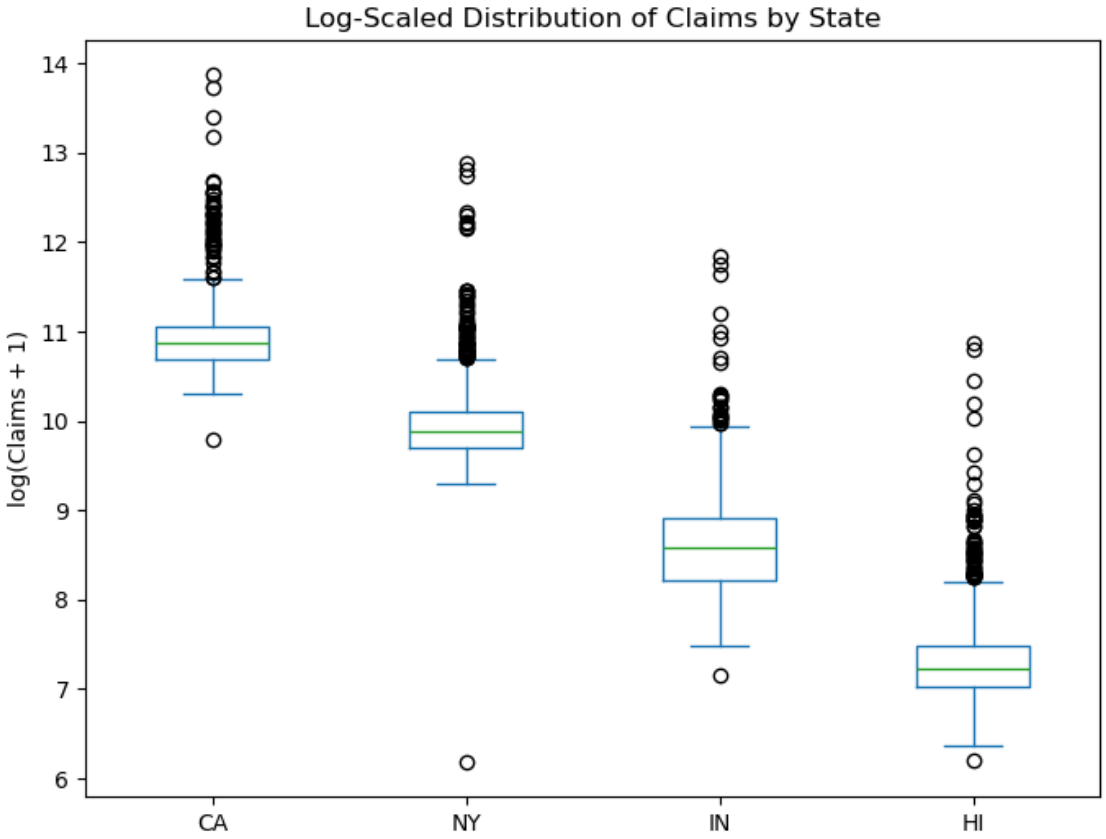


The summary statistics for `df_all` show the distribution of weekly unemployment claims by state. California has the highest median and maximum number of unemployment claims, and also has

the widest spread and the highest outliers. But California's large number of unemployment claims makes it difficult to see the plots for other graphs. Let's graph these plots on a log scale, as seen below.

Code cell purpose: What: This cell creates a log-transformed boxplot of weekly claims across states. Why: The log scale makes the cross-state distribution easier to compare when California's much larger claim counts otherwise dominate the raw-scale plot.

[9]:
```
## Log scale
df_all_log = np.log1p(df_all)
df_all_log.plot(kind="box", figsize=(8,6))

## Setting up plot
plt.title("Log-Scaled Distribution of Claims by State")
plt.ylabel("log(Claims + 1)")
plt.show()
```



California dominates due to much higher absolute claim counts, but the log scale reveals underlying distribution patterns more clearly. Some states show slightly wider boxes and longer whiskers, showing greater relative volatility in claims. Outliers still remain after log-scaling, highlighting extreme economic shocks on a macro scale (ex. COVID-19 pandemic).

Code cell purpose: What: This cell checks the data types in the wide claims dataframe. Why: Verifying the column types confirms that the state series are numeric and ready for summary statistics, plots, and time-series modeling.

[10]:
```
df_all.dtypes
```

[10]:
```
CA    float64
NY    float64
IN    float64
HI    float64
dtype: object
```

The number of claims for each state is a float, which means it can be easily manipulated for computations.

2.7 7. Producing Summary Statistics

Here, we work to produce summary statistics, those of which are useful as they provide us with a scope of the basic scale of claims existing in each state. Prior to any analysis, we naturally expect states like California and New York to have much larger claims as compared to Hawaii and Indiana, as the latter states have much smaller labor markets.

Code cell purpose: Below, we have a cell which calculates relevant summary statistics per state, including spread, raw missing counts, etc. This matters because these statistics provide a compact comparison of each state's claims scale and volatility, which helps interpret forecast RMSE later.

[11]:
```
## Simply getting summary statistics
summary_stats = (
    combined_claims
    .groupby("State")
    .agg(
        observations=("Claims", "size"),
        raw_missing_claims=("Claims_Was_Missing", "sum"),
        mean_claims=("Claims", "mean"),
        median_claims=("Claims", "median"),
        sd_claims=("Claims", "std"),
        min_claims=("Claims", "min"),
        p25_claims=("Claims", lambda x: x.quantile(0.25)),
        p75_claims=("Claims", lambda x: x.quantile(0.75)),
        max_claims=("Claims", "max")
    )
    .reset_index()
)

## Adding a coefficient of variation to compare the volatility across␣
 ↪differently sized states
summary_stats["cv_claims"] = summary_stats["sd_claims"] /␣
 ↪summary_stats["mean_claims"]

## Displaying
display(summary_stats.round(2))
```

	State	observations	raw_missing_claims	mean_claims	median_claims	\
0	California	2097	5	58,606.31	52,384.00	
1	Hawaii	2141	48	1,673.52	1,385.00	
2	Indiana	2096	0	6,547.56	5,379.00	
3	New York	2096	0	22,871.85	19,493.50	

	sd_claims	min_claims	p25_claims	p75_claims	max_claims	cv_claims
0	42,137.74	18,046.00	44,001.00	63,122.00	1,058,325.00	0.72
1	2,028.86	495.00	1,118.00	1,790.00	53,102.00	1.21
2	6,415.86	1,279.00	3,685.75	7,384.50	139,174.00	0.98
3	19,048.16	481.00	16,377.50	24,292.75	394,701.00	0.83

2.8 8. Analyzing Annual Trends

Here, we look at an annual summary for each of the state's claims, that of which provides good insight into how the average and maximum claims have changed progressively. What is most helpful to pull from these results include data for years close to recession periods/the COVID spike, helping us with later model design/validation below.

Code cell purpose: Below, we have a cell which summarizes the claims data per year for each state, helping reveal certain long-term trends. This matters because annual aggregation makes long-run trends and major macro shocks easier to see than weekly values alone.

[12]:
```
## Generating the annual summary
annual_summary = (
    combined_claims
    .groupby(["State", "Year"])
    .agg(
        avg_claims=("Claims", "mean"),
        total_claims=("Claims", "sum"),
        max_claims=("Claims", "max"),
        weeks_observed=("Claims", "size")
    )
    .reset_index()
)

## Displaying most recent years first as these are the most relevant for␣
 ↪forecasting
for code in datasets:
        display(annual_summary[annual_summary["State"] == state_names[code]].
 ↪sort_values("Year", ascending=False).head(20).round(2))
```

	State	Year	avg_claims	total_claims	max_claims	weeks_observed
40	California	2026	43,231.33	648,470.00	55,731.00	15
39	California	2025	43,392.56	2,256,413.00	59,969.00	52
38	California	2024	43,437.54	2,258,752.00	52,368.00	52
37	California	2023	44,741.08	2,326,536.00	63,697.00	52
36	California	2022	43,004.45	2,279,236.00	62,274.00	53
35	California	2021	76,080.29	3,956,175.00	182,625.00	52
34	California	2020	221,880.15	11,537,768.00	1,058,325.00	52
33	California	2019	40,371.40	2,099,313.00	56,886.00	52
32	California	2018	40,750.71	2,119,037.00	58,962.00	52
31	California	2017	43,307.54	2,251,992.00	61,914.00	52
30	California	2016	45,280.87	2,399,886.00	74,705.00	53
29	California	2015	46,389.08	2,412,232.00	66,906.00	52
28	California	2014	55,471.42	2,884,514.00	74,898.00	52
27	California	2013	56,447.40	2,935,265.00	83,383.00	52
26	California	2012	58,277.92	3,030,452.00	84,249.00	52
25	California	2011	64,577.19	3,422,591.00	90,829.00	53
24	California	2010	74,068.27	3,851,550.00	115,462.00	52
23	California	2009	74,601.17	3,879,261.00	95,705.00	52

22	California	2008	57,523.23	2,991,208.00	87,877.00	52
21	California	2007	43,961.94	2,286,021.00	56,753.00	52

	State	Year	avg_claims	total_claims	max_claims	weeks_observed
164	New York	2026	20,277.80	304,167.00	32,714.00	15
163	New York	2025	16,557.25	860,977.00	37,298.00	52
162	New York	2024	15,962.19	830,034.00	37,984.00	52
161	New York	2023	16,369.79	851,229.00	39,050.00	52
160	New York	2022	16,165.94	856,795.00	37,109.00	53
159	New York	2021	29,802.06	1,549,707.00	77,390.00	52
158	New York	2020	90,551.27	4,708,666.00	394,701.00	52
157	New York	2019	15,844.73	823,926.00	45,065.00	52
156	New York	2018	16,302.25	847,717.00	49,361.00	52
155	New York	2017	17,872.71	929,381.00	46,325.00	52
154	New York	2016	18,785.21	995,616.00	41,160.00	53
153	New York	2015	19,848.08	1,032,100.00	41,489.00	52
152	New York	2014	21,618.27	1,124,150.00	52,096.00	52
151	New York	2013	23,564.46	1,225,352.00	62,285.00	52
150	New York	2012	26,172.46	1,360,968.00	63,292.00	52
149	New York	2011	26,332.45	1,395,620.00	61,983.00	53
148	New York	2010	27,214.19	1,415,138.00	55,930.00	52
147	New York	2009	30,140.31	1,567,296.00	54,805.00	52
146	New York	2008	22,958.40	1,193,837.00	53,095.00	52
145	New York	2007	19,000.92	988,048.00	51,348.00	52

	State	Year	avg_claims	total_claims	max_claims	weeks_observed
123	Indiana	2026	3,403.47	51,052.00	7,494.00	15
122	Indiana	2025	2,966.54	154,260.00	5,904.00	52
121	Indiana	2024	3,376.79	175,593.00	5,910.00	52
120	Indiana	2023	3,673.87	191,041.00	6,934.00	52
119	Indiana	2022	4,798.74	254,333.00	11,926.00	53
118	Indiana	2021	7,606.96	395,562.00	17,724.00	52
117	Indiana	2020	24,371.67	1,267,327.00	139,174.00	52
116	Indiana	2019	2,635.83	137,063.00	4,835.00	52
115	Indiana	2018	2,625.15	136,508.00	6,372.00	52
114	Indiana	2017	2,950.92	153,448.00	6,168.00	52
113	Indiana	2016	3,650.38	193,470.00	11,082.00	53
112	Indiana	2015	4,136.67	215,107.00	13,722.00	52
111	Indiana	2014	4,869.83	253,231.00	14,668.00	52
110	Indiana	2013	6,050.40	314,621.00	15,990.00	52
109	Indiana	2012	6,799.29	353,563.00	14,189.00	52
108	Indiana	2011	7,841.75	415,613.00	20,075.00	53
107	Indiana	2010	8,502.52	442,131.00	17,155.00	52
106	Indiana	2009	13,256.38	689,332.00	28,616.00	52
105	Indiana	2008	11,566.79	601,473.00	27,937.00	52
104	Indiana	2007	7,850.71	408,237.00	17,366.00	52

	State	Year	avg_claims	total_claims	max_claims	weeks_observed
82	Hawaii	2026	1,198.80	17,982.00	1,747.00	15

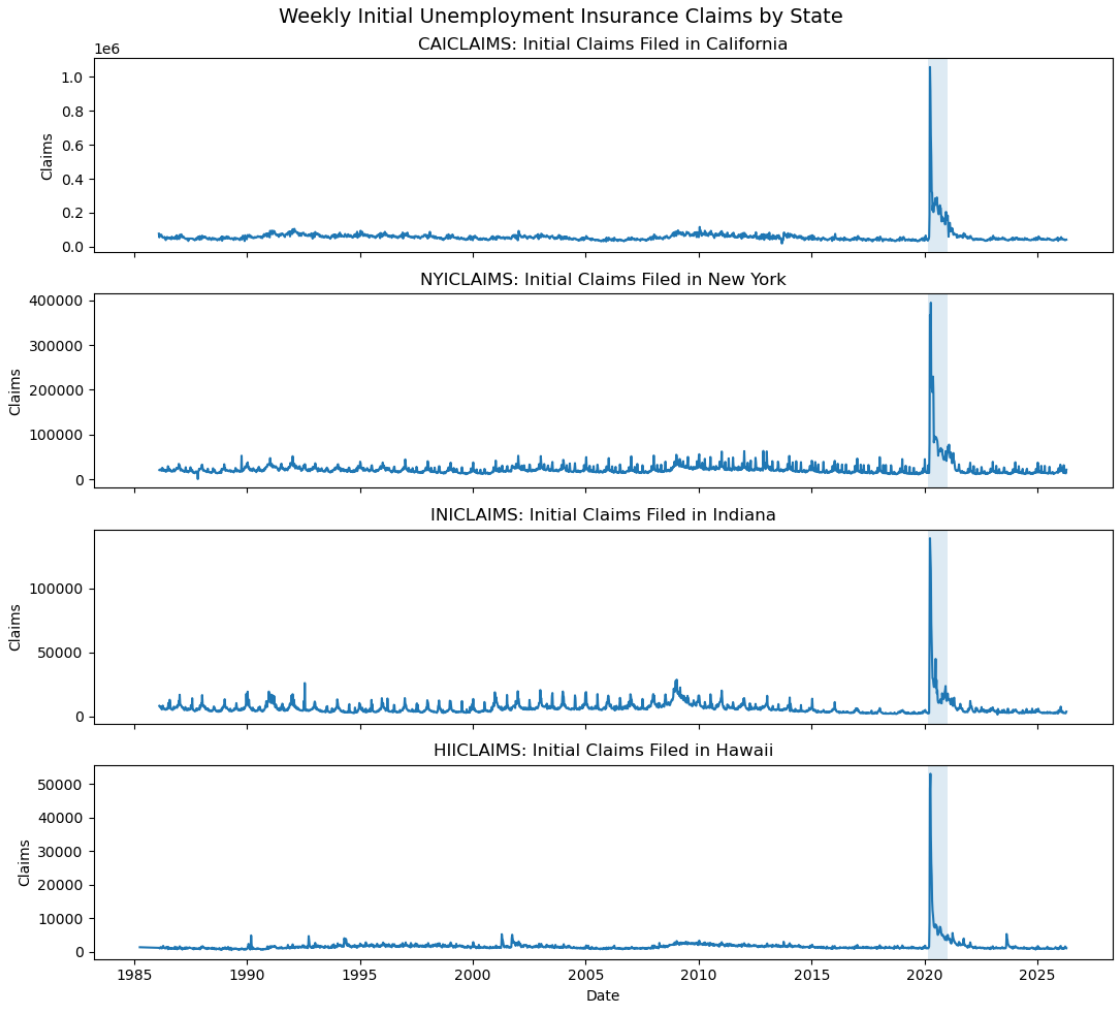
81	Hawaii	2025	1,026.79	53,393.00	1,718.00	52
80	Hawaii	2024	1,081.06	56,215.00	1,729.00	52
79	Hawaii	2023	1,433.62	74,548.00	5,301.00	52
78	Hawaii	2022	1,254.64	66,496.00	2,836.00	53
77	Hawaii	2021	2,694.88	140,134.00	5,598.00	52
76	Hawaii	2020	8,376.58	435,582.00	53,102.00	52
75	Hawaii	2019	1,187.52	61,751.00	1,903.00	52
74	Hawaii	2018	1,250.37	65,019.00	1,956.00	52
73	Hawaii	2017	1,250.31	65,016.00	2,058.00	52
72	Hawaii	2016	1,218.47	64,579.00	1,714.00	53
71	Hawaii	2015	1,296.31	67,408.00	2,123.00	52
70	Hawaii	2014	1,586.35	82,490.00	2,558.00	52
69	Hawaii	2013	1,784.06	92,771.00	3,005.00	52
68	Hawaii	2012	1,769.83	92,031.00	2,566.00	52
67	Hawaii	2011	2,004.06	106,215.00	2,875.00	53
66	Hawaii	2010	2,221.31	115,508.00	3,358.00	52
65	Hawaii	2009	2,509.06	130,471.00	3,211.00	52
64	Hawaii	2008	1,758.21	91,427.00	2,809.00	52
63	Hawaii	2007	1,112.04	57,826.00	1,869.00	52

2.9 9. Generating Time-Series Plots by State

Here, we generate time-series plots for each state, with the first plot showcasing each state separately (do not want California's large scale taking over visualization of smaller states). The second plot has all states overlaid in levels, allowing us to compare timing of spikes.

Code cell purpose: Here, we have both separated and overlaid visualizations of time series for the states. Directly below is separate. This matters because separate plots keep smaller states readable while still showing each state's overall time-series pattern.

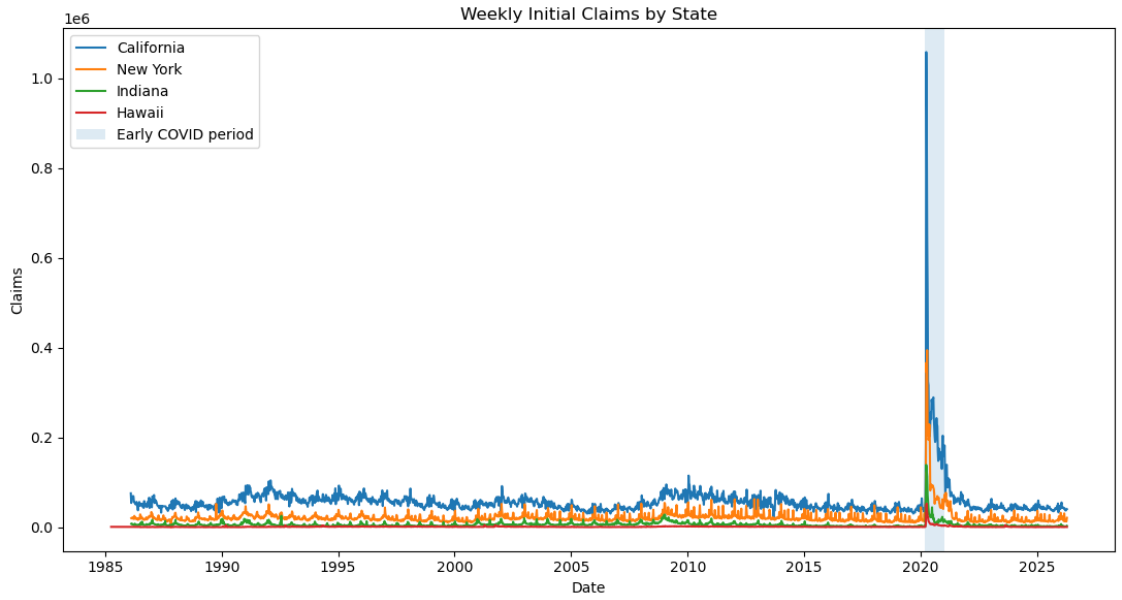
[13]:
```
## Setting figure specifications
fig, axes = plt.subplots(4, 1, figsize=(11, 10), sharex=True)

## Generating the figures
for ax, code in zip(axes, datasets):
    df = claims_dfs[code]
    ax.plot(df["Date"], df["Claims"])
    ax.set_title(f"{code}: Initial Claims Filed in {state_names[code]}")
    ax.set_ylabel("Claims")

    ## Marking the early COVID period as it is the largest structural break in␣
 ↪the series
    ax.axvspan(pd.Timestamp("2020-03-01"), pd.Timestamp("2021-01-01"), alpha=0.
 ↪15)

axes[-1].set_xlabel("Date")
fig.suptitle("Weekly Initial Unemployment Insurance Claims by State",␣
 ↪fontsize=14)
plt.tight_layout()
plt.show();
```



Code cell purpose: Directly below is overlaid. This matters because the overlaid plot shows the relative timing of spikes across states, even though raw levels are not directly comparable.

[14]:
```
## Setting figure specifications
fig, ax = plt.subplots(figsize=(11, 6))

## Generating the figures
for code in datasets:
    df = claims_dfs[code]
    ax.plot(df["Date"], df["Claims"], label=state_names[code])

ax.axvspan(pd.Timestamp("2020-03-01"), pd.Timestamp("2021-01-01"), alpha=0.15,␣
 ↪label="Early COVID period")
ax.set_xlabel("Date")
ax.set_ylabel("Claims")
ax.set_title("Weekly Initial Claims by State")
```

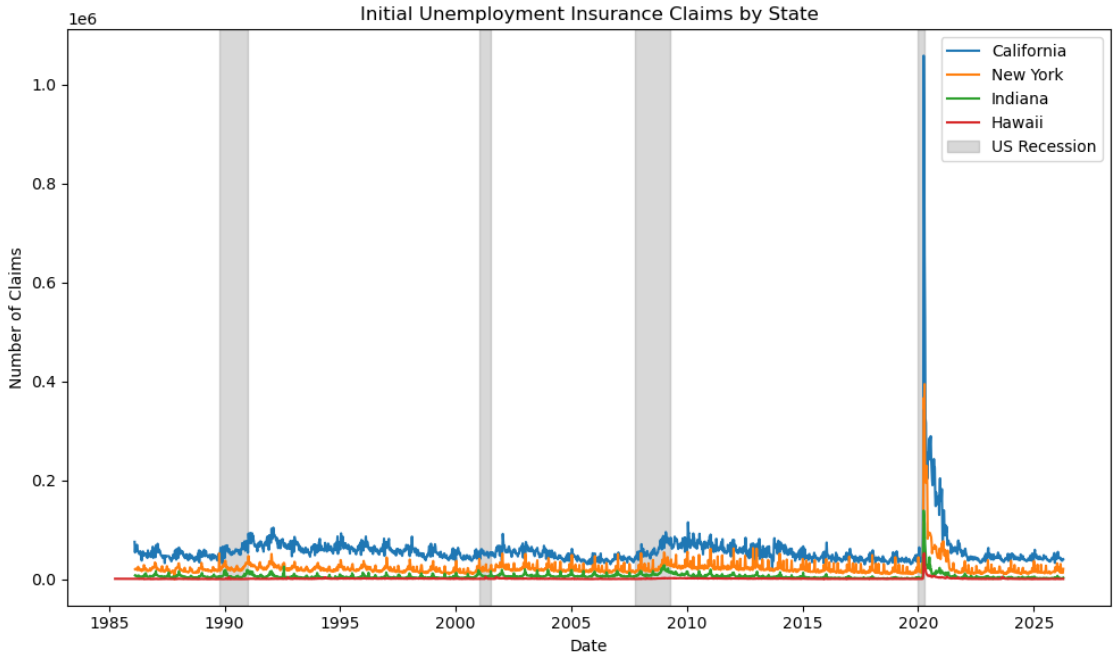
```
ax.legend()
plt.tight_layout()
plt.show();
```



Code cell purpose: Here, we have a visualization in which we can clearly see times of recessions shaded. This matters because recession shading connects claims spikes to national macroeconomic downturns and gives context for later forecast difficulty.

[15]:
```
## Looking at distinct visualization of recessions shaded

recessions = pd.read_csv("Data/JHDUSRGDPBR.csv")
recessions["observation_date"] = pd.to_datetime(recessions["observation_date"])

## Setting the recession dataset to use dates as its index
recessions = recessions.set_index("observation_date")

## Getting the full date range from the claims data using the Date column, not␣
 ↪the row index
start_date = min(claims_dfs[code]["Date"].min() for code in datasets)
end_date = max(claims_dfs[code]["Date"].max() for code in datasets)

## Keeping only recession data within the claims sample period
recessions = recessions.loc[start_date:end_date]

fig, ax = plt.subplots(figsize=(10, 6))

## Plotting weekly initial claims for each state
```

```
for code in datasets:
    df = claims_dfs[code]
    ax.plot(df["Date"], df["Claims"], label=state_names[code])

## Shading weeks/months classified as US recession periods
ax.fill_between(
    recessions.index,
    0,
    1,
    where=recessions["JHDUSRGDPBR"] == 1,
    color="gray",
    alpha=0.3,
    transform=ax.get_xaxis_transform(),
    label="US Recession"
)

## Formatting the figure
ax.set_title("Initial Unemployment Insurance Claims by State")
ax.set_xlabel("Date")
ax.set_ylabel("Number of Claims")
ax.legend()
plt.tight_layout()
plt.show()
```

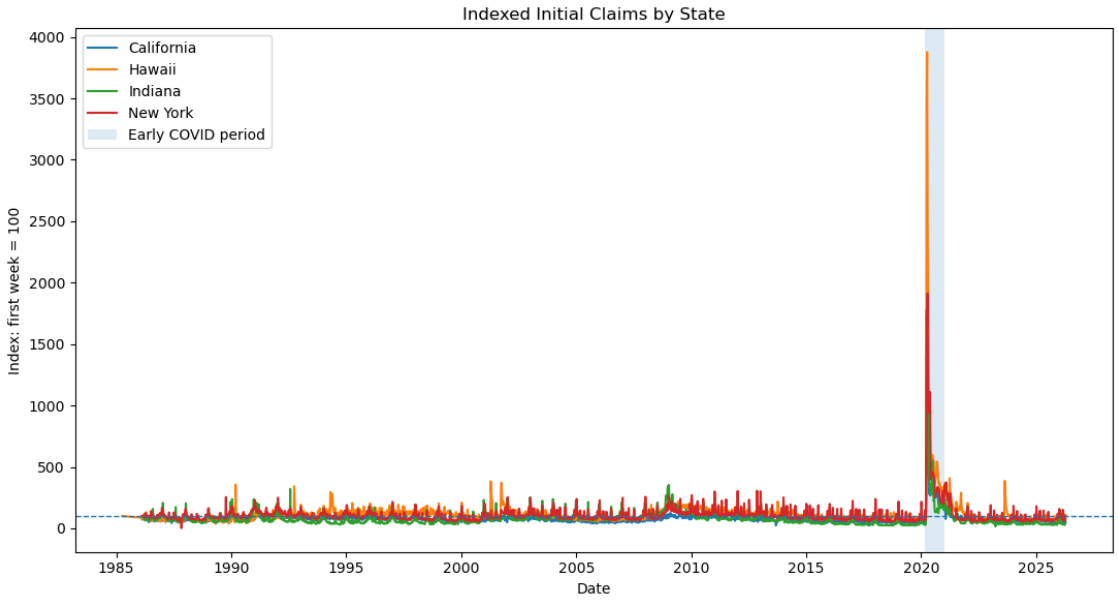


2.10 10. Having Indexed Comparison Across States

Each of the states has varied population sizes which are not directly comparable. Therefore, simply comparing raw claims would not be beneficial. Instead, we have an indexed plot which sets each state's first observed week equivalent to 100, such that the graph makes comparisons of relative changes rather than raw levels.

Code cell purpose: Below, we have a cell generating indexed claims such that states with varied baseline levels can still be directly compared on similar scales. This matters because indexing makes the states comparable by relative change rather than by raw population-size differences.

[16]:
```
## Generating the indexed claims below
indexed_claims = combined_claims.copy()
indexed_claims["Claims_Index"] = (
    indexed_claims
    .groupby("State")["Claims"]
    .transform(lambda x: 100 * x / x.iloc[0])
)

## Setting up figure
fig, ax = plt.subplots(figsize=(11, 6))

## Plot
for state, df in indexed_claims.groupby("State"):
    ax.plot(df["Date"], df["Claims_Index"], label=state)

ax.axhline(100, linestyle="--", linewidth=1)
ax.axvspan(pd.Timestamp("2020-03-01"), pd.Timestamp("2021-01-01"), alpha=0.15,␣
 ↪label="Early COVID period")
ax.set_xlabel("Date")
ax.set_ylabel("Index: first week = 100")
ax.set_title("Indexed Initial Claims by State")
ax.legend()
plt.tight_layout()
plt.show();
```

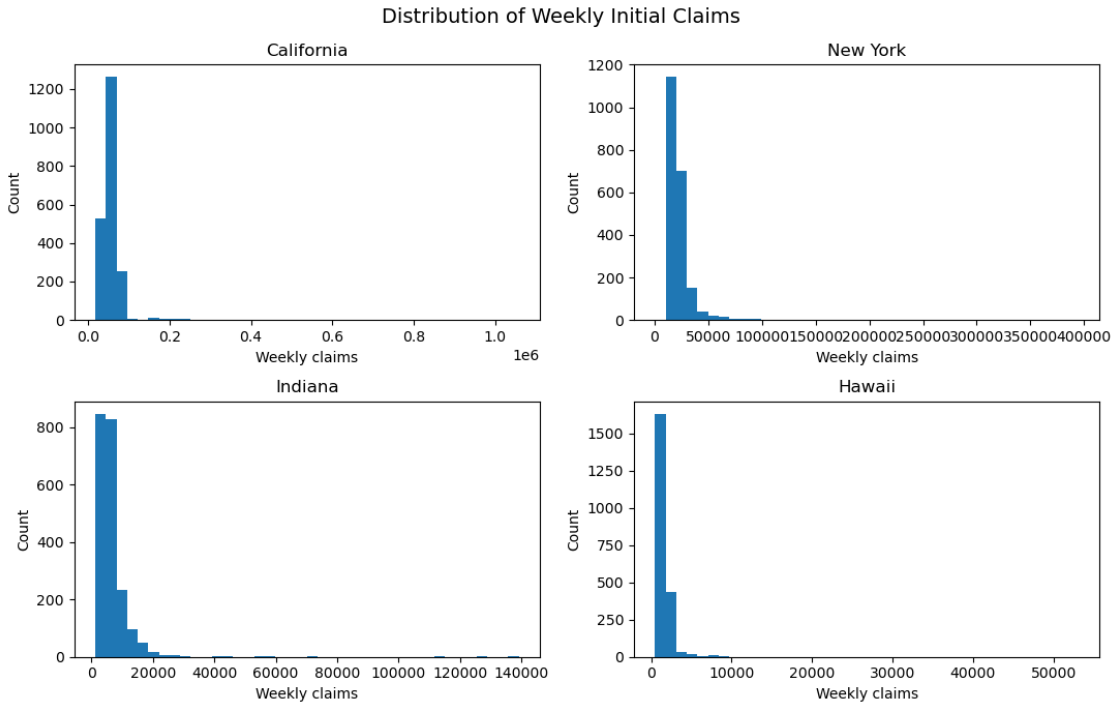

Indexed Initial Claims by State
California
Hawaii
Indiana
New York
Early COVID period
Index: first week = 100
Date
4000
3500
3000
2500
2000
1500
1000
500
0
1985
1990
1995
2000
2005
2010
2015
2020
2025

2.11 11. Reviewing the Distribution and Checking Outliers

Here, we felt it useful to have histograms generated that let us see the distribution of weekly claims for each state. In doing so, we can see things like heavy large right tails, something that checks out with basic sanity as unemployment claims tend to spike during recessions and times of stress like COVID (so occasionally we see singular major rises). Understanding these spikes is important so we are not confused later when modeling.

Code cell purpose: Below, the cell generates visualizations for each state's claims distribution, showcasing potential skews. This matters because distributional skew and extreme observations affect model fit, residual behavior, and forecast-error interpretation.

[17]:
```
## Setting up the figure
fig, axes = plt.subplots(2, 2, figsize=(11, 7))
axes = axes.flatten()

## Generating plot
for ax, code in zip(axes, datasets):
    df = claims_dfs[code]
    ax.hist(df["Claims"], bins=40)
    ax.set_title(state_names[code])
    ax.set_xlabel("Weekly claims")
    ax.set_ylabel("Count")

fig.suptitle("Distribution of Weekly Initial Claims", fontsize=14)
plt.tight_layout()
plt.show();
```



2.12 12. Looking Into the Largest Weekly Claims Observations

As the above distributions have a pretty distinct right-skew, we felt it important to directly identify the weeks with biggest claims for each state. Again, such rows are not problem rows or errors, per say, instead corresponding to things like recessions or COVID shocks. Again, this helps us begin thinking about how to approach structural breaks.

Code cell purpose: Cell below works to identify largest weekly claims observations in each state, separation between data errors from actual shocks. This matters because the largest observations are usually substantive economic shocks, especially COVID, rather than simple data mistakes.

[18]:
```
## Determining largest claims
top_claim_weeks = (
    combined_claims
    .sort_values("Claims", ascending=False)
    .loc[:, ["State", "StateCode", "Date", "Claims", "Year"]]
    .head(50)
)

## Displaying, and seeing that years tend to be COVID times and other recession␣
 ↪periods
display(top_claim_weeks)
```

	State	StateCode	Date	Claims	Year
1781	California	CAICLAIMS	2020-03-28	1,058,325.00	2020
1782	California	CAICLAIMS	2020-04-04	918,814.00	2020
1783	California	CAICLAIMS	2020-04-11	655,472.00	2020
1784	California	CAICLAIMS	2020-04-18	528,360.00	2020
3879	New York	NYICLAIMS	2020-04-11	394,701.00	2020
3877	New York	NYICLAIMS	2020-03-28	366,595.00	2020
3878	New York	NYICLAIMS	2020-04-04	344,451.00	2020
1785	California	CAICLAIMS	2020-04-25	325,343.00	2020
1786	California	CAICLAIMS	2020-05-02	316,257.00	2020
1797	California	CAICLAIMS	2020-07-18	289,594.00	2020
1796	California	CAICLAIMS	2020-07-11	284,914.00	2020
1793	California	CAICLAIMS	2020-06-20	284,494.00	2020
1794	California	CAICLAIMS	2020-06-27	277,362.00	2020
1795	California	CAICLAIMS	2020-07-04	264,791.00	2020
1791	California	CAICLAIMS	2020-06-06	255,836.00	2020
1798	California	CAICLAIMS	2020-07-25	244,653.00	2020
1788	California	CAICLAIMS	2020-05-16	244,431.00	2020
1804	California	CAICLAIMS	2020-09-05	243,404.00	2020
1792	California	CAICLAIMS	2020-06-13	241,424.00	2020
3884	New York	NYICLAIMS	2020-05-16	229,524.00	2020
1790	California	CAICLAIMS	2020-05-30	228,634.00	2020
1806	California	CAICLAIMS	2020-09-19	226,179.00	2020
1805	California	CAICLAIMS	2020-09-12	226,004.00	2020
1799	California	CAICLAIMS	2020-08-01	222,043.00	2020
1803	California	CAICLAIMS	2020-08-29	219,563.00	2020

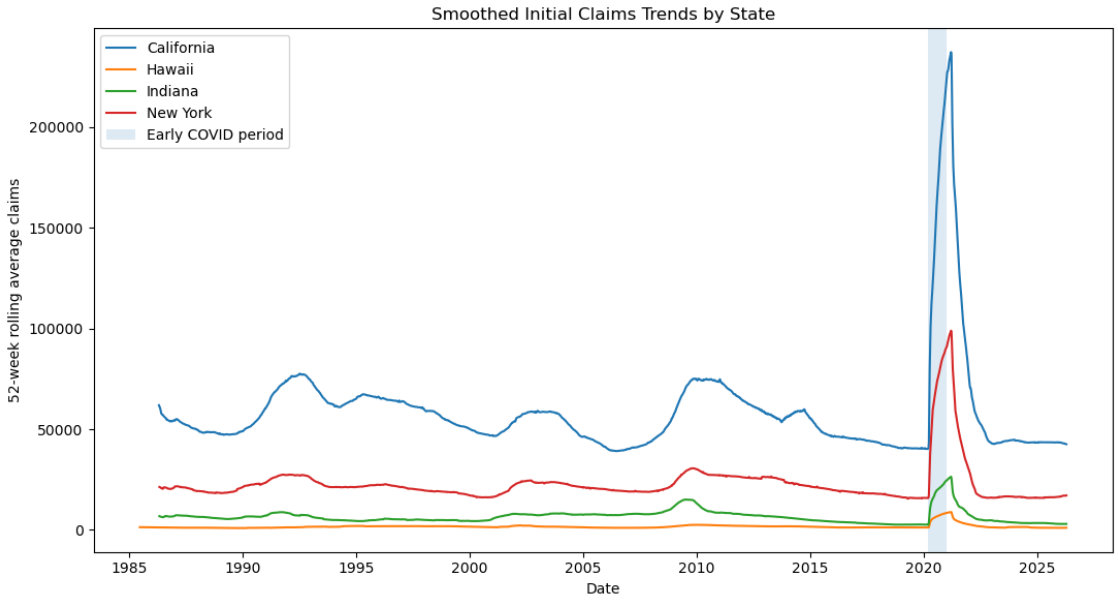
3881	New York	NYICLAIMS	2020-04-25	219,413.00	2020
1787	California	CAICLAIMS	2020-05-09	212,667.00	2020
3880	New York	NYICLAIMS	2020-04-18	205,184.00	2020
1818	California	CAICLAIMS	2020-12-12	204,388.00	2020
1789	California	CAICLAIMS	2020-05-23	203,262.00	2020
1800	California	CAICLAIMS	2020-08-08	202,509.00	2020
3883	New York	NYICLAIMS	2020-05-09	199,419.00	2020
1802	California	CAICLAIMS	2020-08-22	196,916.00	2020
3882	New York	NYICLAIMS	2020-05-02	195,110.00	2020
1801	California	CAICLAIMS	2020-08-15	190,354.00	2020
3885	New York	NYICLAIMS	2020-05-23	189,698.00	2020
1780	California	CAICLAIMS	2020-03-21	186,333.00	2020
1822	California	CAICLAIMS	2021-01-09	182,625.00	2021
1817	California	CAICLAIMS	2020-12-05	178,724.00	2020
1809	California	CAICLAIMS	2020-10-10	176,083.00	2020
1807	California	CAICLAIMS	2020-09-26	171,220.00	2020
1820	California	CAICLAIMS	2020-12-26	168,732.00	2020
1815	California	CAICLAIMS	2020-11-21	168,186.00	2020
1821	California	CAICLAIMS	2021-01-02	160,989.00	2021
1810	California	CAICLAIMS	2020-10-17	159,876.00	2020
1814	California	CAICLAIMS	2020-11-14	158,825.00	2020
1819	California	CAICLAIMS	2020-12-19	158,661.00	2020
1813	California	CAICLAIMS	2020-11-07	157,773.00	2020
1812	California	CAICLAIMS	2020-10-31	152,480.00	2020
1811	California	CAICLAIMS	2020-10-24	152,176.00	2020

2.13 13. Looking at Rolling-Average Trends

By looking at rolling average over 52 weeks rather than individual gaps, we can smooth out minor moments of volatility and instead look at where there are major changes in the long-run. Mainly for building understanding of the data we have prior to intense forecasting.

Code cell purpose: Cell below computes and plots 52-week rolling averages for each state, smoothing weekly volatility prior to modeling. This matters because rolling averages reveal medium-run claims patterns while reducing week-to-week noise before formal forecasting.

[19]:
```
## Generating rolling claims
rolling_claims = combined_claims.sort_values(["State", "Date"]).copy()
rolling_claims["Claims_52wk_MA"] = (
    rolling_claims
    .groupby("State")["Claims"]
    .transform(lambda x: x.rolling(window=52, min_periods=12).mean())
)

## Setting up figure
fig, ax = plt.subplots(figsize=(11, 6))

## Plot
for state, df in rolling_claims.groupby("State"):
    ax.plot(df["Date"], df["Claims_52wk_MA"], label=state)

ax.axvspan(pd.Timestamp("2020-03-01"), pd.Timestamp("2021-01-01"), alpha=0.15,␣
 ↪label="Early COVID period")
ax.set_xlabel("Date")
ax.set_ylabel("52-week rolling average claims")
ax.set_title("Smoothed Initial Claims Trends by State")
ax.legend()
plt.tight_layout()
plt.show();
```


Smoothed Initial Claims Trends by State
California
Hawaii
Indiana
New York
Early COVID period
52-week rolling average claims
200000
150000
100000
50000
0
1985
1990
1995
2000
2005
2010
2015
2020
2025
Date

2.14 14. Major Takeaways to Carry Forward

- We see that there are some missing values for states like Hawaii which needed to be interpolated. Details about process/logic in report
- No issue of duplicate, negative, or oddly-gapped claims data
- Clear difference in the level of claims for each state, matches with basic logic of larger states with larger labor markets have larger claims levels
- Understanding of spikes during shocks like COVID, periods of recession, etc. do appear quite clearly, basic logic checks out here

3 CAICLAIMS Model Testing - Determining Model Efficacy Prior to Scaling

This entire section works with California data as the initial example for forecasting, prior to scaling up to all of the four states we have. We utilized the final two years of data as a test set, fitting a basic ARIMA baseline, and afterwards adding SARIMAX specifications with Fourier seasonality and COVID indicators to deal with the structural break. RMSE is utilized as a measure of out-of-sample prediction accuracy, and we additionally have AIC/BIC as model-selection diagnostics.

3.1 1. Helper Funcrtions and Setup

3.1.1 Generic Helper Functions

Code cell purpose: What: This cell defines shared helper functions for RMSE, MAE, forecast tables, cache loading, rolling cross-validation slices, and SARIMAX grid-search summaries. Why: These utilities keep repeated modeling steps modular, reproducible, and scalable across the CA-only and all-state sections.

[20]:
```
## Shared modeling helper functions used in both CA-only and all-state sections
def calc_rmse(actual, predicted):
    return np.sqrt(mean_squared_error(actual, predicted))

def calc_mae(actual, predicted):
    return np.mean(np.abs(np.asarray(actual) - np.asarray(predicted)))

def forecast_frame(dates, actual, prediction):
    return pd.DataFrame({
        "Date": pd.Series(dates).reset_index(drop=True),
        "Actual": pd.Series(actual).reset_index(drop=True),
        "Prediction": pd.Series(prediction).reset_index(drop=True)
    })

def load_valid_cache(cache_path, required_cols):
    if not Path(cache_path).exists():
        return pd.DataFrame()
    cached = pd.read_csv(cache_path)
    return cached if required_cols.issubset(cached.columns) else pd.DataFrame()

def cv_slices(y, X, split, horizon):
    y_train = y.iloc[:split]
    y_test = y.iloc[split:split + horizon]
    X_train = X.iloc[:split]
    X_test = X.iloc[split:split + horizon][X_train.columns]
    return y_train, y_test, X_train, X_test

def cv_row(order, split, horizon, rmse, aic, bic, status):
    p, d, q = map(int, order)
    return {"train_end": split, "test_start": split, "test_end": split +␣
 ↪horizon - 1,
            "p": p, "d": d, "q": q, "RMSE": rmse, "AIC": aic, "BIC": bic,␣
 ↪"status": status}

def cv_fit_one(y, X, order, split, horizon):
    y_tr, y_te, X_tr, X_te = cv_slices(y, X, split, horizon)
    try:
        model = SARIMAX(y_tr, exog=X_tr, order=tuple(map(int, order)),
```

```
                        enforce_stationarity=False,␣
 ↪enforce_invertibility=False).fit(disp=False)
        pred = model.get_forecast(steps=horizon, exog=X_te).predicted_mean
        return cv_row(order, split, horizon, calc_rmse(y_te, pred), model.aic,␣
 ↪model.bic, "success")
    except Exception:
        return cv_row(order, split, horizon, np.nan, np.nan, np.nan, "failed")

def cv_summary(order, rows):
    p, d, q = map(int, order)
    df = pd.DataFrame(rows)
    return {"p": p, "d": d, "q": q, "CV_RMSE": df["RMSE"].mean(skipna=True),
            "CV_AIC": df["AIC"].mean(skipna=True), "CV_BIC": df["BIC"].
 ↪mean(skipna=True),
            "n_success": df["RMSE"].notna().sum(), "n_splits": len(df)}

def rolling_cv_sarimax(y, X, order, initial_train_size=600, horizon=52,␣
 ↪step=26):
    y = pd.Series(y).reset_index(drop=True)
    X = X.reset_index(drop=True)
    splits = range(initial_train_size, len(y) - horizon + 1, step)
    rows = [cv_fit_one(y, X, order, split, horizon) for split in splits]
    return cv_summary(order, rows)
```

3.1.2 Google Trends Helper Functions

Code cell purpose: What: This cell defines helper functions for reading, cleaning, weekly resampling, aligning, and renaming Google Trends variables. Why: Google Trends data must be placed on the same weekly date structure as the FRED claims data before it can be used as an exogenous forecasting feature.

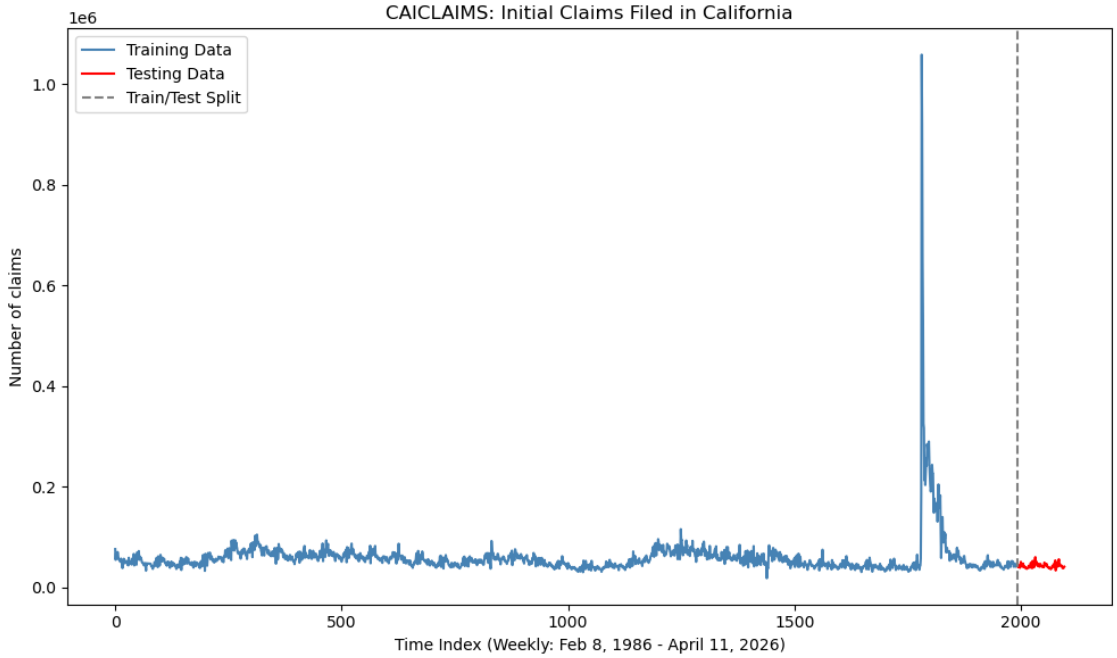
[21]:
```
## Loading, cleaning, and aligning Google Trends to FRED dates
def read_google_trends_state(state_abbrev):
    path = data_dir / f"google_trends_{state_abbrev}.csv"
    gt = pd.read_csv(path)
    gt = gt.rename(columns={gt.columns[0]: "Date"})
    gt["Date"] = pd.to_datetime(gt["Date"])
    return gt

def clean_google_trends_cols(gt):
    gt = gt.drop(columns=["isPartial"], errors="ignore").copy()
    trend_cols = [c for c in gt.columns if c != "Date"]
    gt[trend_cols] = gt[trend_cols].apply(pd.to_numeric, errors="coerce")
    return gt

def convert_google_dates_to_claims_week(gt):
    gt = gt.set_index("Date").sort_index()
    gt = gt.resample("W-SAT").mean()
    return gt.reset_index()

def align_google_trends(dates, gt):
    out = pd.DataFrame({"Date": pd.to_datetime(dates)})
    out = out.merge(gt, on="Date", how="left")
    trend_cols = [c for c in out.columns if c != "Date"]
    out[trend_cols] = out[trend_cols].interpolate(limit_direction="both")
    return out

def rename_google_trends_features(gt_aligned):
    trend_cols = [c for c in gt_aligned.columns if c != "Date"]
    X_gt = gt_aligned[trend_cols].copy()
    X_gt.columns = [f"gt_{c.replace(' ', '_')}" for c in trend_cols]
    return X_gt
```

3.2 2. ARIMA Baseline Model

Code cell purpose: The cell below generates the California modeling series, additionally defining the training and two-year holdout test periods. This matters because the same two-year holdout logic becomes the evaluation standard for comparing forecasts out of sample.

[22]:
```
## Choosing just California
y_CA = claims_dfs["CAICLAIMS"]["Claims"].reset_index(drop=True)
date_CA = claims_dfs["CAICLAIMS"]["Date"].reset_index(drop=True)

## Separating the test set
test_horizon = 104  ## final 2 years
train_end = len(y_CA) - test_horizon

## Numeric time indexes are kept because the original modeling code was built␣
 ↪around them
ttrain = np.arange(train_end)
ttest = np.arange(train_end, len(y_CA))
tpred = np.arange(train_end, len(y_CA) + test_horizon)  ## testing period plus␣
 ↪2-year future forecast

## Training modeling
ytrain_CA = y_CA.iloc[:train_end]
ytest_CA = y_CA.iloc[train_end:]

train_start_date = date_CA.iloc[0]
train_end_date = date_CA.iloc[train_end - 1]
test_start_date = date_CA.iloc[train_end]
test_end_date = date_CA.iloc[-1]

## Setting up the figures
print(f"Training period: {train_start_date.date()} to {train_end_date.date()}␣
 ↪({len(ytrain_CA):,} weeks)")
print(f"Testing period: {test_start_date.date()} to {test_end_date.date()}␣
 ↪({len(ytest_CA):,} weeks)")

plt.figure(figsize=(10, 6))
plt.plot(ttrain, ytrain_CA, label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Testing Data", color="red")
plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.title("CAICLAIMS: Initial Claims Filed in California")
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();
```

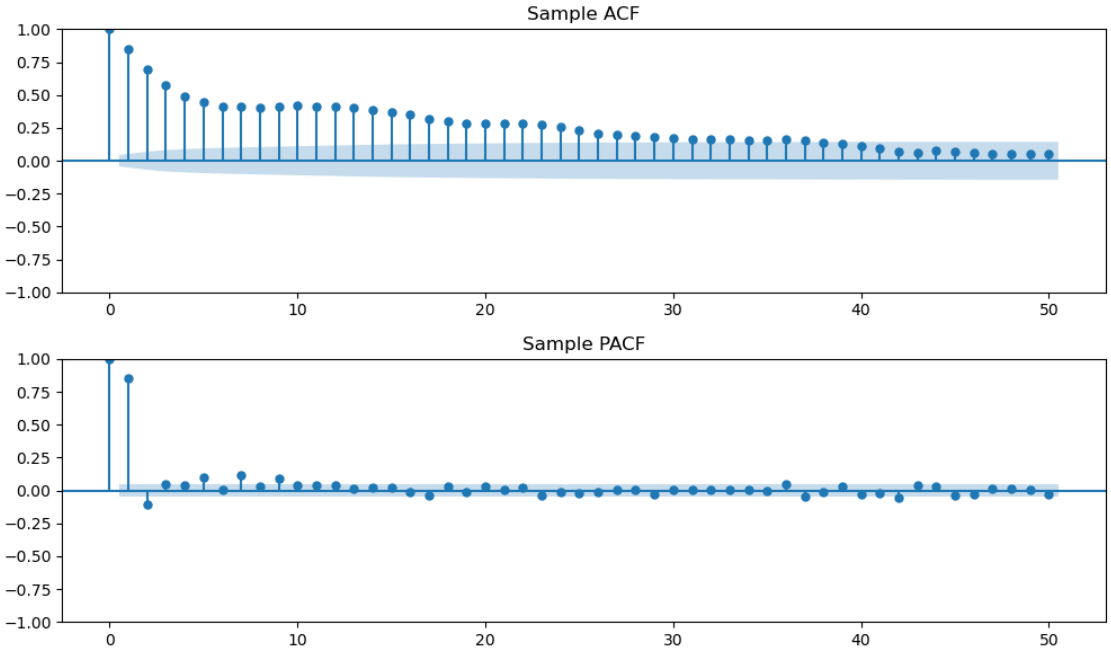
```
Training period: 1986-02-08 to 2024-04-13 (1,993 weeks)
Testing period: 2024-04-20 to 2026-04-11 (104 weeks)
```


CAICLAIMS: Initial Claims Filed in California
1e6
Training Data
Testing Data
Train/Test Split
Number of claims
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
2000
Time Index (Weekly: Feb 8, 1986 - April 11, 2026)

3.2.1 Analysis of the Training Data Time Series

Code cell purpose: Below, we have a cell which plots ACF and PACF of training level series, allowing us to inspect persistence prior to actual ARIMA modeling. This matters because ACF/PACF plots help motivate candidate AR and MA orders before running a formal grid search.

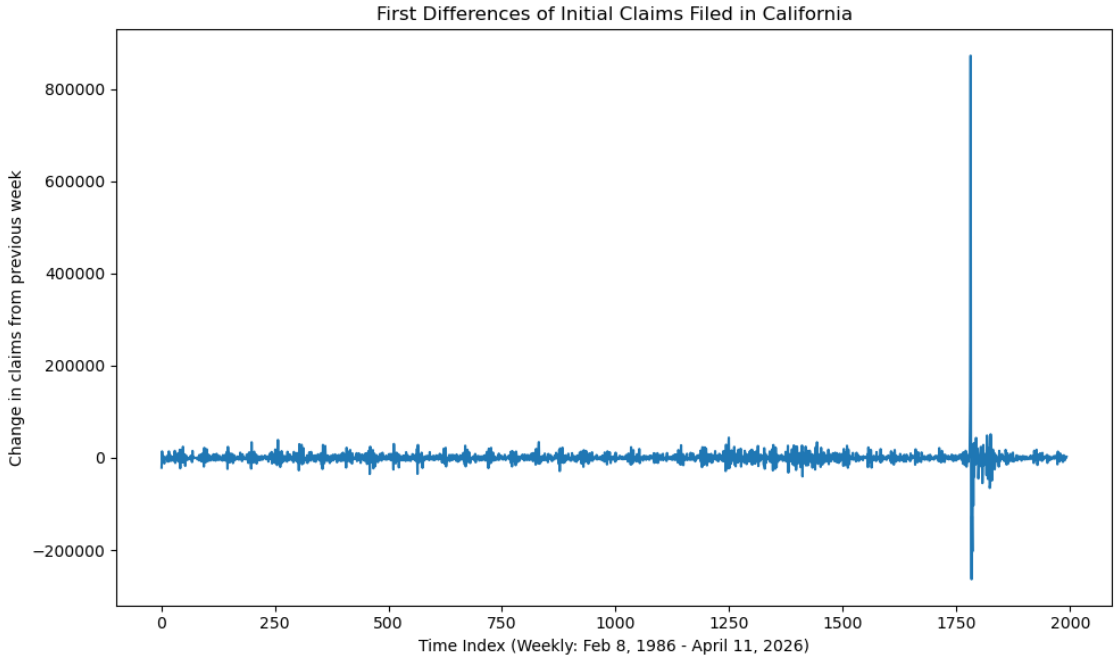
[23]:
```
## Setting p_max
p_max = 50

## ACF/PACF on the level series helps show persistence before differencing
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 6))
plot_acf(ytrain_CA.dropna(), lags=p_max, ax=ax1, title="Sample ACF")
plot_pacf(ytrain_CA.dropna(), lags=p_max, ax=ax2, title="Sample PACF")
plt.tight_layout()
plt.show();
```



Code cell purpose: Cell below completes differencing of training series and plots weekly changes, allowing us to asses whether differencing actualy improves stationarity. This matters because differencing is one of the main ways ARIMA handles non-stationary claims levels.

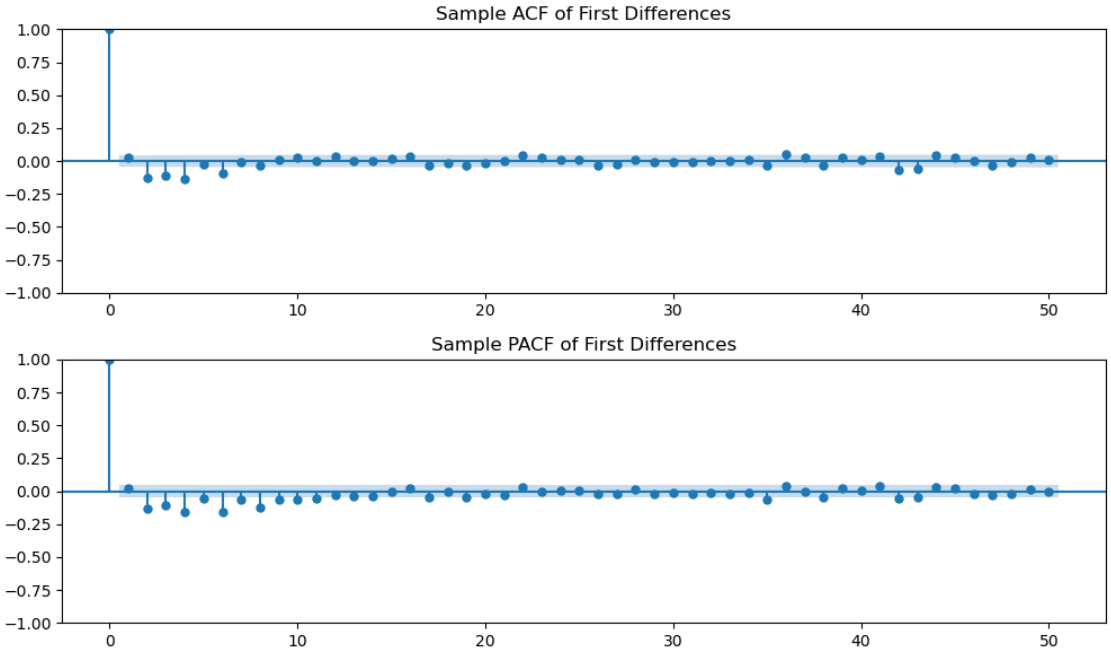
[24]:
```
## Differenced training set
ytrain_CA_diff = ytrain_CA.diff()

## First differences to stationarize the data
plt.figure(figsize=(10,6))
plt.plot(ytrain_CA_diff.dropna())
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - April 11, 2026)")
```

```
plt.ylabel("Change in claims from previous week")
plt.title("First Differences of Initial Claims Filed in California")
plt.tight_layout()
plt.show();
```



Code cell purpose: Cell below plots the ACF/PACF of differenced series, again for ARIMA order choice logic. This matters because the differenced ACF/PACF gives a cleaner view of short-run dependence after removing slow-moving level behavior.

[25]:
```
## Setting p_max
p_max = 50

## ACF/PACF after the differencing gives us a practical guide for candidate AR␣
 ↪and MA orders
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 6))
plot_acf(ytrain_CA_diff.dropna(), lags=p_max, ax=ax1, title="Sample ACF of␣
 ↪First Differences")
plot_pacf(ytrain_CA_diff.dropna(), lags=p_max, ax=ax2, title="Sample PACF of␣
 ↪First Differences")
plt.tight_layout()
plt.show();
```


Sample ACF of First Differences
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
10
20
30
40
50
Sample PACF of First Differences
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
10
20
30
40
50

3.2.2 Checking Stationarity

Utilizing the ADF to formally check for a unit root (which would make our time series non-stationary). Basically, this is to inform our decision to potentially include the parameter of `d=1` in later ARIMA/SARIMAX grid work when we have a very prevalent level series.

Code cell purpose: Cell below runs ADF tests on level and differenced training series, want to justify choice of differencing. This matters because the ADF results provide a formal check supporting whether the ARIMA/SARIMAX models should include differencing.

[26]:
```
## Brief function for ADF work
def adf_summary(series, label):
    clean_series = pd.Series(series).dropna()
    stat, pvalue, used_lag, nobs, crit_vals, _ = adfuller(clean_series,␣
 ↪autolag="AIC")
    return {
        "series": label, "adf_statistic": stat, "p_value": pvalue,
        "used_lag": used_lag, "n_obs": nobs,
        "critical_value_5pct": crit_vals["5%"]
    }

## Getting the ADF results
adf_results = pd.DataFrame([
    adf_summary(ytrain_CA, "CA claims, levels"),
    adf_summary(ytrain_CA_diff, "CA claims, first differences")
])

display(adf_results.round(4))
```

```
                       series  adf_statistic  p_value  used_lag  n_obs  \
0            CA claims, levels          -6.15     0.00        11   1981
1  CA claims, first differences         -14.23     0.00        18   1973

   critical_value_5pct
0                -2.86
1                -2.86
```

3.2.3 Parameter Selection (Grid Search)

Code cell purpose: Below cell works to define ARIMA grid-search helper, function is brief and only goal is to return AIC, BIC, and holdout RMSE here. This matters because the helper standardizes how every ARIMA candidate is scored using train-fit diagnostics and holdout RMSE.

[27]:
```
## Setting parameters
pmax, dmax, qmax = 8, 1, 6

## Defining the ARIMA fitting function with parameters
def fit_arima(params):
    p, d, q = map(int, params)
    try:
        model = ARIMA(ytrain_CA, order=(p, d, q)).fit()
        yhat = model.get_forecast(steps=len(ytest_CA)).predicted_mean
        yhat.index = ytest_CA.index
        rmse = np.sqrt(mean_squared_error(ytest_CA, yhat))
        return {"p": p, "d": d, "q": q, "AIC": model.aic, "BIC": model.bic,␣
 ↪"RMSE": rmse}
    except Exception:
        return None
```

Code cell purpose: Cell below actually runs/loads ARIMA grid search, gives best candidate models based on the holdout RMSE. This matters because caching prevents the same expensive model grid from being refit every time the notebook is rerun.

[28]:
```
## Setting distinct path
cache_path_arima = "CAICLAIMS_Model_Cache/arima_grid_results_CAICLAIMS.csv"

## Building grid for parameter
param_grid = list(itertools.product(
    range(pmax + 1),
    range(dmax + 1),
    range(qmax + 1)
))

## Necessary columns
required_arima_cols = {"p", "d", "q", "AIC", "BIC", "RMSE"}

## Check for cached results, otherwise run ARIMA grid search
if os.path.exists(cache_path_arima):
    print(f"Loading cached ARIMA grid search results from {cache_path_arima}")
    results_arima_df = pd.read_csv(cache_path_arima)

    ## Re-run if the cache is not in the expected format
    if not required_arima_cols.issubset(results_arima_df.columns):
        print("The cached ARIMA results are missing required columns, grid␣
 ↪search must be run again.")
```

```
        results_arima_df = pd.DataFrame()

else:
    results_arima_df = pd.DataFrame()

if results_arima_df.empty:
    print("Running ARIMA grid search...")

    results_arima = Parallel(n_jobs=-1, verbose=10)(
        delayed(fit_arima)(params) for params in param_grid
    )

    results_arima = [r for r in results_arima if r is not None]
    results_arima_df = pd.DataFrame(results_arima)

    if results_arima_df.empty:
        raise ValueError("ARIMA grid search failed, no models were successfully␣
 ↪fit.")

    results_arima_df.to_csv(cache_path_arima, index=False)
    print(f"Saved ARIMA grid search results to {cache_path_arima}")

## Showing the best few models by holdout RMSE for transparency
display(results_arima_df.sort_values("RMSE").head(10).round(3))
```

```
Loading cached ARIMA grid search results from
CAICLAIMS_Model_Cache/arima_grid_results_CAICLAIMS.csv
```

	p	d	q	AIC	BIC	RMSE
119	8	1	0	45,577.51	45,627.88	4,599.13
63	4	1	0	45,662.47	45,690.45	4,602.98
77	5	1	0	45,658.34	45,691.92	4,621.72
105	7	1	0	45,604.59	45,649.36	4,635.10
91	6	1	0	45,609.98	45,649.16	4,664.12
10	0	1	3	45,644.79	45,667.18	4,681.73
9	0	1	2	45,710.97	45,727.76	4,682.99
49	3	1	0	45,709.97	45,732.36	4,688.87
11	0	1	4	45,560.18	45,588.16	4,756.92
37	2	1	2	45,571.39	45,599.37	4,764.09

3.2.4 Determining the Best Models by Evaluation Metrics

Code cell purpose: Cell below extracts the best ARIMA specifications as according to our three desired metrics of AIC, BIC, and holdout RMSE. This matters because the notebook compares selection by fit diagnostics against selection by out-of-sample forecast accuracy.

[29]:
```
## Getting best values
best_aic = results_arima_df.loc[results_arima_df["AIC"].idxmin()]
best_bic = results_arima_df.loc[results_arima_df["BIC"].idxmin()]
best_rmse = results_arima_df.loc[results_arima_df["RMSE"].idxmin()]

## Printing it out legibly below
print(
    f"Best model by AIC: ARIMA({int(best_aic['p'])}, {int(best_aic['d'])},␣
 ↪{int(best_aic['q'])}), "
    f"AIC: {best_aic['AIC']:.3f}, BIC: {best_aic['BIC']:.3f}, RMSE:␣
 ↪{best_aic['RMSE']:.3f}"
)

print(
    f"Best model by BIC: ARIMA({int(best_bic['p'])}, {int(best_bic['d'])},␣
 ↪{int(best_bic['q'])}), "
    f"AIC: {best_bic['AIC']:.3f}, BIC: {best_bic['BIC']:.3f}, RMSE:␣
 ↪{best_bic['RMSE']:.3f}"
)

print(
    f"Best model by RMSE: ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
 ↪{int(best_rmse['q'])}), "
    f"AIC: {best_rmse['AIC']:.3f}, BIC: {best_rmse['BIC']:.3f}, RMSE:␣
 ↪{best_rmse['RMSE']:.3f}"
)
```

```
Best model by AIC: ARIMA(6, 0, 4), AIC: 45543.538, BIC: 45610.707, RMSE:
14423.869
Best model by BIC: ARIMA(0, 1, 6), AIC: 45545.010, BIC: 45584.189, RMSE:
4809.309
Best model by RMSE: ARIMA(8, 1, 0), AIC: 45577.506, BIC: 45627.878, RMSE:
4599.134
```

Code cell purpose: Cell below completes fitting of ARIMA model selected using AIC, prints out summary. This matters because the AIC-selected model summary lets us inspect the model that is preferred by in-sample information criteria.

[30]:
```
## Getting the AIC summary results
best_aic_model = ARIMA(
    ytrain_CA,
    order=(int(best_aic["p"]), int(best_aic["d"]), int(best_aic["q"]))
```

```
).fit()

print(best_aic_model.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:                 ARIMA(6, 0, 4)   Log Likelihood              -22759.757
Date:                Mon, 04 May 2026   AIC                          45543.513
Time:                        16:51:59   BIC                          45610.682
Sample:                             0   HQIC                         45568.180
                                 - 1993
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       5.942e+04   1.18e+04      5.020      0.000    3.62e+04    8.26e+04
ar.L1          0.4586      0.182      2.518      0.012       0.102       0.816
ar.L2          1.5480      0.060     25.873      0.000       1.431       1.665
ar.L3         -0.8610      0.271     -3.173      0.002      -1.393      -0.329
ar.L4         -0.6723      0.053    -12.654      0.000      -0.776      -0.568
ar.L5          0.5020      0.120      4.192      0.000       0.267       0.737
ar.L6         -0.0058      0.036     -0.160      0.873      -0.077       0.065
ma.L1          0.4571      0.182      2.514      0.012       0.101       0.814
ma.L2         -1.2891      0.177     -7.286      0.000      -1.636      -0.942
ma.L3         -0.3437      0.152     -2.257      0.024      -0.642      -0.045
ma.L4          0.5040      0.158      3.191      0.001       0.194       0.813
sigma2      4.904e+08     27.878   1.76e+07      0.000     4.9e+08     4.9e+08
===================================================================================
===
Ljung-Box (L1) (Q):                   0.23   Jarque-Bera (JB):
140456405.96
Prob(Q):                              0.63   Prob(JB):
0.00
Heteroskedasticity (H):              20.01   Skew:
32.54
Prob(H) (two-sided):                  0.00   Kurtosis:
1301.91
===================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
6.92e+20. Standard errors may be unstable.
```

Code cell purpose: Cell below completes fitting of ARIMA model selected using BIC, prints out

summary. This matters because the BIC-selected model summary provides a more parsimonious benchmark for comparison.

[31]:
```
## Getting the BIC summary results
best_bic_model = ARIMA(
    ytrain_CA,
    order=(int(best_bic["p"]), int(best_bic["d"]), int(best_bic["q"]))
).fit()

print(best_bic_model.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:                 ARIMA(0, 1, 6)   Log Likelihood              -22765.505
Date:                Mon, 04 May 2026   AIC                          45545.010
Time:                        16:52:01   BIC                          45584.189
Sample:                             0   HQIC                         45559.399
                                 - 1993
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1         -0.0635      0.005    -13.003      0.000      -0.073      -0.054
ma.L2         -0.2219      0.012    -17.911      0.000      -0.246      -0.198
ma.L3         -0.1540      0.029     -5.395      0.000      -0.210      -0.098
ma.L4         -0.1784      0.011    -16.333      0.000      -0.200      -0.157
ma.L5         -0.0328      0.026     -1.255      0.209      -0.084       0.018
ma.L6         -0.0918      0.014     -6.384      0.000      -0.120      -0.064
sigma2      5.007e+08   1.05e-10   4.79e+18      0.000    5.01e+08    5.01e+08
===================================================================================
===
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):
131445191.44
Prob(Q):                              0.96   Prob(JB):
0.00
Heteroskedasticity (H):              19.30   Skew:
31.66
Prob(H) (two-sided):                  0.00   Kurtosis:
1259.85
===================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
4.02e+32. Standard errors may be unstable.
```

Code cell purpose. Cell below completes fitting of ARIMA model selected using holdout RMSE, prints out summary. This matters because the RMSE-selected model is the direct forecasting benchmark used in later plots and tables.

[32]:
```
## Getting the RMSE summary results
best_rmse_model = ARIMA(
    ytrain_CA,
    order=(int(best_rmse["p"]), int(best_rmse["d"]), int(best_rmse["q"]))
).fit()

print(best_rmse_model.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:                 ARIMA(8, 1, 0)   Log Likelihood              -22779.753
Date:                Mon, 04 May 2026   AIC                          45577.506
Time:                        16:52:03   BIC                          45627.878
Sample:                             0   HQIC                         45596.005
                                 - 1993
Covariance Type:                   opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.0415      0.004    -10.140      0.000      -0.050      -0.033
ar.L2         -0.2007      0.010    -20.251      0.000      -0.220      -0.181
ar.L3         -0.1481      0.042     -3.536      0.000      -0.230      -0.066
ar.L4         -0.2117      0.009    -22.386      0.000      -0.230      -0.193
ar.L5         -0.0853      0.030     -2.826      0.005      -0.144      -0.026
ar.L6         -0.1809      0.014    -13.248      0.000      -0.208      -0.154
ar.L7         -0.0646      0.031     -2.102      0.036      -0.125      -0.004
ar.L8         -0.1200      0.009    -14.030      0.000      -0.137      -0.103
sigma2      5.028e+08   2.31e-10   2.18e+18      0.000    5.03e+08    5.03e+08
===================================================================================
===
Ljung-Box (L1) (Q):                   0.07   Jarque-Bera (JB):
120183048.50
Prob(Q):                              0.80   Prob(JB):
0.00
Heteroskedasticity (H):              18.68   Skew:
30.54
Prob(H) (two-sided):                  0.00   Kurtosis:
1204.77
===================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
```

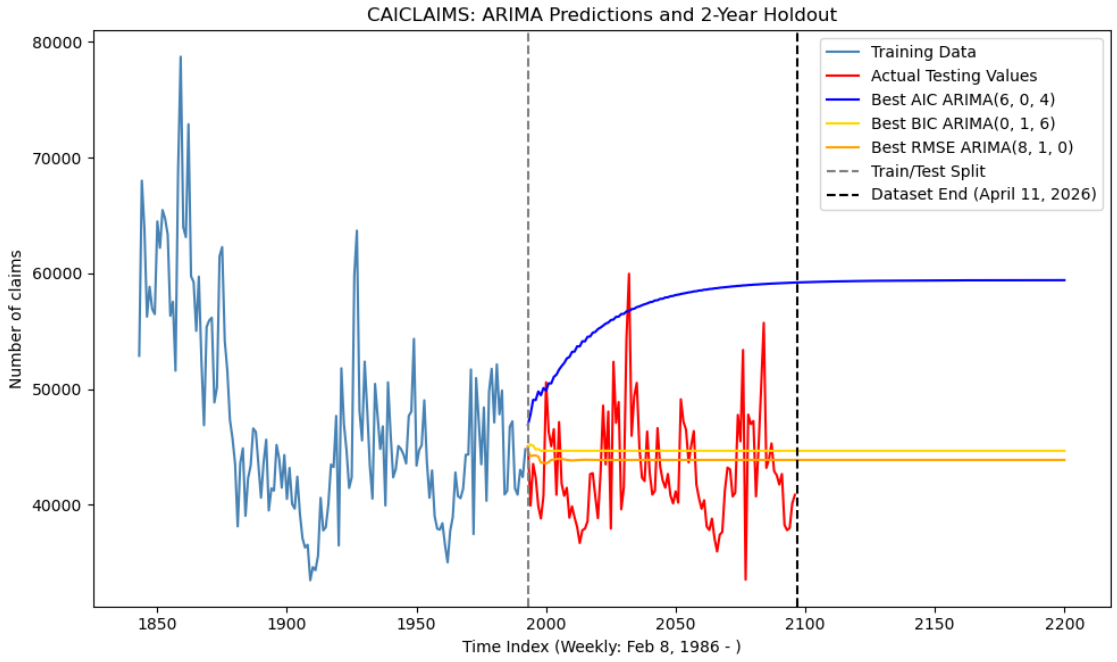
```
step).
[2] Covariance matrix is singular or near-singular, with condition number
4.17e+33. Standard errors may be unstable.
```

3.2.5 Plotted ARIMA Predictions

Code cell purpose: Cell below performs forecasting with selected ARIMA models, additionally plotting their predictions against holdout data. This matters because plotting forecasts against the holdout set shows not only the RMSE value but also when the model misses most visibly.

[33]:
```
## Best results from AIC
best_aic_fcast = best_aic_model.get_prediction(start=train_end, end=len(y_CA) +␣
 ↪test_horizon - 1)
best_aic_yhat = best_aic_fcast.predicted_mean
best_aic_rmse = np.sqrt(mean_squared_error(ytest_CA, best_aic_yhat.iloc[:
 ↪len(ytest_CA)]))

## Best results from BIC
best_bic_fcast = best_bic_model.get_prediction(start=train_end, end=len(y_CA) +␣
 ↪test_horizon - 1)
best_bic_yhat = best_bic_fcast.predicted_mean
best_bic_rmse = np.sqrt(mean_squared_error(ytest_CA, best_bic_yhat.iloc[:
 ↪len(ytest_CA)]))

## Best results from holdout RMSE
best_rmse_fcast = best_rmse_model.get_prediction(start=train_end, end=len(y_CA)␣
 ↪+ test_horizon - 1)
best_rmse_yhat = best_rmse_fcast.predicted_mean
best_rmse_rmse = np.sqrt(mean_squared_error(ytest_CA, best_rmse_yhat.iloc[:
 ↪len(ytest_CA)]))

## Getting the figures/plots set up
plt.figure(figsize=(10,6))
plt.plot(ttrain[(train_end - 150):], ytrain_CA.iloc[(train_end - 150):],␣
 ↪color="steelblue", label="Training Data")
plt.plot(ttest, ytest_CA, color="red", label="Actual Testing Values")

plt.plot(tpred, best_aic_yhat, color="blue",
         label=f"Best AIC ARIMA({int(best_aic['p'])}, {int(best_aic['d'])},␣
 ↪{int(best_aic['q'])})")
plt.plot(tpred, best_bic_yhat, color="gold",
         label=f"Best BIC ARIMA({int(best_bic['p'])}, {int(best_bic['d'])},␣
 ↪{int(best_bic['q'])})")
plt.plot(tpred, best_rmse_yhat, color="orange",
         label=f"Best RMSE ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
 ↪{int(best_rmse['q'])})")

plt.axvline(x=train_end, color="grey", linestyle="--", label="Train/Test Split")
plt.axvline(x=len(y_CA), color="black", linestyle="--", label="Dataset End␣
 ↪(April 11, 2026)")
```

```
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - )")
plt.ylabel("Number of claims")
plt.title("CAICLAIMS: ARIMA Predictions and 2-Year Holdout")
plt.legend(); plt.tight_layout(); plt.show();

print(f"Root Mean Squared Error of Best AIC ARIMA({int(best_aic['p'])},␣
  ↪{int(best_aic['d'])}, {int(best_aic['q'])}) Model: {best_aic_rmse:.3f}")
print(f"Root Mean Squared Error of Best BIC ARIMA({int(best_bic['p'])},␣
  ↪{int(best_bic['d'])}, {int(best_bic['q'])}) Model: {best_bic_rmse:.3f}")
print(f"Root Mean Squared Error of Best Holdout RMSE␣
  ↪ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])}, {int(best_rmse['q'])})␣
  ↪Model: {best_rmse_rmse:.3f}")
```

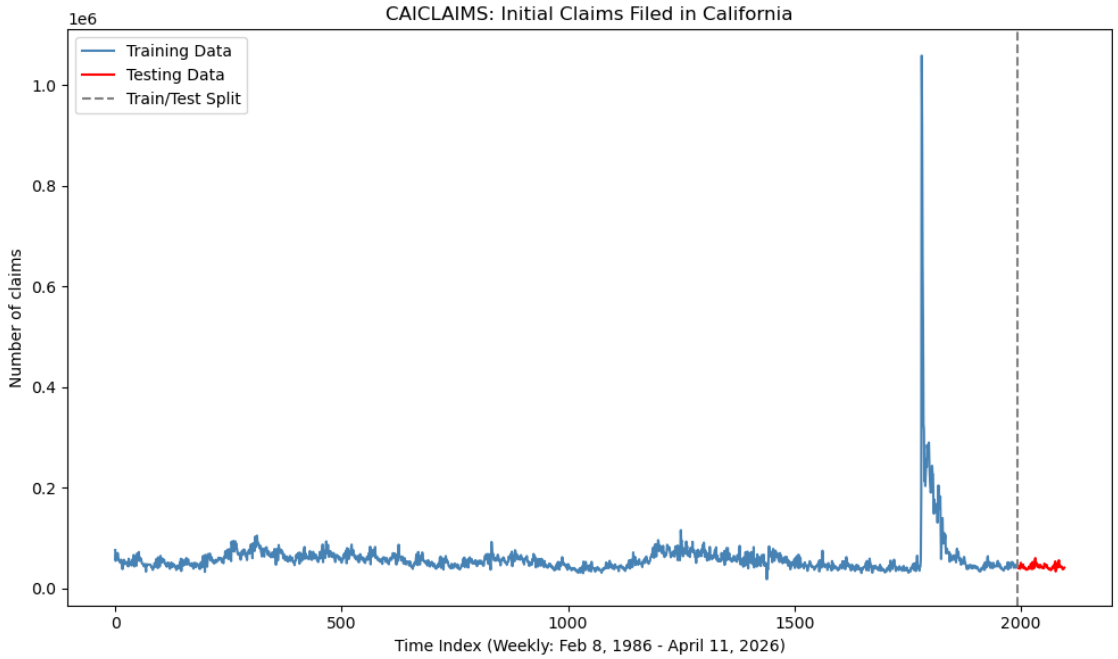



```
Root Mean Squared Error of Best AIC ARIMA(6, 0, 4) Model: 14433.620
Root Mean Squared Error of Best BIC ARIMA(0, 1, 6) Model: 4809.309
Root Mean Squared Error of Best Holdout RMSE ARIMA(8, 1, 0) Model: 4599.134
```

3.3 3. SARIMAX with Fourier Frequencies, Before/After COVID Dummy

Code cell purpose: Cell below plots entire California split between train and test, allows forecasting setup to be clear visually. This matters because the visual split confirms which observations are used for estimation versus out-of-sample evaluation.

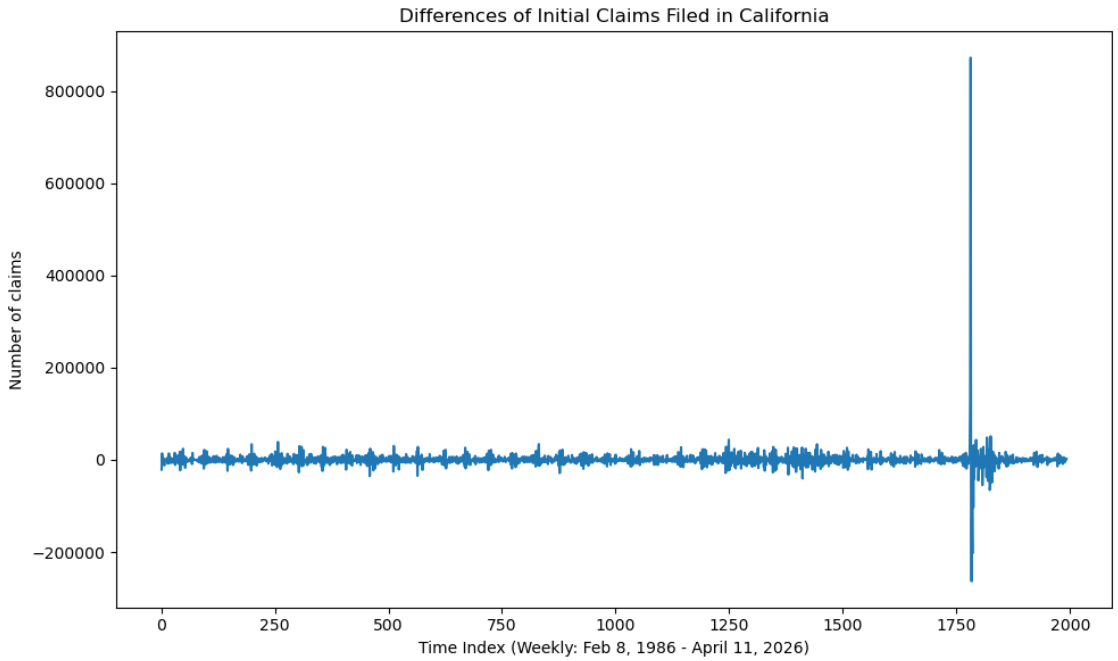
[34]:
```
## Plot
plt.figure(figsize=(10, 6))
plt.plot(ttrain, ytrain_CA, label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Testing Data", color="red")
plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.title("CAICLAIMS: Initial Claims Filed in California")
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();
```



Code cell purpose: Cell below re-plots the first differences prior to SARIMAX section, meant to keep model diagnostic close to the model code. This matters because keeping the diagnostic plot near the SARIMAX code helps connect the model specification to the series behavior.

[35]:
```
## Plot
plt.figure(figsize=(10,6))
plt.plot(ytrain_CA_diff.dropna())
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.title("Differences of Initial Claims Filed in California")
plt.tight_layout()
```

```
plt.show();
```


Differences of Initial Claims Filed in California
800000
600000
400000
200000
0
−200000
Number of claims
0
250
500
750
1000
1250
1500
1750
2000
Time Index (Weekly: Feb 8, 1986 - April 11, 2026)

3.3.1 Design Matrix Construction

Code cell purpose: Cell below has function to briefly define exogenous SARIMAX features, specifically Fourier seasonality, trend, and post-COVID indicator. This matters because the SARIMAX model needs interpretable exogenous features to capture seasonality, trend, and COVID-related structural change.

[36]:
```
def make_ca_exog_after(n, covid_start=1780):
    t = np.arange(n)
    X = pd.DataFrame({"trend": t}, index=np.arange(n))
    for k in range(1, 4):
        X[f"sin_{k}"] = np.sin(2 * np.pi * k * t / 52)
        X[f"cos_{k}"] = np.cos(2 * np.pi * k * t / 52)
    X["covid_after"] = (t >= covid_start).astype(int)
    return X
```

3.3.2 SARIMAX Model Testing

Code cell purpose: Building the actual design matrices for SARIMAX, fitting baseline exogenous model utilizing best ARIMA order. This matters because it creates the actual exogenous matrices used to estimate and forecast the SARIMAX baseline.

[37]:
```
## Exog features through observed data only (for holdout RMSE/grid search)
X_CA = make_ca_exog_after(len(y_CA), covid_start=1780)

Xtrain_CA = X_CA.iloc[:train_end]
Xtest_CA = X_CA.iloc[train_end:len(y_CA)]
Xtest_CA = Xtest_CA[Xtrain_CA.columns]

## Exog through observed data and future period (for extended future forecasts)
X_CA_extended = make_ca_exog_after(len(y_CA) + test_horizon, covid_start=1780)

Xtrain_CA_extended = X_CA_extended.iloc[:train_end]
Xtest_CA_extended = X_CA_extended.iloc[train_end : len(y_CA) + test_horizon]
Xtest_CA_extended = Xtest_CA_extended[Xtrain_CA_extended.columns]


## Baseline SARIMAX uses the best holdout-RMSE ARIMA order, then adds the␣
 ↪exogenous terms
sarimax_model = SARIMAX(
    ytrain_CA,
    exog=Xtrain_CA_extended,
    order=(int(best_rmse["p"]), int(best_rmse["d"]), int(best_rmse["q"])),
    enforce_stationarity=False,
    enforce_invertibility=False
).fit(disp=False)

print(sarimax_model.summary())
```

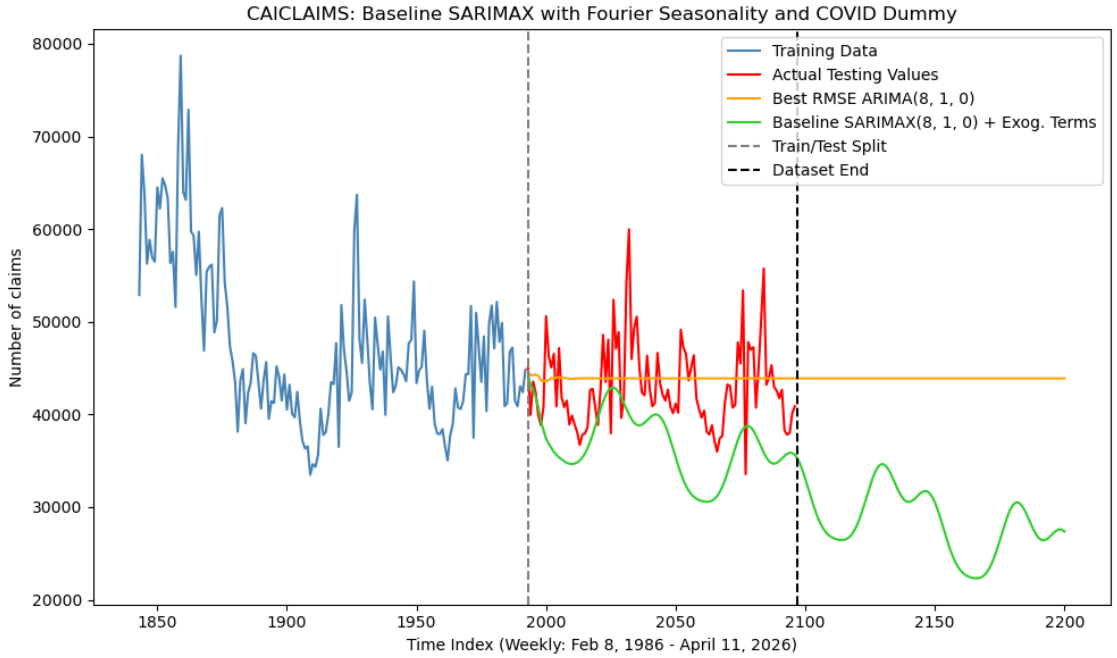
```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:               SARIMAX(8, 1, 0)   Log Likelihood              -22661.539
Date:                Mon, 04 May 2026   AIC                          45357.079
Time:                        16:52:04   BIC                          45452.157
Sample:                             0   HQIC                         45392.004
                                 - 1993
Covariance Type:                   opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
trend         -79.4505    614.994     -0.129      0.897   -1284.816    1125.915
sin_1        2431.3356   7173.627      0.339      0.735   -1.16e+04    1.65e+04
cos_1        3340.7817   7575.653      0.441      0.659   -1.15e+04    1.82e+04
```

```
sin_2       -1744.7635   4270.352     -0.409      0.683   -1.01e+04    6624.973
cos_2         274.1402   3985.192      0.069      0.945   -7536.693    8084.974
sin_3        -659.5435   3330.824     -0.198      0.843   -7187.838    5868.751
cos_3         673.9158   3668.786      0.184      0.854   -6516.773    7864.604
covid_after  1.284e+05    2.5e+04      5.135      0.000    7.94e+04    1.77e+05
ar.L1          -0.1654      0.035     -4.751      0.000      -0.234      -0.097
ar.L2          -0.2212      0.008    -28.458      0.000      -0.236      -0.206
ar.L3          -0.1807      0.014    -13.069      0.000      -0.208      -0.154
ar.L4          -0.2376      0.009    -27.249      0.000      -0.255      -0.221
ar.L5          -0.1203      0.013     -9.221      0.000      -0.146      -0.095
ar.L6          -0.2006      0.008    -23.717      0.000      -0.217      -0.184
ar.L7          -0.0942      0.014     -6.593      0.000      -0.122      -0.066
ar.L8          -0.1298      0.008    -17.072      0.000      -0.145      -0.115
sigma2       4.883e+08    124.401   3.93e+06      0.000    4.88e+08    4.88e+08
===================================================================================
===
Ljung-Box (L1) (Q):                   0.19   Jarque-Bera (JB):
127847120.25
Prob(Q):                              0.66   Prob(JB):
0.00
Heteroskedasticity (H):              20.88   Skew:
31.17
Prob(H) (two-sided):                  0.00   Kurtosis:
1245.04
===================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
5.09e+21. Standard errors may be unstable.
```

Code cell purpose: Here, having the entire California train-test split plotted such that forecasting setup is clear visually. This matters because the plot verifies that the SARIMAX forecast is being evaluated over the intended holdout window.

[38]:
```
## Forecast the test period plus an additional 2-year future window.
sarimax_fcast = sarimax_model.get_forecast(steps=len(ytest_CA) + test_horizon,␣
 ↪exog=Xtest_CA_extended)
sarimax_yhat = sarimax_fcast.predicted_mean
sarimax_yhat.index = tpred
sarimax_rmse = np.sqrt(mean_squared_error(ytest_CA, sarimax_yhat.iloc[:
 ↪len(ytest_CA)]))


## Plot predictions against ARIMA baselines and actual values.
plt.figure(figsize=(10, 6))
```

```
plt.plot(ttrain[(train_end - 150):], ytrain_CA.iloc[(train_end - 150):],␣
 ↪label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Actual Testing Values", color="red")

plt.plot(tpred, best_rmse_yhat, color="orange",
         label=f"Best RMSE ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
 ↪{int(best_rmse['q'])})")
plt.plot(tpred, sarimax_yhat, color="limegreen",
         label=f"Baseline SARIMAX({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
 ↪{int(best_rmse['q'])}) + Exog. Terms")

plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split"␣
 ↪)
plt.axvline(x=len(y_CA), color="black", linestyle="--", label="Dataset End")

plt.title("CAICLAIMS: Baseline SARIMAX with Fourier Seasonality and COVID␣
 ↪Dummy")
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();


print(f"Root Mean Squared Error of Best RMSE ARIMA({int(best_rmse['p'])},␣
 ↪{int(best_rmse['d'])}, {int(best_rmse['q'])}) Model: {best_rmse_rmse:.3f}")
print(f"Root Mean Squared Error of Baseline SARIMAX({int(best_rmse['p'])},␣
 ↪{int(best_rmse['d'])}, {int(best_rmse['q'])}) Model: {sarimax_rmse:.3f}")
```



```
Root Mean Squared Error of Best RMSE ARIMA(8, 1, 0) Model: 4599.134
Root Mean Squared Error of Baseline SARIMAX(8, 1, 0) Model: 7840.432
```

3.3.3 Parameter Selection (Grid Search)

Code cell purpose: Cell below defines the SARIMAX grid-search helper for post-COVID specification. This matters because a consistent helper lets the SARIMAX grid compare many orders using the same exogenous controls and RMSE metric.

[39]:
```
## Parameters
pmax, dmax, qmax = 8, 1, 6

## Brief function
def fit_sarimax_exog(params):
    p, d, q = map(int, params)
    try:
        model = SARIMAX(ytrain_CA, exog=Xtrain_CA, order=(p, d, q),␣
 ↪enforce_stationarity=False, enforce_invertibility=False).fit(disp=False)
        y_hat = model.get_forecast(steps=len(ytest_CA), exog=Xtest_CA).
 ↪predicted_mean
        y_hat.index = ytest_CA.index
        rmse = np.sqrt(mean_squared_error(ytest_CA, y_hat))
        return {"p": p, "d": d, "q": q, "AIC": model.aic, "BIC": model.bic,␣
 ↪"RMSE": rmse}
    except Exception:
        return None
```

Code cell purpose: Running/loading the SARIMAX grid search for post-COVID specification. This matters because caching makes the SARIMAX model search reproducible without forcing long reruns after results are saved.

[40]:
```
## Setting up the path
cache_path_sarimax = "CAICLAIMS_Model_Cache/sarimax_grid_results_CAICLAIMS.csv"

## Grid for parameters
param_grid = list(itertools.product(
    range(pmax + 1),
    range(dmax + 1),
    range(qmax + 1)
))

## Necessary columns
required_sarimax_cols = {"p", "d", "q", "AIC", "BIC", "RMSE"}

## Setting up the process
if os.path.exists(cache_path_sarimax):
    print(f"Loading cached SARIMAX grid search results from␣
 ↪{cache_path_sarimax}")
    results_sarimax_df = pd.read_csv(cache_path_sarimax)

    ## Re-run if the cache is not in the expected format
```

```
    if not required_sarimax_cols.issubset(results_sarimax_df.columns):
        print("Cached SARIMAX results are missing the required columns, grid␣
 ↪search must be run again")
        results_sarimax_df = pd.DataFrame()

else:
    results_sarimax_df = pd.DataFrame()

if results_sarimax_df.empty:
    print("Running SARIMAX grid search...")

    results_sarimax = Parallel(n_jobs=-1, verbose=10)(
        delayed(fit_sarimax_exog)(params) for params in param_grid
    )

    results_sarimax = [r for r in results_sarimax if r is not None]
    results_sarimax_df = pd.DataFrame(results_sarimax)

    if results_sarimax_df.empty:
        raise ValueError(
            "SARIMAX grid search failed, no models were successfully fit."
            "Please check that len(Xtest_CA) equals len(ytest_CA)."
        )

    results_sarimax_df.to_csv(cache_path_sarimax, index=False)
    print(f"Saved SARIMAX grid search results to {cache_path_sarimax}")

## Showing the best few models by holdout RMSE for transparency
display(results_sarimax_df.sort_values("RMSE").head(10).round(3))
```

```
Loading cached SARIMAX grid search results from
CAICLAIMS_Model_Cache/sarimax_grid_results_CAICLAIMS.csv
```

	p	d	q	AIC	BIC	RMSE
7	0	1	0	58,858.05	58,908.42	7,120.88
120	8	1	1	45,310.04	45,410.71	7,139.81
109	7	1	4	45,332.17	45,444.03	7,156.28
97	6	1	6	45,336.91	45,454.37	7,161.53
83	5	1	6	45,334.92	45,446.79	7,163.80
69	4	1	6	45,332.86	45,439.14	7,166.92
95	6	1	4	45,355.52	45,461.80	7,167.27
121	8	1	2	45,310.80	45,417.06	7,171.70
122	8	1	3	45,312.05	45,423.91	7,172.84
82	5	1	5	45,354.87	45,461.15	7,175.86

3.3.4 Best Models by Evaluation Metrics

Code cell purpose: Extracting the best SARIMAX specifications according to the metrics of AIC, BIC, and holdout RMSE. This matters because it separates the best model by information criteria from the best model by forecast accuracy.

[41]:
```
## BEst specifications
best_aic_sarimax = results_sarimax_df.loc[results_sarimax_df["AIC"].idxmin()]
best_bic_sarimax = results_sarimax_df.loc[results_sarimax_df["BIC"].idxmin()]
best_rmse_sarimax = results_sarimax_df.loc[results_sarimax_df["RMSE"].idxmin()]

## Printing out respective messages for metrics
print(
    f"Best model by AIC: SARIMAX({int(best_aic_sarimax['p'])}, "
    f"{int(best_aic_sarimax['d'])}, {int(best_aic_sarimax['q'])}), "
    f"AIC: {best_aic_sarimax['AIC']:.3f}, "
    f"BIC: {best_aic_sarimax['BIC']:.3f}, "
    f"RMSE: {best_aic_sarimax['RMSE']:.3f}"
)

print(
    f"Best model by BIC: SARIMAX({int(best_bic_sarimax['p'])}, "
    f"{int(best_bic_sarimax['d'])}, {int(best_bic_sarimax['q'])}), "
    f"AIC: {best_bic_sarimax['AIC']:.3f}, "
    f"BIC: {best_bic_sarimax['BIC']:.3f}, "
    f"RMSE: {best_bic_sarimax['RMSE']:.3f}"
)

print(
    f"Best model by RMSE: SARIMAX({int(best_rmse_sarimax['p'])}, "
    f"{int(best_rmse_sarimax['d'])}, {int(best_rmse_sarimax['q'])}), "
    f"AIC: {best_rmse_sarimax['AIC']:.3f}, "
    f"BIC: {best_rmse_sarimax['BIC']:.3f}, "
    f"RMSE: {best_rmse_sarimax['RMSE']:.3f}"
)
```

```
Best model by AIC: SARIMAX(8, 1, 1), AIC: 45310.035, BIC: 45410.707, RMSE:
7139.811
Best model by BIC: SARIMAX(8, 1, 1), AIC: 45310.035, BIC: 45410.707, RMSE:
7139.811
Best model by RMSE: SARIMAX(0, 1, 0), AIC: 58858.051, BIC: 58908.419, RMSE:
7120.883
```

Code cell purpose: Cell fits best holdout-RMSE SARIMAX model, also prints out summary. This matters because the summary gives coefficient and diagnostic detail for the strongest holdout-RMSE SARIMAX specification.

[42]:
```
## Best values
best_p = int(best_rmse_sarimax["p"])
best_d = int(best_rmse_sarimax["d"])
best_q = int(best_rmse_sarimax["q"])

## Best model
best_rmse_sarimax_model = SARIMAX(
    ytrain_CA,
    exog=Xtrain_CA_extended,
    order=(best_p, best_d, best_q),
    enforce_stationarity=False,
    enforce_invertibility=False
).fit(disp=False)

print(best_rmse_sarimax_model.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:               SARIMAX(0, 1, 0)   Log Likelihood              -29420.026
Date:                Mon, 04 May 2026   AIC                          58858.051
Time:                        16:52:05   BIC                          58908.419
Sample:                             0   HQIC                         58876.550
                                 - 1993
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
trend         -79.4505   2.55e+06  -3.11e-05      1.000   -5.01e+06    5.01e+06
sin_1        2431.3356    3.2e+07    7.6e-05      1.000   -6.27e+07    6.27e+07
cos_1        3340.7817   2.83e+07      0.000      1.000   -5.54e+07    5.54e+07
sin_2       -1744.7635   1.34e+07     -0.000      1.000   -2.62e+07    2.62e+07
cos_2         274.1402   1.44e+07   1.91e-05      1.000   -2.82e+07    2.82e+07
sin_3        -659.5435   8.77e+06  -7.52e-05      1.000   -1.72e+07    1.72e+07
cos_3         673.9158   8.58e+06   7.85e-05      1.000   -1.68e+07    1.68e+07
covid_after  1.284e+05      0.007   1.73e+07      0.000    1.28e+05    1.28e+05
sigma2       1.087e+12   4.88e+10     22.299      0.000    9.92e+11    1.18e+12
==============================================================================
===
Ljung-Box (L1) (Q):                  13.62   Jarque-Bera (JB):
80368288.19
Prob(Q):                              0.00   Prob(JB):
0.00
Heteroskedasticity (H):              22.22   Skew:
24.80
Prob(H) (two-sided):                  0.00   Kurtosis:
986.01
```

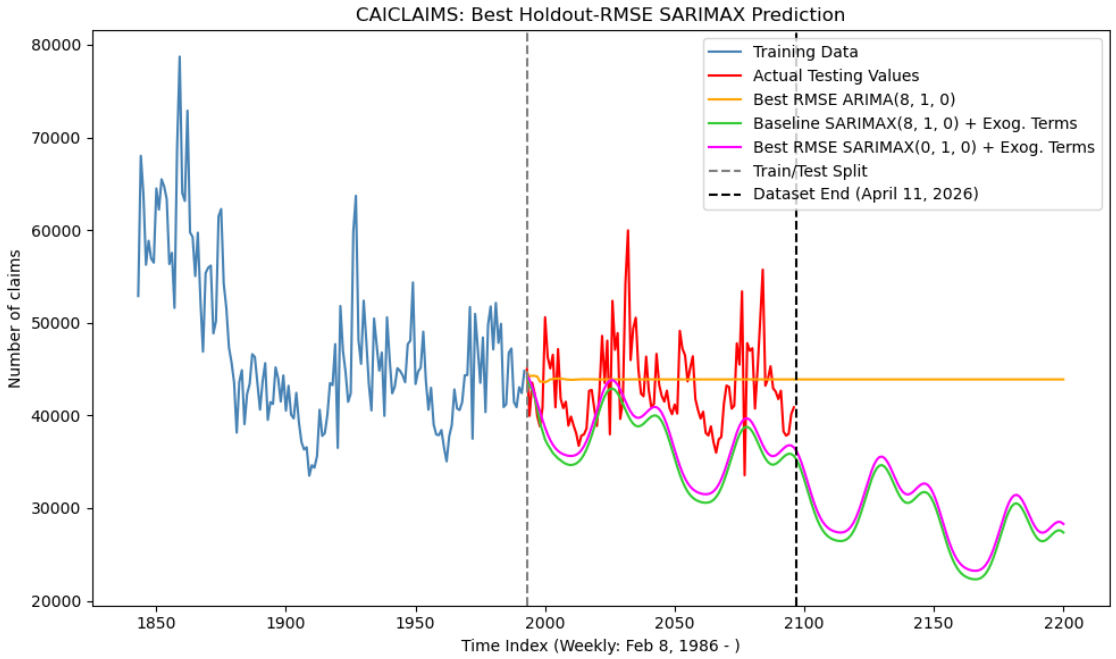
```
================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
2.14e+37. Standard errors may be unstable.
```

3.3.5 Plotted SARIMAX Predictions

Code cell purpose: Plotting entire California train-test split to have clear visualization of the forecasting setup. This matters because the plot directly compares ARIMA and SARIMAX forecasts over the same testing period.

[43]:
```
## Our best forcasting model
best_rmse_sarimax_fcast = best_rmse_sarimax_model.get_forecast(
    steps=len(ytest_CA) + test_horizon,
    exog=Xtest_CA_extended
)

## Best predictors
best_rmse_sarimax_yhat = best_rmse_sarimax_fcast.predicted_mean
best_rmse_sarimax_yhat.index = tpred
best_sarimax_rmse = np.sqrt(mean_squared_error(ytest_CA, best_rmse_sarimax_yhat.
 ↪iloc[:len(ytest_CA)]))

## Plotting
plt.figure(figsize=(10, 6))
plt.plot(ttrain[(train_end - 150):], ytrain_CA.iloc[(train_end - 150):],␣
 ↪label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Actual Testing Values", color="red")

plt.plot(tpred, best_rmse_yhat, color="orange",
         label=f"Best RMSE ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
 ↪{int(best_rmse['q'])})")
plt.plot(tpred, sarimax_yhat, color="limegreen",
         label=f"Baseline SARIMAX({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
 ↪{int(best_rmse['q'])}) + Exog. Terms")
plt.plot(tpred, best_rmse_sarimax_yhat, color="magenta",
         label=f"Best RMSE SARIMAX({best_p}, {best_d}, {best_q}) + Exog. Terms")

plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.axvline(x=len(y_CA), color="black", linestyle="--", label="Dataset End␣
 ↪(April 11, 2026)")

plt.title("CAICLAIMS: Best Holdout-RMSE SARIMAX Prediction")
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - )")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();


print(f"Root Mean Squared Error of Best Holdout RMSE␣
 ↪ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])}, {int(best_rmse['q'])})␣
 ↪Model: {best_rmse_rmse:.3f}")
```

```
print(f"Root Mean Squared Error of Baseline SARIMAX({int(best_rmse['p'])},␣
  ↪{int(best_rmse['d'])}, {int(best_rmse['q'])}) Model: {sarimax_rmse:.3f}")
print(f"Root Mean Squared Error of Best Holdout RMSE SARIMAX({best_p},␣
  ↪{best_d}, {best_q}) Model: {best_sarimax_rmse:.3f}")
```

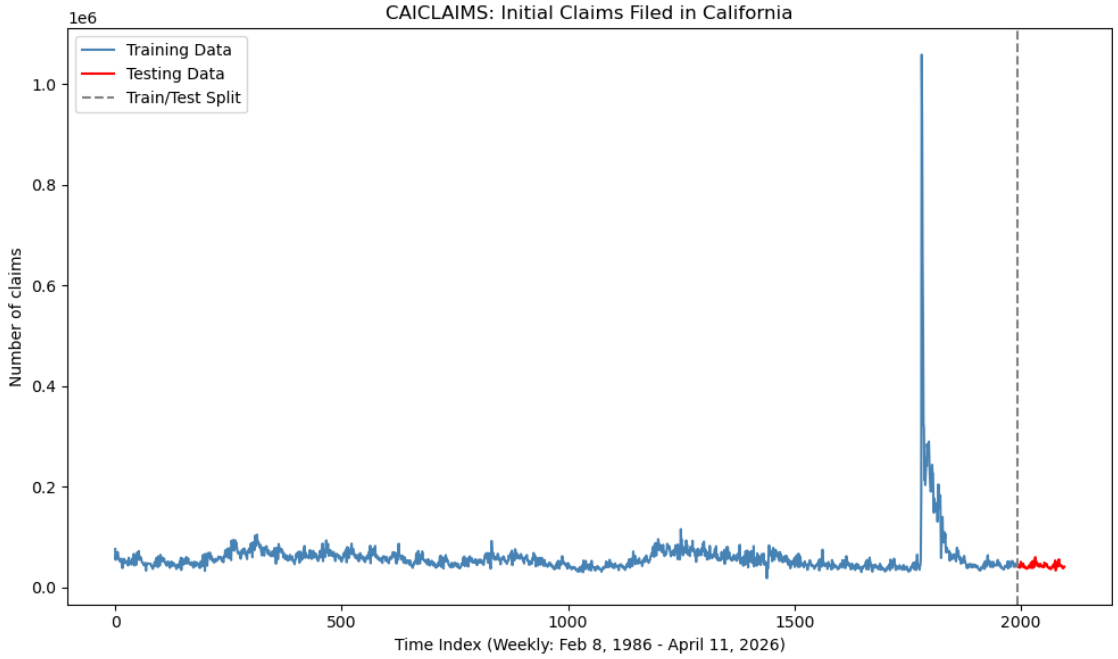



```
Root Mean Squared Error of Best Holdout RMSE ARIMA(8, 1, 0) Model: 4599.134
Root Mean Squared Error of Baseline SARIMAX(8, 1, 0) Model: 7840.432
Root Mean Squared Error of Best Holdout RMSE SARIMAX(0, 1, 0) Model: 7120.883
```

3.4 4. Rolling-CV SARIMAX with Fourier Frequencies, During COVID Dummy

Code cell purpose: Plotting entire California train-test split so forecasting setup is visually clear. This matters because the visualization anchors the second SARIMAX specification to the same train/test forecasting setup.

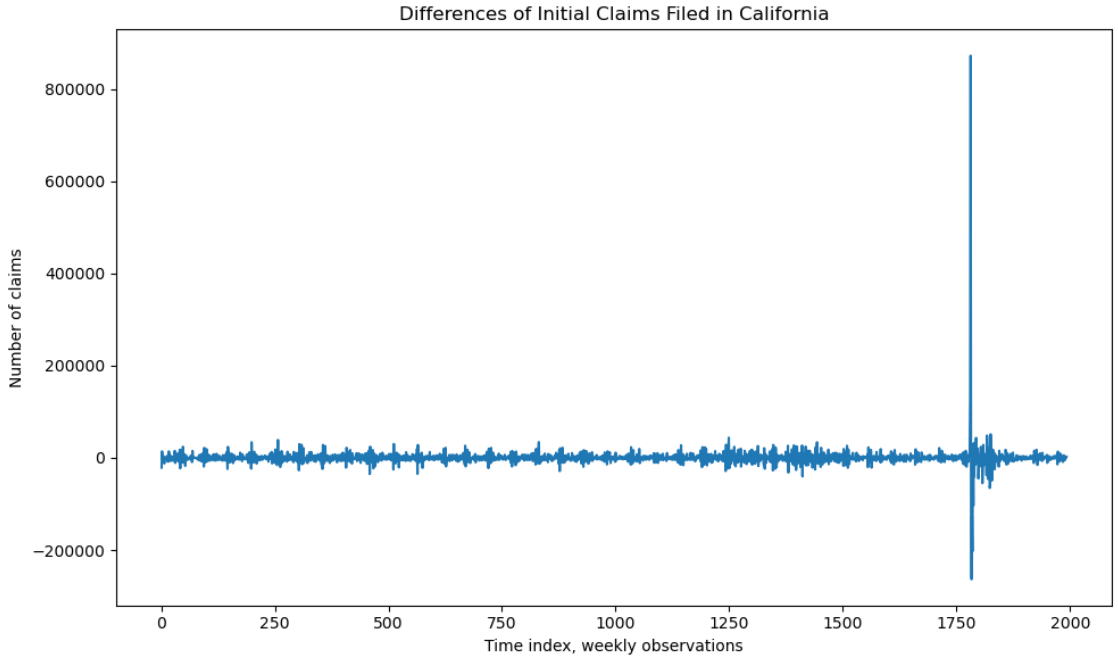
[44]:
```
## Plotting
plt.figure(figsize=(10, 6))
plt.plot(ttrain, ytrain_CA, label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Testing Data", color="red")
plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.title("CAICLAIMS: Initial Claims Filed in California")
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - April 11, 2026)")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();
```



Code cell purpose: Replotting initial differences prior to SARIMAX section so that we keep the model diagnostics similar to the model code. This matters because it keeps the stationarity and differencing logic visible before the alternative COVID specification is fit.

[45]:
```
## Plotting
plt.figure(figsize=(10,6))
plt.plot(ytrain_CA_diff.dropna())
plt.xlabel("Time index, weekly observations")
plt.ylabel("Number of claims")
plt.title("Differences of Initial Claims Filed in California")
```

```
plt.tight_layout()
plt.show();
```



3.4.1 Design Matrix Construction

Code cell purpose: Defines alternative SARIMAX feature set with a during-COVID indicator instead of a post-COVID indicator. This matters because the during-COVID indicator treats the pandemic shock as a temporary disruption rather than a permanent post-COVID shift.

[46]:
```
## Brief function
def make_ca_exog_during(n, covid_start=1775, covid_end=1860, n_harmonics=3):
    t = np.arange(n)
    X = pd.DataFrame({"trend": t}, index=np.arange(n))
    for k in range(1, n_harmonics + 1):
        X[f"sin_{k}"] = np.sin(2 * np.pi * k * t / 52)
        X[f"cos_{k}"] = np.cos(2 * np.pi * k * t / 52)
    X["during_covid"] = ((t >= covid_start) & (t <= covid_end)).astype(int)
    return X
```

Code cell purpose: Builds observed and extended feature matrices for the during-COVID SARIMAX specification. This matters because SARIMAX requires the same exogenous columns to exist for both the observed sample and forecast horizon.

[47]:
```
## Exog features through observed data only (for holdout RMSE/grid search)
X_CA_new = make_ca_exog_during(n=len(y_CA), covid_start=1775, covid_end=1860,␣
 ↪n_harmonics=3)

## Training/Test set
Xtrain_CA_new = X_CA_new.iloc[:train_end]
Xtest_CA_new = X_CA_new.iloc[train_end:len(y_CA)]
Xtest_CA_new = Xtest_CA_new[Xtrain_CA_new.columns]

## Exog through observed data and future period (for extended future forecasts)
X_CA_new_extended = make_ca_exog_during(n=len(y_CA) + test_horizon,␣
 ↪covid_start=1775, covid_end=1860, n_harmonics=3)

## Extended model
Xtrain_CA_new_extended = X_CA_new_extended.iloc[:train_end]
Xtest_CA_new_extended = X_CA_new_extended.iloc[train_end : len(y_CA) +␣
 ↪test_horizon]
Xtest_CA_new_extended = Xtest_CA_new_extended[Xtrain_CA_new_extended.columns]
```

3.4.2 Parameter Selection (Grid Search)

Code cell purpose: Cell below sets up a couple of brief functions to define short rolling-origin cross-validation helpers, specifically for the SARIMAX (during-COVID) specification. This matters because rolling-origin validation evaluates whether a SARIMAX order forecasts well across multiple historical splits.

Code cell purpose: Runs/loads the rolling-origin cross-validation for SARIMAX during-COVID grid. This matters because it uses cached validation results when available and otherwise builds a more robust SARIMAX order comparison.

[48]:
```
## Path
cache_path = "CAICLAIMS_Model_Cache/sarimax_cv_grid_results_CAICLAIMS.csv"

## Parameters
pmax, dmax, qmax = 8, 1, 6

## Parameter grid
param_grid = list(itertools.product(
    range(pmax + 1),
    range(dmax + 1),
    range(qmax + 1)
))

## Initial setups
initial_train_size = 1500
horizon = 52
step = 26

## Brief function for order evaluation
def evaluate_order_cv(order):
    return rolling_cv_sarimax(
        y=ytrain_CA, X=Xtrain_CA_new, order=order,
        initial_train_size=initial_train_size, horizon=horizon, step=step
    )

## Necessary columns
required_cv_cols = {"p", "d", "q", "CV_RMSE", "CV_AIC", "CV_BIC", "n_success",␣
 ↪"n_splits"}

## Running everything
if os.path.exists(cache_path):
    print(f"Loading cached grid search results from {cache_path}")
    results_sarimax_new_df = pd.read_csv(cache_path)

    ## Re-run if the cache is not in the expected format
    if not required_cv_cols.issubset(results_sarimax_new_df.columns):
```

```
        print("Cached rolling-CV SARIMAX results are missing required columns,␣
 ↪grid search must be run again.")
        results_sarimax_new_df = pd.DataFrame()

else:
    results_sarimax_new_df = pd.DataFrame()

if results_sarimax_new_df.empty:
    print("Running SARIMAX rolling-CV grid search...")

    results_sarimax_new = Parallel(n_jobs=-1, verbose=10)(
        delayed(evaluate_order_cv)(order) for order in param_grid
    )

    results_sarimax_new_df = pd.DataFrame(results_sarimax_new)

    ## Saving cache so the notebook can be run again quickly
    results_sarimax_new_df.to_csv(cache_path, index=False)
    print(f"Saved grid search results to {cache_path}")
```

```
Loading cached grid search results from
CAICLAIMS_Model_Cache/sarimax_cv_grid_results_CAICLAIMS.csv
```

3.4.3 Best Models by Evaluation Metrics

Code cell purpose: Cell below completes cleaning and ranking of the rolling-CV SARIMAX grid results by cross-validation RMSE. This matters because sorting the cross-validation results identifies the SARIMAX order with the strongest repeated forecasting performance.

[49]:
```
## Clean and sort results
results_sarimax_new_df = results_sarimax_new_df.dropna(subset=["CV_RMSE"])
results_sarimax_new_df = results_sarimax_new_df.sort_values("CV_RMSE").
 ↪reset_index(drop=True)

print("Top SARIMAX models by rolling-origin CV RMSE:")
display(results_sarimax_new_df.head(10))
```

```
Top SARIMAX models by rolling-origin CV RMSE:
```

	p	d	q	CV_RMSE	CV_AIC	CV_BIC	n_success	n_splits
0	0	1	1	47,712.63	36,588.57	36,642.95	17	17
1	1	1	0	47,925.95	36,835.75	36,890.14	17	17
2	0	1	0	48,233.20	48,112.16	48,161.11	17	17
3	0	1	5	48,303.72	36,424.06	36,500.17	17	17
4	2	1	3	48,985.66	36,554.06	36,630.18	17	17
5	2	1	6	49,353.00	36,380.53	36,472.94	17	17
6	2	1	5	49,509.13	36,401.95	36,488.93	17	17
7	1	1	3	49,546.19	36,467.11	36,537.79	17	17
8	1	1	2	49,645.12	36,497.52	36,562.78	17	17
9	1	1	6	49,664.54	36,398.65	36,485.62	17	17

Code cell purpose: Cell stores the best rolling-CV SARIMAX order, also prints its CV diagnostics. This matters because the chosen rolling-CV order is carried into the final holdout forecast and comparison table.

[50]:
```
## Generating new object
best_rmse_sarimax_new = results_sarimax_new_df.iloc[0]

## Best of different parameters
best_p_new = int(best_rmse_sarimax_new['p'])
best_d_new = int(best_rmse_sarimax_new['d'])
best_q_new = int(best_rmse_sarimax_new['q'])

## Printing
print(f"Best SARIMAX by rolling CV: SARIMAX({best_p_new}, {best_d_new},␣
 ↪{best_q_new})")
print(f"CV RMSE: {best_rmse_sarimax_new['CV_RMSE']:.3f}")
print(f"CV AIC: {best_rmse_sarimax_new['CV_AIC']:.3f}")
print(f"CV BIC: {best_rmse_sarimax_new['CV_BIC']:.3f}")
print(f"Successful CV splits: {int(best_rmse_sarimax_new['n_success'])} /␣
 ↪{int(best_rmse_sarimax_new['n_splits'])}")
```

```
Best SARIMAX by rolling CV: SARIMAX(0, 1, 1)
CV RMSE: 47712.634
CV AIC: 36588.570
CV BIC: 36642.955
Successful CV splits: 17 / 17
```

3.4.4 Final Holdout Evaluation

Code cell purpose: Forecasting from the baseline SARIMAX model and makes direct comparison with the ARIMA baseline. This matters because it converts the selected SARIMAX specification into a final out-of-sample forecast on the same test window.

[51]:
```
## Best SARIMAX model
best_rmse_sarimax_model_new = SARIMAX(
    ytrain_CA,
    exog=Xtrain_CA_new_extended,
    order=(best_p_new, best_d_new, best_q_new),
    enforce_stationarity=False,
    enforce_invertibility=False
).fit(disp=False)

## Summary
print(best_rmse_sarimax_model_new.summary())

## Forecasting
best_rmse_sarimax_fcast_new = best_rmse_sarimax_model_new.get_forecast(
    steps=len(ytest_CA) + test_horizon,
    exog=Xtest_CA_new_extended
)

best_rmse_sarimax_yhat_new = best_rmse_sarimax_fcast_new.predicted_mean
best_rmse_sarimax_yhat_new.index = tpred
best_sarimax_rmse_new = np.sqrt(mean_squared_error(ytest_CA,␣
 ↪best_rmse_sarimax_yhat_new.iloc[:len(ytest_CA)]))

print(f"\nFinal 2-year holdout RMSE for Best Rolling-CV SARIMAX({best_p_new},␣
 ↪{best_d_new}, {best_q_new}) Model: {best_sarimax_rmse_new:.3f}")
```

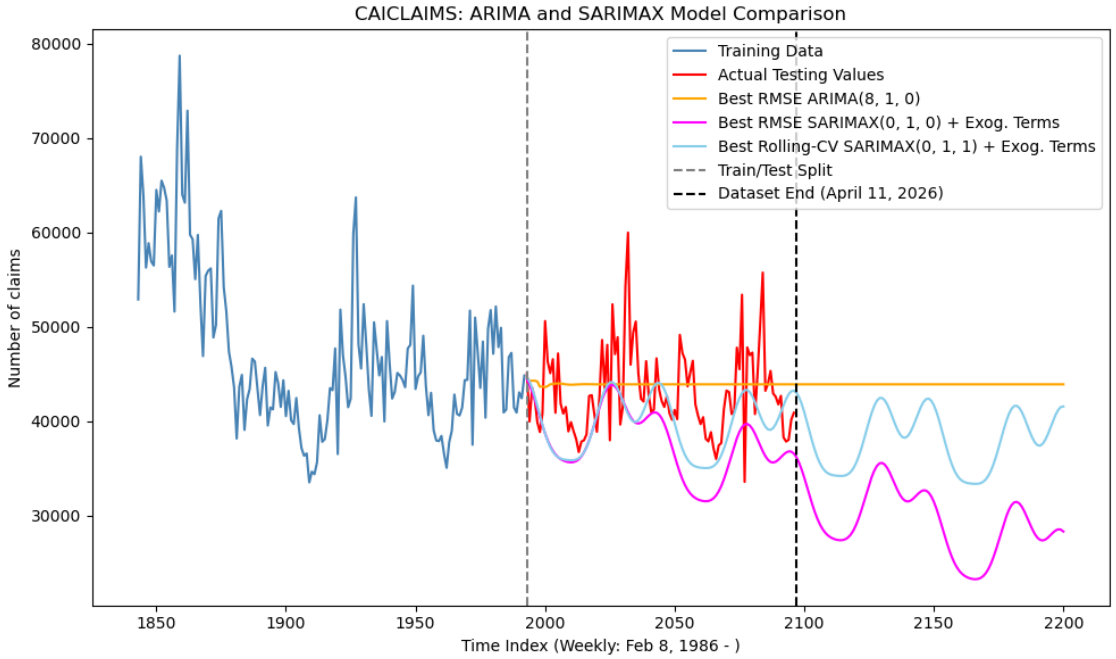
```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1993
Model:               SARIMAX(0, 1, 1)   Log Likelihood              -22855.555
Date:                Mon, 04 May 2026   AIC                          45731.110
Time:                        16:52:06   BIC                          45787.069
Sample:                             0   HQIC                         45751.662
                                 - 1993
Covariance Type:                   opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
trend        -16.0757   1329.037     -0.012      0.990   -2620.940    2588.788
sin_1       2611.3769   1.67e+04      0.157      0.875      -3e+04    3.53e+04
cos_1       2341.5993   1.47e+04      0.159      0.874   -2.65e+04    3.12e+04
sin_2      -2246.1760   6939.342     -0.324      0.746   -1.58e+04    1.14e+04
```

```
cos_2          100.8307   7572.613      0.013      0.989   -1.47e+04    1.49e+04
sin_3         -840.5430   4602.707     -0.183      0.855   -9861.682    8180.596
cos_3          976.4103   4486.557      0.218      0.828   -7817.080    9769.901
during_covid  2322.6661      1.445   1607.125      0.000    2319.834    2325.499
ma.L1            0.0290      0.004      7.802      0.000       0.022       0.036
sigma2        5.548e+08     30.400   1.82e+07      0.000    5.55e+08    5.55e+08
==============================================================================
===
Ljung-Box (L1) (Q):                   0.03   Jarque-Bera (JB):
73053920.16
Prob(Q):                              0.87   Prob(JB):
0.00
Heteroskedasticity (H):              21.96   Skew:
23.99
Prob(H) (two-sided):                  0.00   Kurtosis:
940.42
==============================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
6.87e+23. Standard errors may be unstable.

Final 2-year holdout RMSE for Best Rolling-CV SARIMAX(0, 1, 1) Model: 5764.098
```

3.4.5 Plotted SARIMAX Predictions

Code cell purpose: This cell plots the entire California train-test split so the forecasting setup is visually clear. This matters because the final plot shows how the main ARIMA and SARIMAX forecasts differ in timing and magnitude over the holdout period.

[52]:
```
## Setting up figure
plt.figure(figsize=(10, 6))
plt.plot(ttrain[(train_end - 150):], ytrain_CA.iloc[(train_end - 150):],␣
 ↪label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Actual Testing Values", color="red")

## Plotting
plt.plot(tpred, best_rmse_yhat, color="orange",
         label=f"Best RMSE ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
 ↪{int(best_rmse['q'])})")
plt.plot(tpred, best_rmse_sarimax_yhat, color="magenta",
         label=f"Best RMSE SARIMAX({best_p}, {best_d}, {best_q}) + Exog. Terms")
plt.plot(tpred, best_rmse_sarimax_yhat_new, color="skyblue",
         label=f"Best Rolling-CV SARIMAX({best_p_new}, {best_d_new},␣
 ↪{best_q_new}) + Exog. Terms")

plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.axvline(x=len(y_CA), color="black", linestyle="--", label="Dataset End␣
 ↪(April 11, 2026)")

plt.title("CAICLAIMS: ARIMA and SARIMAX Model Comparison")
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - )")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();

## Printing
print(f"Root Mean Squared Error of Best Holdout RMSE␣
 ↪ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])}, {int(best_rmse['q'])})␣
 ↪Model: {best_rmse_rmse:.3f}")
print(f"Root Mean Squared Error of Best Holdout RMSE SARIMAX({best_p},␣
 ↪{best_d}, {best_q}) Model: {best_sarimax_rmse:.3f}")
print(f"Root Mean Squared Error of Best Rolling-CV SARIMAX({best_p_new},␣
 ↪{best_d_new}, {best_q_new}) Model: {best_sarimax_rmse_new:.3f}")
```




```
Root Mean Squared Error of Best Holdout RMSE ARIMA(8, 1, 0) Model: 4599.134
Root Mean Squared Error of Best Holdout RMSE SARIMAX(0, 1, 0) Model: 7120.883
Root Mean Squared Error of Best Rolling-CV SARIMAX(0, 1, 1) Model: 5764.098
```

3.5 5. Rolling-CV SARIMAX with Fourier Frequencies, During COVID Dummy, Google Trends Data

3.5.1 Pull and Process Google Trends Data

Code cell purpose: What: This cell pulls or loads cached Google Trends data for California using unemployment-related search terms. Why: Caching the Trends file makes the notebook reproducible and avoids repeated API calls while still allowing the SARIMAX model to include external labor-market search information.

[53]:
```
CA_trend_geos = {
    "CAICLAIMS": "US-CA"
}

CA_trend_state_abbrev = {
    "CAICLAIMS": "CA"
}

trend_keywords = [
    "unemployment",
    "unemployment benefits",
    "file for unemployment",
    "unemployment office",
    "apply for unemployment",
]

## Pull Google Trends data for CA
pytrends = TrendReq(hl="en-US", tz=360)

for code, geo in CA_trend_geos.items():
    abbrev = CA_trend_state_abbrev[code]
    trends_path = data_dir / f"google_trends_{abbrev}.csv"

    if trends_path.exists():
        gt_pull = pd.read_csv(trends_path, index_col=0, parse_dates=True)
        print(f"Loaded cached Google Trends data for {code}")

    else:
        attempts = 0
        success = False

        while not success and attempts < 5:
            try:
                pytrends.build_payload(
                    kw_list=trend_keywords,
                    timeframe="2004-01-01 2026-04-01",
                    geo=geo
                )
```

```
                gt_pull = pytrends.interest_over_time()
                gt_pull = gt_pull.drop(columns=["isPartial"], errors="ignore")
                gt_pull.index = pd.to_datetime(gt_pull.index)
                gt_pull = gt_pull.resample("W-SAT").mean()

                gt_pull.to_csv(trends_path)

                print(f"Saved Google Trends data for {code}")
                success = True

                time.sleep(30)

            except TooManyRequestsError:
                attempts += 1
                wait_time = 60 * attempts
                print(f"Too many requests for {code}. Waiting {wait_time}␣
↪seconds before retrying...")
                time.sleep(wait_time)

        if not success:
            print(f"Skipping {code}: Google Trends request failed after 5␣
↪attempts.")
```

```
Loaded cached Google Trends data for CAICLAIMS
```

Code cell purpose: What: This cell reads the cached California Google Trends data, cleans it, converts it to weekly dates, aligns it to the CAICLAIMS series, and displays the resulting frames. Why: This verifies that the external Trends features are correctly prepared before they are added to the forecasting design matrix.

[54]:
```
## Loading, cleaning, and aligning Google Trends to CAICLAIMS dates
gt_CA_raw = read_google_trends_state("CA")
gt_CA_clean = clean_google_trends_cols(gt_CA_raw)
gt_CA_weekly = convert_google_dates_to_claims_week(gt_CA_clean)
gt_CA_aligned = align_google_trends(date_CA, gt_CA_weekly)

display(gt_CA_weekly.head())
display(gt_CA_aligned.loc[gt_CA_aligned["Date"] >= "2004-01-01"].head())
display(gt_CA_aligned.tail())
```

```
        Date  unemployment  unemployment benefits  file for unemployment  \
0 2004-01-03          8.00                   1.00                   0.00
1 2004-01-10           NaN                    NaN                    NaN
2 2004-01-17           NaN                    NaN                    NaN
3 2004-01-24           NaN                    NaN                    NaN
4 2004-01-31           NaN                    NaN                    NaN

   unemployment office  apply for unemployment
```

```
0                 1.00                    0.00
1                  NaN                     NaN
2                  NaN                     NaN
3                  NaN                     NaN
4                  NaN                     NaN
```

```
          Date  unemployment  unemployment benefits  file for unemployment  \
934 2004-01-03          8.00                    1.00                   0.00
935 2004-01-10          7.80                    1.00                   0.00
936 2004-01-17          7.60                    1.00                   0.00
937 2004-01-24          7.40                    1.00                   0.00
938 2004-01-31          7.20                    1.00                   0.00

     unemployment office  apply for unemployment
934                 1.00                    0.00
935                 1.00                    0.00
936                 1.00                    0.00
937                 1.00                    0.00
938                 1.00                    0.00
```

```
           Date  unemployment  unemployment benefits  file for unemployment  \
2092 2026-03-14          7.75                    1.00                   0.00
2093 2026-03-21          7.50                    1.00                   0.00
2094 2026-03-28          7.25                    1.00                   0.00
2095 2026-04-04          7.00                    1.00                   0.00
2096 2026-04-11          7.00                    1.00                   0.00

      unemployment office  apply for unemployment
2092                 0.00                    0.00
2093                 0.00                    0.00
2094                 0.00                    0.00
2095                 0.00                    0.00
2096                 0.00                    0.00
```

Code cell purpose: What: This cell prints date and missingness checks for the aligned California Google Trends variables. Why: These checks confirm that the Trends data line up with the claims dates and that interpolation has handled missing values appropriately for modeling.

[55]:
```
## Quick alignment checks
trend_cols = [c for c in gt_CA_aligned.columns if c != "Date"]

print("First Google Trends weekly date:")
print(gt_CA_weekly["Date"].min())

print("\nFirst CA claims dates in 2004:")
print(date_CA[date_CA >= pd.Timestamp("2004-01-01")].head())

print("\nMissing share after alignment, full CA claims period:")
print(gt_CA_aligned[trend_cols].isna().mean())
```

```
print("\nMissing share after 2004:")
print(gt_CA_aligned.loc[gt_CA_aligned["Date"] >= "2004-01-01", trend_cols].
  ↪isna().mean())
```

```
First Google Trends weekly date:
2004-01-03 00:00:00

First CA claims dates in 2004:
934   2004-01-03
935   2004-01-10
936   2004-01-17
937   2004-01-24
938   2004-01-31
Name: Date, dtype: datetime64[ns]

Missing share after alignment, full CA claims period:
unemployment             0.00
unemployment benefits    0.00
file for unemployment    0.00
unemployment office      0.00
apply for unemployment   0.00
dtype: float64

Missing share after 2004:
unemployment             0.00
unemployment benefits    0.00
file for unemployment    0.00
unemployment office      0.00
apply for unemployment   0.00
dtype: float64
```

3.5.2 Design Matrix Construction

Code cell purpose: What: This cell builds the California SARIMAX + Google Trends model sample, including trend terms, Fourier seasonality, a during-COVID indicator, and aligned Trends features. Why: The model needs a clean design matrix and consistent train/test split so the added Google Trends variables can be evaluated fairly against earlier baselines.

[56]:
```
## Building the Google Trends model sample
def make_ca_gt_base_features(dates, n_harmonics=3):
    dates = pd.to_datetime(pd.Series(dates)).reset_index(drop=True)
    t = np.arange(len(dates))
    X = pd.DataFrame({"trend": t})
    for k in range(1, n_harmonics + 1):
        X[f"sin_{k}"] = np.sin(2 * np.pi * k * t / 52)
        X[f"cos_{k}"] = np.cos(2 * np.pi * k * t / 52)
    X["during_covid"] = dates.between("2020-03-14", "2021-10-02").astype(int)
    return X

def make_ca_gt_exog(dates, gt_aligned):
    X_base = make_ca_gt_base_features(dates)
    X_gt = rename_google_trends_features(gt_aligned)
    return pd.concat([X_base, X_gt.reset_index(drop=True)], axis=1)

gt_start_date_CA = gt_CA_weekly["Date"].min()
gt_start_idx_CA = int(np.where(date_CA >= gt_start_date_CA)[0][0])

y_CA_gt = y_CA.iloc[gt_start_idx_CA:].reset_index(drop=True)
date_CA_gt = date_CA.iloc[gt_start_idx_CA:].reset_index(drop=True)
gt_CA_model = gt_CA_aligned.iloc[gt_start_idx_CA:].reset_index(drop=True)

test_horizon_gt = 104
train_end_gt = len(y_CA_gt) - test_horizon_gt

ytrain_CA_gt = y_CA_gt.iloc[:train_end_gt]
ytest_CA_gt = y_CA_gt.iloc[train_end_gt:]

date_train_CA_gt = date_CA_gt.iloc[:train_end_gt]
date_test_CA_gt = date_CA_gt.iloc[train_end_gt:]

X_CA_gt = make_ca_gt_exog(date_CA_gt, gt_CA_model)
Xtrain_CA_gt = X_CA_gt.iloc[:train_end_gt]
Xtest_CA_gt = X_CA_gt.iloc[train_end_gt:][Xtrain_CA_gt.columns]

print(f"Google Trends model period: {date_CA_gt.iloc[0].date()} to {date_CA_gt.
 ↪iloc[-1].date()}")
print(f"Training period: {date_train_CA_gt.iloc[0].date()} to {date_train_CA_gt.
 ↪iloc[-1].date()}")
```

```
print(f"Testing period: {date_test_CA_gt.iloc[0].date()} to {date_test_CA_gt.
  ↪iloc[-1].date()}")
print(f"Number of exog columns: {X_CA_gt.shape[1]}")
display(X_CA_gt.head())
```

```
Google Trends model period: 2004-01-03 to 2026-04-11
Training period: 2004-01-03 to 2024-04-13
Testing period: 2024-04-20 to 2026-04-11
Number of exog columns: 13
```

	trend	sin_1	cos_1	sin_2	cos_2	sin_3	cos_3	during_covid	\
0	0	0.00	1.00	0.00	1.00	0.00	1.00	0	
1	1	0.12	0.99	0.24	0.97	0.35	0.94	0	
2	2	0.24	0.97	0.46	0.89	0.66	0.75	0	
3	3	0.35	0.94	0.66	0.75	0.89	0.46	0	
4	4	0.46	0.89	0.82	0.57	0.99	0.12	0	

	gt_unemployment	gt_unemployment_benefits	gt_file_for_unemployment	\
0	8.00	1.00	0.00	
1	7.80	1.00	0.00	
2	7.60	1.00	0.00	
3	7.40	1.00	0.00	
4	7.20	1.00	0.00	

	gt_unemployment_office	gt_apply_for_unemployment
0	1.00	0.00
1	1.00	0.00
2	1.00	0.00
3	1.00	0.00
4	1.00	0.00

3.5.3 Parameter Selection (Grid Search)

Code cell purpose: What: This cell sets the SARIMAX + Google Trends parameter grid, loads cached rolling-CV results when available, and runs the grid search if needed. Why: Rolling cross-validation selects a time-series specification using repeated out-of-sample windows rather than relying only on one holdout split.

[57]:
```
cache_path_gt = "CAICLAIMS_Model_Cache/sarimax_cv_gt_grid_results_CAICLAIMS.csv"

pmax, dmax, qmax = 8, 1, 6

param_grid_gt = list(itertools.product(
    range(pmax + 1),
    range(dmax + 1),
    range(qmax + 1)
))

initial_train_size_gt = 600
horizon_gt = 52
step_gt = 26

required_gt_cv_cols = {
    "p", "d", "q", "CV_RMSE", "CV_AIC", "CV_BIC", "n_success", "n_splits"
}

def evaluate_order_cv_gt(order):
    return rolling_cv_sarimax(
        y=ytrain_CA_gt,
        X=Xtrain_CA_gt,
        order=order,
        initial_train_size=initial_train_size_gt,
        horizon=horizon_gt,
        step=step_gt
    )

if os.path.exists(cache_path_gt):
    print(f"Loading cached Google Trends SARIMAX CV results from␣
 ↪{cache_path_gt}")
    results_sarimax_gt_df = pd.read_csv(cache_path_gt)
    if not required_gt_cv_cols.issubset(results_sarimax_gt_df.columns):
        results_sarimax_gt_df = pd.DataFrame()
else:
    results_sarimax_gt_df = pd.DataFrame()

if results_sarimax_gt_df.empty:
    print("Running SARIMAX + Google Trends rolling-CV grid search...")
    results_sarimax_gt = Parallel(n_jobs=-1, verbose=10)(
        delayed(evaluate_order_cv_gt)(order) for order in param_grid_gt
```

```
    )
    results_sarimax_gt_df = pd.DataFrame(results_sarimax_gt)
    results_sarimax_gt_df.to_csv(cache_path_gt, index=False)
    print(f"Saved Google Trends SARIMAX CV results to {cache_path_gt}")
```

```
Loading cached Google Trends SARIMAX CV results from
CAICLAIMS_Model_Cache/sarimax_cv_gt_grid_results_CAICLAIMS.csv
```

3.5.4 Best Models by Evaluation Metrics

Code cell purpose: What: This cell sorts the SARIMAX + Google Trends grid-search results by rolling-CV RMSE and displays the top candidates. Why: Showing the best candidate models makes the parameter-selection step transparent and lets us verify which specifications performed best during validation.

[58]:
```
## Selecting the best SARIMAX + Google Trends model by rolling-CV RMSE
results_sarimax_gt_df = results_sarimax_gt_df.dropna(subset=["CV_RMSE"])
results_sarimax_gt_df = results_sarimax_gt_df.sort_values("CV_RMSE").
 ↪reset_index(drop=True)

print("Top SARIMAX + Google Trends models by rolling-origin CV RMSE:")
display(results_sarimax_gt_df.head(10).round(3))
```

```
Top SARIMAX + Google Trends models by rolling-origin CV RMSE:
```

	p	d	q	CV_RMSE	CV_AIC	CV_BIC	n_success	n_splits
0	5	0	2	30,633.85	17,271.85	17,369.72	16	16
1	1	0	3	30,707.57	17,297.20	17,381.11	16	16
2	1	0	1	30,710.22	17,345.55	17,420.18	16	16
3	1	0	2	30,728.18	17,324.09	17,403.36	16	16
4	5	0	1	30,752.57	17,273.12	17,366.33	16	16
5	1	0	0	30,757.91	17,426.96	17,496.94	16	16
6	4	0	1	30,778.14	17,294.74	17,383.31	16	16
7	2	0	6	30,796.02	17,230.47	17,332.94	16	16
8	2	0	0	30,813.16	17,354.94	17,429.56	16	16
9	0	0	3	30,821.83	17,367.46	17,446.71	16	16

Code cell purpose: What: This cell extracts the best SARIMAX + Google Trends order from the rolling-CV results and prints its main validation metrics. Why: These selected parameters are reused in the final holdout evaluation, so the notebook needs to clearly record the chosen model specification.

[59]:
```
best_rmse_sarimax_gt = results_sarimax_gt_df.iloc[0]

best_p_gt = int(best_rmse_sarimax_gt["p"])
best_d_gt = int(best_rmse_sarimax_gt["d"])
best_q_gt = int(best_rmse_sarimax_gt["q"])

print(f"Best SARIMAX + Google Trends by rolling CV: SARIMAX({best_p_gt},␣
 ↪{best_d_gt}, {best_q_gt})")
print(f"CV RMSE: {best_rmse_sarimax_gt['CV_RMSE']:.3f}")
print(f"CV AIC: {best_rmse_sarimax_gt['CV_AIC']:.3f}")
print(f"CV BIC: {best_rmse_sarimax_gt['CV_BIC']:.3f}")
print(f"Successful CV splits: {int(best_rmse_sarimax_gt['n_success'])} /␣
 ↪{int(best_rmse_sarimax_gt['n_splits'])}")
```

```
Best SARIMAX + Google Trends by rolling CV: SARIMAX(5, 0, 2)
```

```
CV RMSE: 30633.854
CV AIC: 17271.851
CV BIC: 17369.717
Successful CV splits: 16 / 16
```

3.5.5 Final Holdout Evaluation

Code cell purpose: What: This cell fits the selected SARIMAX + Google Trends model on the training data and evaluates it on the final two-year holdout. Why: The final holdout RMSE gives the main out-of-sample test of whether Google Trends improves the time-series forecast.

[60]:
```
## Final two-year holdout evaluation, no future forecast due to Google Trends␣
 ↪data only being available through the end of the observed period.
best_rmse_sarimax_gt_model = SARIMAX(
    ytrain_CA_gt,
    exog=Xtrain_CA_gt,
    order=(best_p_gt, best_d_gt, best_q_gt),
    enforce_stationarity=False,
    enforce_invertibility=False
).fit(disp=False)

print(best_rmse_sarimax_gt_model.summary())

best_rmse_sarimax_gt_fcast = best_rmse_sarimax_gt_model.get_forecast(
    steps=len(ytest_CA_gt),
    exog=Xtest_CA_gt
)

best_rmse_sarimax_gt_yhat = best_rmse_sarimax_gt_fcast.predicted_mean
best_rmse_sarimax_gt_yhat.index = ytest_CA_gt.index
best_sarimax_rmse_gt = np.sqrt(mean_squared_error(ytest_CA_gt,␣
 ↪best_rmse_sarimax_gt_yhat))

print(f"\nFinal 2-year holdout RMSE for Best Rolling-CV SARIMAX({best_p_gt},␣
 ↪{best_d_gt}, {best_q_gt}) + Google Trends Model: {best_sarimax_rmse_gt:.3f}")
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                 Claims   No. Observations:                 1059
Model:               SARIMAX(5, 0, 2)   Log Likelihood              -12180.957
Date:                Mon, 04 May 2026   AIC                         24403.913
Time:                        16:52:09   BIC                         24508.081
Sample:                             0   HQIC                        24443.403
                                - 1059
Covariance Type:                  opg
==============================================================================
=============
                                 coef    std err          z      P>|z|
[0.025      0.975]
------------------------------------------------------------------------------
-------------
trend                         -1.5480     12.722     -0.122      0.903
-26.482      23.386
```

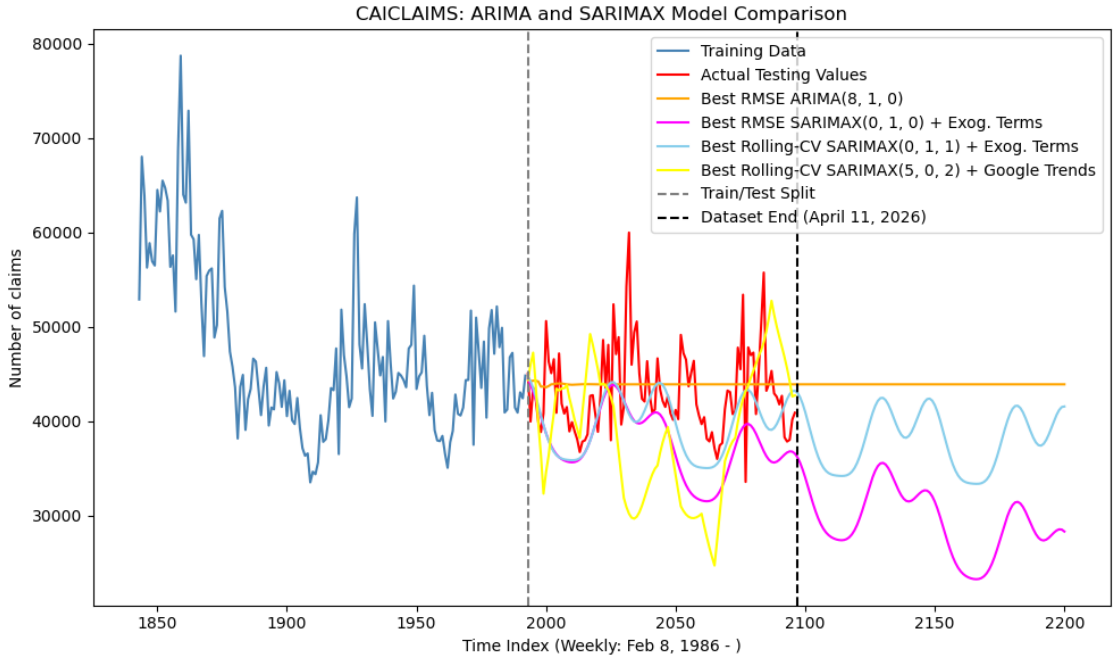
```
sin_1                     -547.8992   5729.566     -0.096      0.924
-1.18e+04    1.07e+04
cos_1                     1758.2966   4975.446      0.353      0.724
-7993.399    1.15e+04
sin_2                    -3310.1167   5619.924     -0.589      0.556
-1.43e+04    7704.732
cos_2                      490.3588   5563.931      0.088      0.930
-1.04e+04    1.14e+04
sin_3                     -524.4306   5246.064     -0.100      0.920
-1.08e+04    9757.666
cos_3                      873.6865   5056.030      0.173      0.863
-9035.950    1.08e+04
during_covid             -6.744e+04   9584.072     -7.037      0.000
-8.62e+04   -4.87e+04
gt_unemployment           7817.7637    685.923     11.397      0.000
6473.379    9162.149
gt_unemployment_benefits  -1.28e+04   8180.678     -1.564      0.118
-2.88e+04    3235.890
gt_file_for_unemployment  -2.391e+04   3681.568     -6.493      0.000
-3.11e+04   -1.67e+04
gt_unemployment_office    -5828.2465   7974.408     -0.731      0.465
-2.15e+04    9801.306
gt_apply_for_unemployment  6290.4625   3601.250      1.747      0.081
-767.858    1.33e+04
ar.L1                         0.6661      0.113      5.903      0.000
0.445       0.887
ar.L2                         0.8014      0.053     15.151      0.000
0.698       0.905
ar.L3                        -0.6267      0.068     -9.246      0.000
-0.760      -0.494
ar.L4                         0.0229      0.038      0.602      0.547
-0.052       0.097
ar.L5                         0.1209      0.029      4.238      0.000
0.065       0.177
ma.L1                         0.0946      0.113      0.836      0.403
-0.127       0.316
ma.L2                        -1.0551      0.117     -9.001      0.000
-1.285      -0.825
sigma2                    6.256e+08      0.273   2.29e+09      0.000
6.26e+08    6.26e+08
===================================================================================
===
Ljung-Box (L1) (Q):                   0.26   Jarque-Bera (JB):
11320437.69
Prob(Q):                              0.61   Prob(JB):
0.00
Heteroskedasticity (H):              18.24   Skew:
18.32
```

```
Prob(H) (two-sided):                  0.00   Kurtosis:
509.39
===============================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
[2] Covariance matrix is singular or near-singular, with condition number
7.31e+24. Standard errors may be unstable.

Final 2-year holdout RMSE for Best Rolling-CV SARIMAX(5, 0, 2) + Google Trends
Model: 9244.610
```

3.5.6 Plotted SARIMAX Predictions

Code cell purpose: What: This cell plots the CAICLAIMS ARIMA and SARIMAX variants together and prints their RMSE values. Why: Comparing the forecasts visually and numerically shows whether the added exogenous terms, COVID controls, and Google Trends features improve performance relative to the baseline.

[61]:
```
## Setting up figure
plt.figure(figsize=(10, 6))
plt.plot(ttrain[(train_end - 150):], ytrain_CA.iloc[(train_end - 150):],␣
 ↪label="Training Data", color="steelblue")
plt.plot(ttest, ytest_CA, label="Actual Testing Values", color="red")

## Plotting
plt.plot(tpred, best_rmse_yhat, color="orange",
         label=f"Best RMSE ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
 ↪{int(best_rmse['q'])})")
plt.plot(tpred, best_rmse_sarimax_yhat, color="magenta",
         label=f"Best RMSE SARIMAX({best_p}, {best_d}, {best_q}) + Exog. Terms")
plt.plot(tpred, best_rmse_sarimax_yhat_new, color="skyblue",
         label=f"Best Rolling-CV SARIMAX({best_p_new}, {best_d_new},␣
 ↪{best_q_new}) + Exog. Terms")
plt.plot(ttest, best_rmse_sarimax_gt_yhat, color="yellow",
         label=f"Best Rolling-CV SARIMAX({best_p_gt}, {best_d_gt}, {best_q_gt})␣
 ↪+ Google Trends")

plt.axvline(x=train_end, color="gray", linestyle="--", label="Train/Test Split")
plt.axvline(x=len(y_CA), color="black", linestyle="--", label="Dataset End␣
 ↪(April 11, 2026)")

plt.title("CAICLAIMS: ARIMA and SARIMAX Model Comparison")
plt.xlabel("Time Index (Weekly: Feb 8, 1986 - )")
plt.ylabel("Number of claims")
plt.legend(); plt.tight_layout(); plt.show();

## Printing
print(f"Root Mean Squared Error of Best Holdout RMSE␣
 ↪ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])}, {int(best_rmse['q'])})␣
 ↪Model: {best_rmse_rmse:.3f}")
print(f"Root Mean Squared Error of Best Holdout RMSE SARIMAX({best_p},␣
 ↪{best_d}, {best_q}) Model: {best_sarimax_rmse:.3f}")
print(f"Root Mean Squared Error of Best Rolling-CV SARIMAX({best_p_new},␣
 ↪{best_d_new}, {best_q_new}) Model: {best_sarimax_rmse_new:.3f}")
print(f"Root Mean Squared Error of Best Rolling-CV SARIMAX({best_p_gt},␣
 ↪{best_d_gt}, {best_q_gt}) + Google Trends Model: {best_sarimax_rmse_gt:.3f}")
```




```
Root Mean Squared Error of Best Holdout RMSE ARIMA(8, 1, 0) Model: 4599.134
Root Mean Squared Error of Best Holdout RMSE SARIMAX(0, 1, 0) Model: 7120.883
Root Mean Squared Error of Best Rolling-CV SARIMAX(0, 1, 1) Model: 5764.098
Root Mean Squared Error of Best Rolling-CV SARIMAX(5, 0, 2) + Google Trends
Model: 9244.610
```

3.6 6. Model Comparison Table

Table below provides a succint summary of the main models utilized for prediction here. What is most relevant for report/discussion is the 2-year holdout RMSE, that of which evaluates how effective any given model is at predicting observations which were not utilized in training.

Code cell purpose: Cell below generates concise comparison table for the main ARIMA and SARIMAX specifications. This matters because the comparison table turns the modeling results into one concise object for interpretation and reporting.

[62]:
```
model_comparison = pd.DataFrame([
    {
        "model": f"ARIMA({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
↪{int(best_rmse['q'])})",
        "selection_rule": "Best holdout RMSE among ARIMA grid",
        "holdout_rmse": best_rmse_rmse
    },
    {
        "model": f"SARIMAX({int(best_rmse['p'])}, {int(best_rmse['d'])},␣
↪{int(best_rmse['q'])}) + Fourier + after-COVID dummy",
        "selection_rule": "Uses best ARIMA order, then adds exog terms",
        "holdout_rmse": sarimax_rmse
    },
    {
        "model": f"SARIMAX({best_p}, {best_d}, {best_q}) + Fourier +␣
↪after-COVID dummy",
        "selection_rule": "Best holdout RMSE among SARIMAX grid",
        "holdout_rmse": best_sarimax_rmse
    },
    {
        "model": f"SARIMAX({best_p_new}, {best_d_new}, {best_q_new}) + Fourier␣
↪+ during-COVID dummy",
        "selection_rule": "Best rolling-CV RMSE among SARIMAX grid",
        "holdout_rmse": best_sarimax_rmse_new
    },
    {
        "model": f"SARIMAX({best_p_gt}, {best_d_gt}, {best_q_gt}) + Fourier +␣
↪during-COVID dummy + Google Trends",
        "selection_rule": "Best rolling-CV RMSE among SARIMAX + Google Trends␣
↪grid; 2004+ sample only",
        "holdout_rmse": best_sarimax_rmse_gt
    }
])

display(model_comparison.sort_values("holdout_rmse").round(3))
```

```
                                        model  \
0                                ARIMA(8, 1, 0)
```

```
3    SARIMAX(0, 1, 1) + Fourier + during-COVID dummy
2     SARIMAX(0, 1, 0) + Fourier + after-COVID dummy
1     SARIMAX(8, 1, 0) + Fourier + after-COVID dummy
4  SARIMAX(5, 0, 2) + Fourier + during-COVID dumm…
```

	selection_rule	holdout_rmse
0	Best holdout RMSE among ARIMA grid	4,599.13
3	Best rolling-CV RMSE among SARIMAX grid	5,764.10
2	Best holdout RMSE among SARIMAX grid	7,120.88
1	Uses best ARIMA order, then adds exog terms	7,840.43
4	Best rolling-CV RMSE among SARIMAX + Google Tr…	9,244.61

4 Running Forecasting on All the States, Adding A Machine Learning Comparison

The California-only section above where we utilized ARIMA/SARIMAX it useful as it showcases our logic and flow of work that helped guide us to build a concise pipeline. This section now builds upon that to apply the holistic forecasting workflow to all four states of analysis, California, New York, Indiana, and Hawaii.

Additionally, the purpose of the section here is to compare an econometric time-series baseline to ML modeling processes, where both utilize the same exact two-year test data/holdout split. For each state, we perform work to fit ARIMA, SARIMAX with interpretable exogenous controls, and a LightGBM recursive forecasting model. The main shared metric is RMSE, but MAE is also included as an additional readability check. We also account for explicit handling of the COVID period. Here, the pure ARIMA model is kept as the simple benchmark, having no exogenous controls. The SARIMAX models add Fourier seasonality and COVID structural-break controls, with the simple SARIMAX using an after-COVID dummy and the rolling-CV SARIMAX models using a during-COVID dummy. The LightGBM model receives the same COVID indicators as supervised-learning features, with additional lagged claims, rolling averages, and calendar seasonality.

4.1 1. Helper Functions and Setup

Here, we have a couple of functions that serve as helpers, working to generate the common train/test split, compute shared metrics, build SARIMAX exogenous variables, and handle simple cache validation. Keeping these utilities stored in functions makes the later all-state loops easier to read and lets us focus on actual logic of the models.

4.1.1 Hyperparameters for Analyzing All States

In the cell, we have defined our shared forecasting choice to be used across each state/model, ensuring that the two-year set at the end of the period is kept as a test. Allows us to ensure that the ARIMA, SARIMAX, and LightGBM models are evaluated on the same holdout window. We also create one model-cache folder so that grid-search results can be saved and reused instead of recomputed every run.

Code cell purpose: This cell defines the all-state test horizon, COVID structural-break dates, model-cache folder, and compact ARIMA/SARIMAX candidate grid. This matters because every model below needs to use the same split, the same COVID definitions, and the same saved-cache location for a fair and reproducible comparison.

[63]:
```
all_state_test_horizon = 104

covid_break_start = pd.Timestamp("2020-03-14")
covid_break_end = pd.Timestamp("2021-10-02")

model_cache_dir = Path("Model_Cache")
model_cache_dir.mkdir(exist_ok=True)

all_state_pmax, all_state_dmax, all_state_qmax = 8, 1, 6

all_state_param_grid = list(itertools.product(
    range(all_state_pmax + 1),
    range(all_state_dmax + 1),
    range(all_state_qmax + 1)
))

def get_state_split(code, horizon=104):
    df = claims_dfs[code].sort_values("Date").reset_index(drop=True)
    y = df["Claims"].reset_index(drop=True)
    dates = df["Date"].reset_index(drop=True)
    return df, y, dates, len(y) - horizon

print(f"All-state holdout horizon: {all_state_test_horizon} weeks")
print(f"Candidate ARIMA/SARIMAX orders: {len(all_state_param_grid)}")
```

```
All-state holdout horizon: 104 weeks
Candidate ARIMA/SARIMAX orders: 126
```

4.1.2 SARIMAX Exogenous Feature Constructors

Code cell purpose: This cell defines some helper functions for splitting each state and creating SARIMAX exogenous variables. This matters because the all-state comparison should repeat the same modeling logic across states without copy-pasting inconsistent code. Also, it's just smoother this way.

[64]:
```
def fourier_base(dates, n_harmonics=3):
    dates = pd.to_datetime(pd.Series(dates)).reset_index(drop=True)
    t = np.arange(len(dates))
    X = pd.DataFrame({"trend": t})
    for k in range(1, n_harmonics + 1):
        X[f"sin_{k}"] = np.sin(2 * np.pi * k * t / 52)
        X[f"cos_{k}"] = np.cos(2 * np.pi * k * t / 52)
    return X

def make_exog_after(dates):
    X = fourier_base(dates)
    dates = pd.to_datetime(pd.Series(dates)).reset_index(drop=True)
    X["covid_after"] = (dates > covid_break_end).astype(int)
    return X

def make_exog_during(dates):
    X = fourier_base(dates)
    dates = pd.to_datetime(pd.Series(dates)).reset_index(drop=True)
    X["during_covid"] = dates.between(covid_break_start, covid_break_end).
 ↪astype(int)
    return X
```

4.1.3 Predictions Plot Helper

Code cell purpose: What: This cell defines helper functions for creating future weekly dates and forecast frames that extend beyond observed data. Why: These utilities allow later all-state models to report both the two-year holdout period and an additional future forecast horizon in a consistent format.

[65]:
```
# Helper functions for plotting predictions into the future (beyond observed␣
↪data)
def make_future_dates(dates, horizon=104):
    """
    Creates future weekly dates continuing after the last observed FRED date.
    """
    last_date = pd.to_datetime(dates.iloc[-1])
    return pd.Series(
        pd.date_range(
            start=last_date + pd.Timedelta(weeks=1),
            periods=horizon,
            freq="W-SAT"
        )
    )

def extend_dates(dates, horizon=104):
    """
    Observed dates plus future dates.
    """
    future_dates = make_future_dates(dates, horizon)
    return pd.concat(
        [pd.Series(dates).reset_index(drop=True), future_dates],
        ignore_index=True
    )

def forecast_frame_extended(dates, actual, prediction):
    """
    Allows prediction to be longer than actual.
    Actual values are NaN for future periods beyond the holdout.
    """
    return pd.DataFrame({
        "Date": pd.Series(dates).reset_index(drop=True),
        "Actual": pd.Series(actual).reset_index(drop=True),
        "Prediction": pd.Series(prediction).reset_index(drop=True)
    })
```

Code cell purpose: What: This cell defines helper functions that retrieve model rows, format model-label text, and plot predictions for all four states. Why: Reusing one plotting routine keeps the ARIMA, SARIMAX, Google Trends, and LightGBM forecast comparisons consistent across model families.

[66]:
```
# Helper functions for generating plots
def get_state_model_row(best_df, code):
    return best_df.loc[best_df["StateCode"] == code].iloc[0]


def build_model_info_text(model_sources, code):
    lines = []

    for source in model_sources:
        best_df = source["best_df"]

        if code not in best_df["StateCode"].values:
            continue

        row = get_state_model_row(best_df, code)

        lines.append(
            f"{source['label_prefix']}:\n"
            f"{row['Plot_Label']}\n"
            f"RMSE = {row['RMSE']:,.2f}"
        )

    return "\n\n".join(lines)


def plot_all_state_predictions(model_sources, main_title, train_window=150):
    """
    Plots all-state model predictions using weekly time index, matching the␣
 ↪CAICLAIMS plots

    model_sources should be a list of dictionaries with:
    - forecasts: forecast dictionary keyed by StateCode
    - best_df: model results dataframe with StateCode, Model, RMSE
    - label_prefix: short model label
    - color: plot color
    """

    fig, axes = plt.subplots(4, 1, figsize=(14, 20), sharex=False)

    if not isinstance(axes, np.ndarray):
        axes = np.array([axes])

    for ax, code in zip(axes, datasets):
        df, y, dates, train_end_state = get_state_split(code,␣
 ↪all_state_test_horizon)

        y_train = y.iloc[:train_end_state]
```

```
        y_test = y.iloc[train_end_state:]
        ttrain_state = np.arange(train_end_state)
        ttest_state = np.arange(train_end_state, len(y))

        ax.plot(ttrain_state[-train_window:], y_train.iloc[-train_window:],␣
↪label="Training Data", color="steelblue")
        ax.plot(ttest_state, y_test, label="Actual Future Values", color="red")

        for source in model_sources:
            forecasts = source["forecasts"]

            if code not in forecasts:
                continue

            forecast_df = forecasts[code]
            row = get_state_model_row(source["best_df"], code)

            # Forecasts should align to the holdout weekly index
            forecast_x = np.arange(
                train_end_state,
                train_end_state + len(forecast_df)
            )

            label = f"{source['label_prefix']}:\n{row['Plot_Label']}"

            ax.plot(
                forecast_x,
                forecast_df["Prediction"],
                label=label,
                color=source["color"],
                alpha=source.get("alpha", 0.65)
            )

        ax.axvline(x=train_end_state, color="gray", linestyle="--",␣
↪label="Train/Test Split")

        ax.axvline(x=len(y) - 1, color="black", linestyle="--", label=f"Dataset␣
↪End ({dates.iloc[-1].date()})")

        ax.set_title(f"{code}: {state_names[code]}")
        ax.set_ylabel("Number of claims")

        forecast_end_date = dates.iloc[-1].date()

        visible_start_date = dates.iloc[max(0, train_end_state - train_window)].
↪date()
        observed_end_date = dates.iloc[-1].date()
```

```
        forecast_end_date = observed_end_date
        for source in model_sources:
            forecasts = source["forecasts"]
            if code in forecasts:
                forecast_end_date = max(
                    forecast_end_date,
                    pd.to_datetime(forecasts[code]["Date"]).max().date()
                )

        ax.set_xlabel(f"Weekly Index (Visible Window): {visible_start_date} to␣
↪{forecast_end_date}")

        model_info_text = build_model_info_text(model_sources, code)
        ax.text(
            1.01,
            0.98,
            model_info_text,
            transform=ax.transAxes,
            va="top",
            ha="left",
            fontsize=9,
            bbox=dict(facecolor="white", alpha=0.85, edgecolor="gray")
        )

        ax.legend(fontsize=8, loc="upper left")

    fig.suptitle(main_title, fontsize=16)
    plt.tight_layout(rect=[0, 0, 0.73, 0.98])
    plt.show()
```

4.2 2. ARIMA Baseline Model

ARIMA is the simplest benchmark since it uses only past values of the claims series. It doesn't explicitly receive COVID information, so it tells us how much forecasting power comes from the time-series structure alone. The code below applies the same compact ARIMA grid to all four states and caches each state's grid results. Again, helpful for testing.

Code cell purpose: This cell defines a short ARIMA fitting helper function and applies the ARIMA grid search to every state using the same final two-year test split. This matters because it extends the baseline model beyond California and creates a directly comparable RMSE/MAE benchmark for the later SARIMAX and LightGBM models.

[67]:
```
## Fits one ARIMA order and returns holdout metrics, brief helper function
def arima_grid_row(y_train, y_test, order):
    p, d, q = map(int, order)
    try:
        model = ARIMA(y_train, order=(p, d, q)).fit()
        pred = model.get_forecast(steps=len(y_test)).predicted_mean
        pred.index = y_test.index
        return {"p": p, "d": d, "q": q, "AIC": model.aic, "BIC": model.bic,
                "RMSE": calc_rmse(y_test, pred), "status": "success"}
    except Exception:
        return {"p": p, "d": d, "q": q, "AIC": np.nan, "BIC": np.nan,
                "RMSE": np.nan, "status": "failed"}
```

4.2.1 Parameter Selection (Grid Search)

Code cell purpose: What: This cell runs the ARIMA baseline grid search for all four states, selects each state's best holdout-RMSE specification, fits it, and stores forecasts through the holdout and future horizon. Why: ARIMA provides the main time-series benchmark against which the richer SARIMAX and machine-learning models are compared.

[68]:
```
## Run ARIMA baseline for all states
arima_all_results = {}
arima_all_best = []
arima_all_forecasts = {}

arima_required_cols = {"p", "d", "q", "AIC", "BIC", "RMSE", "status"}

for code in datasets:
    df, y, dates, train_end_state = get_state_split(code,␣
 ↪all_state_test_horizon)

    y_train = y.iloc[:train_end_state]
    y_test = y.iloc[train_end_state:]
    test_dates = dates.iloc[train_end_state:]

    cache_path = model_cache_dir / f"arima_grid_results_{code}.csv"
    results_df = load_valid_cache(cache_path, arima_required_cols)

    if results_df.empty:
        print(f"Running ARIMA grid search for {code}")
        rows = [arima_grid_row(y_train, y_test, order) for order in␣
 ↪all_state_param_grid]
        results_df = pd.DataFrame(rows)
        results_df.to_csv(cache_path, index=False)

    valid_results = results_df.dropna(subset=["RMSE"]).sort_values("RMSE").
 ↪reset_index(drop=True)
    best_row = valid_results.iloc[0]
    best_order = (int(best_row["p"]), int(best_row["d"]), int(best_row["q"]))

    best_model = ARIMA(y_train, order=best_order).fit()

    # Forecasting through the end of the holdout plus the future forecast␣
 ↪horizon
    forecast_steps = len(y_test) + all_state_test_horizon

    future_dates = make_future_dates(dates, all_state_test_horizon)
    pred_dates = pd.concat(
        [test_dates.reset_index(drop=True), future_dates],
        ignore_index=True
    )
```

```
    best_pred = best_model.get_forecast(steps=forecast_steps).predicted_mean
    best_pred.index = np.arange(train_end_state, train_end_state +␣
↪forecast_steps)

    arima_all_results[code] = results_df
    arima_all_forecasts[code] = forecast_frame_extended(pred_dates, y_test,␣
↪best_pred)

    arima_all_best.append({
        "StateCode": code,
        "State": state_names[code],
        "Model": f"ARIMA{best_order}",
        "RMSE": calc_rmse(y_test, best_pred.iloc[:len(y_test)]),
        "MAE": calc_mae(y_test, best_pred.iloc[:len(y_test)]),
        "Selection": "Best holdout RMSE among ARIMA grid",
        "Plot_Label": f"ARIMA{best_order}"
    })
```

4.2.2 Best Models by Two-Year Holdout RMSE

Code cell purpose: What: This cell creates and displays the table of best ARIMA specifications and errors for each state. Why: The table documents the baseline parameter choices and gives a concise benchmark for later model comparisons.

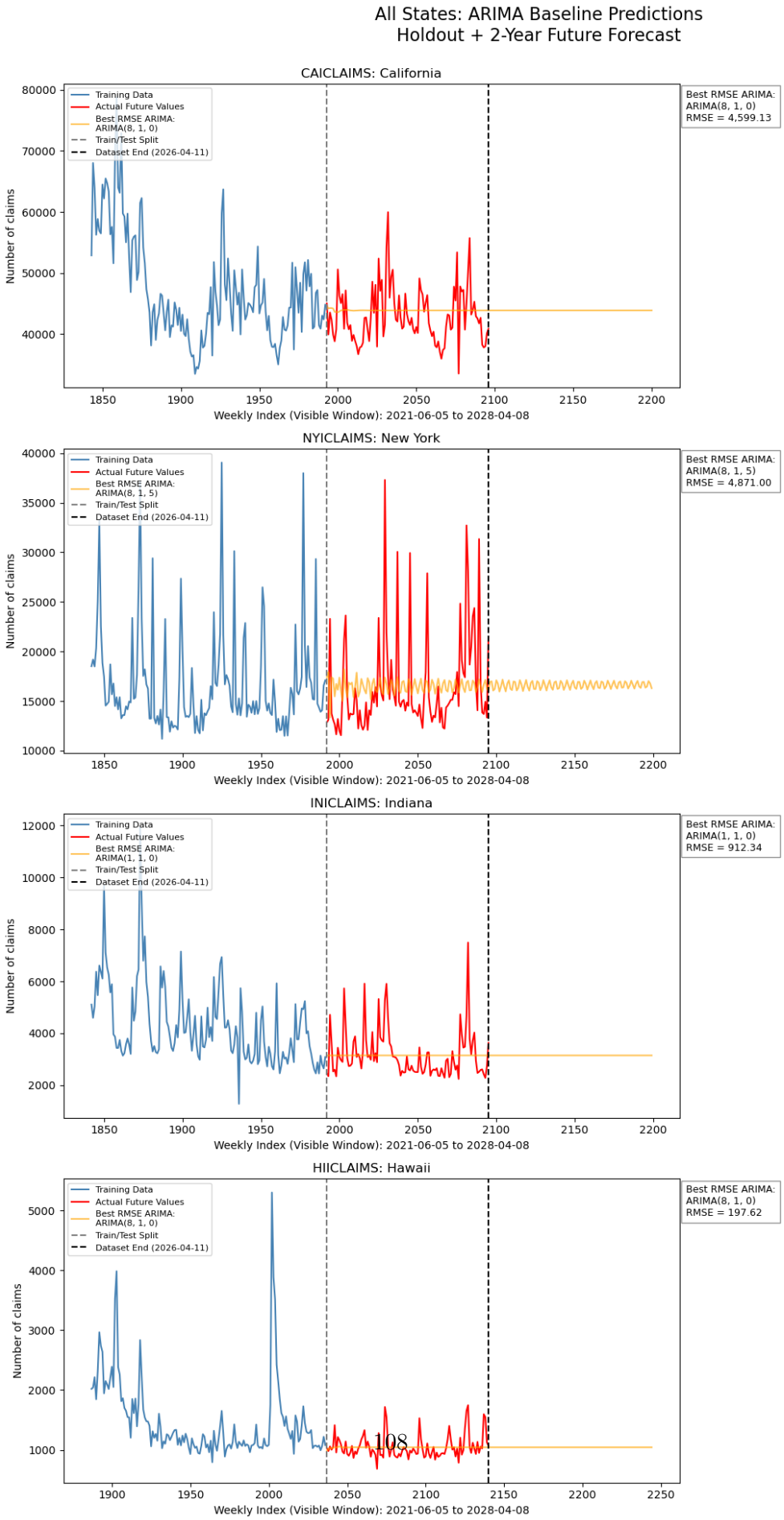
[69]:
```
arima_all_best_df = pd.DataFrame(arima_all_best)
display(arima_all_best_df.iloc[:, :-1].sort_values(["State", "RMSE"]).round(3))
```

```
   StateCode       State           Model     RMSE      MAE  \
0  CAICLAIMS  California  ARIMA(8, 1, 0) 4,599.13 3,754.13
3  HIICLAIMS      Hawaii  ARIMA(8, 1, 0)   197.62   136.54
2  INICLAIMS     Indiana  ARIMA(1, 1, 0)   912.34   641.19
1  NYICLAIMS    New York  ARIMA(8, 1, 5) 4,871.00 3,418.95

                           Selection
0  Best holdout RMSE among ARIMA grid
3  Best holdout RMSE among ARIMA grid
2  Best holdout RMSE among ARIMA grid
1  Best holdout RMSE among ARIMA grid
```

4.2.3 Plotted ARIMA Predictions

Code cell purpose: What: This cell plots the ARIMA baseline forecasts for all four states using the shared all-state plotting helper. Why: The figure makes it easier to see where the baseline model tracks or misses the actual holdout behavior across states.

[70]:
```
# Plot with helper function
plot_all_state_predictions(
    model_sources=[
        {
            "forecasts": arima_all_forecasts,
            "best_df": arima_all_best_df,
            "label_prefix": "Best RMSE ARIMA",
            "color": "orange"
        }
    ],
    main_title="All States: ARIMA Baseline Predictions\nHoldout + 2-Year Future␣
 ↪Forecast"
)
```


All States: ARIMA Baseline Predictions
Holdout + 2-Year Future Forecast
CAICLAIMS: California
Training Data
Actual Future Values
Best RMSE ARIMA: ARIMA(8, 1, 0)
Train/Test Split
Dataset End (2026-04-11)
Best RMSE ARIMA: ARIMA(8, 1, 0) RMSE = 4,599.13
Number of claims
Weekly Index (Visible Window): 2021-06-05 to 2028-04-08
NYICLAIMS: New York
Best RMSE ARIMA: ARIMA(8, 1, 5)
Best RMSE ARIMA: ARIMA(8, 1, 5) RMSE = 4,871.00
INICLAIMS: Indiana
Best RMSE ARIMA: ARIMA(1, 1, 0)
Best RMSE ARIMA: ARIMA(1, 1, 0) RMSE = 912.34
HIICLAIMS: Hawaii
Best RMSE ARIMA: ARIMA(8, 1, 0)
Best RMSE ARIMA: ARIMA(8, 1, 0) RMSE = 197.62

4.3 3. SARIMAX with Fourier Frequencies, Before/After COVID Dummy

SARIMAX adds interpretable exogenous variables to the ARIMA structure. Here, the exogenous variables are a linear trend, annual Fourier terms for weekly seasonality, and a COVID structural break indicator: `covid_after`. This means the baseline econometric model keeps the COVID observations but gives the model an explicit post-COVID level-shift indicator.

Code cell purpose: This cell defines a short SARIMAX fitting helper and applies the SARIMAX grid search to every state using the same split and COVID-break controls. This matters because SARIMAX is our interpretable baseline for handling the COVID structural break while staying comparable to ARIMA and LightGBM by RMSE/MAE.

[71]:
```
## Generic SARIMAX holdout-grid row
def sarimax_grid_row(y_train, y_test, X_train, X_test, order):
    p, d, q = map(int, order)
    try:
        model = SARIMAX(y_train, exog=X_train, order=(p, d, q),
                        enforce_stationarity=False,␣
 ↪enforce_invertibility=False).fit(disp=False)
        pred = model.get_forecast(steps=len(y_test), exog=X_test).predicted_mean
        pred.index = y_test.index
        return {"p": p, "d": d, "q": q, "AIC": model.aic, "BIC": model.bic,
                "RMSE": calc_rmse(y_test, pred), "status": "success"}
    except Exception:
        return {"p": p, "d": d, "q": q, "AIC": np.nan, "BIC": np.nan,
                "RMSE": np.nan, "status": "failed"}
```

4.3.1 Parameter Selection (Grid Search)

Code cell purpose: What: This cell runs the all-state SARIMAX model with an after-COVID dummy, selecting the best specification and generating holdout plus future forecasts for each state. Why: Adding an explicit post-COVID structural indicator tests whether a simple external control improves forecasts relative to the ARIMA baseline.

[72]:
```
## Run simple SARIMAX + after-COVID dummy for all states
sarimax_after_all_results = {}
sarimax_after_all_best = []
sarimax_after_all_forecasts = {}

sarimax_required_cols = {"p", "d", "q", "AIC", "BIC", "RMSE", "status"}

for code in datasets:
    df, y, dates, train_end_state = get_state_split(code,␣
 ↪all_state_test_horizon)

    y_train = y.iloc[:train_end_state]
    y_test = y.iloc[train_end_state:]
    test_dates = dates.iloc[train_end_state:]

    X = make_exog_after(dates)
    X_train = X.iloc[:train_end_state]
    X_test = X.iloc[train_end_state:][X_train.columns]


    # Extend dates and exog for future forecasting beyond the holdout period
    dates_extended = extend_dates(dates, all_state_test_horizon)

    X_extended = make_exog_after(dates_extended)
    X_train_extended = X_extended.iloc[:train_end_state]
    X_forecast_extended = X_extended.iloc[
        train_end_state : len(y) + all_state_test_horizon
    ][X_train_extended.columns]

    future_dates = make_future_dates(dates, all_state_test_horizon)
    pred_dates = pd.concat(
        [test_dates.reset_index(drop=True), future_dates],
        ignore_index=True
    )

    forecast_steps = len(y_test) + all_state_test_horizon


    cache_path = model_cache_dir / f"sarimax_after_grid_results_{code}.csv"
    results_df = load_valid_cache(cache_path, sarimax_required_cols)
```

```
    if results_df.empty:
        print(f"Running SARIMAX after-COVID grid search for {code}")
        rows = [sarimax_grid_row(y_train, y_test, X_train, X_test, order) for␣
↪order in all_state_param_grid]
        results_df = pd.DataFrame(rows)
        results_df.to_csv(cache_path, index=False)

    valid_results = results_df.dropna(subset=["RMSE"]).sort_values("RMSE").
↪reset_index(drop=True)
    best_row = valid_results.iloc[0]
    best_order = (int(best_row["p"]), int(best_row["d"]), int(best_row["q"]))

    best_model = SARIMAX(
        y_train,
        exog=X_train_extended,
        order=best_order,
        enforce_stationarity=False,
        enforce_invertibility=False
    ).fit(disp=False)


    # Forecasting through the end of the holdout plus the future forecast␣
↪horizon
    best_pred = best_model.get_forecast(
        steps=forecast_steps,
        exog=X_forecast_extended
    ).predicted_mean

    best_pred.index = np.arange(train_end_state, train_end_state +␣
↪forecast_steps)

    sarimax_after_all_results[code] = results_df
    sarimax_after_all_forecasts[code] = forecast_frame_extended(pred_dates,␣
↪y_test, best_pred)

    sarimax_after_all_best.append({
        "StateCode": code,
        "State": state_names[code],
        "Model": f"SARIMAX{best_order} + Fourier + after-COVID dummy",
        "RMSE": calc_rmse(y_test, best_pred.iloc[:len(y_test)]),
        "MAE": calc_mae(y_test, best_pred.iloc[:len(y_test)]),
        "Selection": "Best holdout RMSE among simple SARIMAX grid",
        "Plot_Label": f"SARIMAX{best_order} + Fourier\n+ after-COVID dummy"
    })
```

4.3.2 Best Models by Two-Year Holdout RMSE

Code cell purpose: What: This cell creates and displays the best all-state SARIMAX + after-COVID results table. Why: The table summarizes the selected orders and forecast errors so this model family can be directly compared with ARIMA and later specifications.

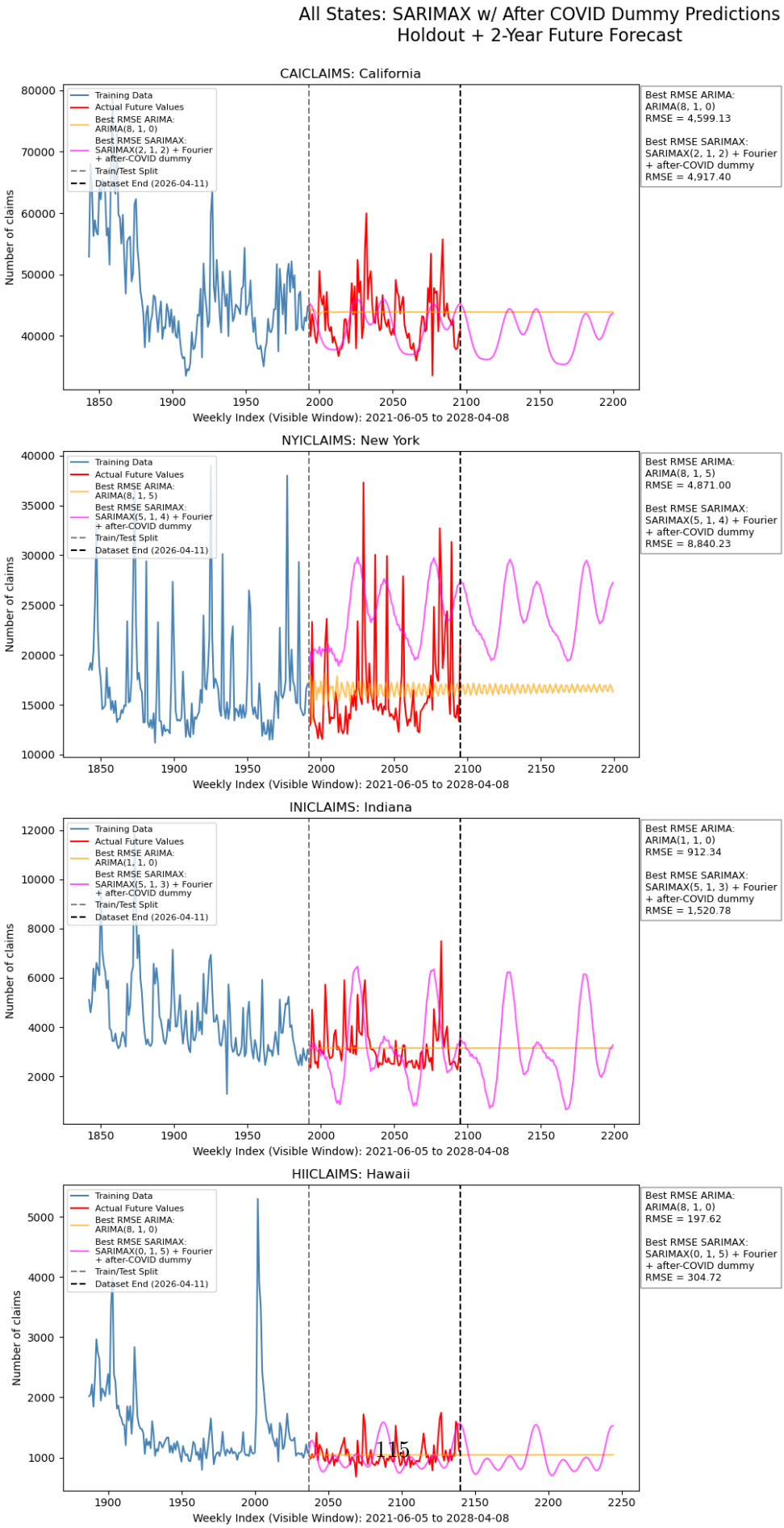
[73]:
```
sarimax_after_all_best_df = pd.DataFrame(sarimax_after_all_best)
display(sarimax_after_all_best_df.iloc[:, :-1].sort_values(["State", "RMSE"]).
  ↪round(3))
```

```
   StateCode       State                                              Model  \
0  CAICLAIMS  California  SARIMAX(2, 1, 2) + Fourier + after-COVID dummy
3  HIICLAIMS      Hawaii  SARIMAX(0, 1, 5) + Fourier + after-COVID dummy
2  INICLAIMS     Indiana  SARIMAX(5, 1, 3) + Fourier + after-COVID dummy
1  NYICLAIMS    New York  SARIMAX(5, 1, 4) + Fourier + after-COVID dummy

       RMSE      MAE                                      Selection
0 4,917.40 3,708.22  Best holdout RMSE among simple SARIMAX grid
3   304.72   232.38  Best holdout RMSE among simple SARIMAX grid
2 1,520.78 1,192.38  Best holdout RMSE among simple SARIMAX grid
1 8,840.23 8,234.57  Best holdout RMSE among simple SARIMAX grid
```

4.3.3 Plotted SARIMAX Predictions

Code cell purpose: What: This cell plots all-state forecasts from ARIMA and SARIMAX with the after-COVID dummy. Why: Putting both model families on the same figure shows whether the added COVID structure changes forecast behavior in a meaningful way.

[74]:
```
# Plot with helper function
plot_all_state_predictions(
    model_sources=[
        {
            "forecasts": arima_all_forecasts,
            "best_df": arima_all_best_df,
            "label_prefix": "Best RMSE ARIMA",
            "color": "orange"
        },
        {
            "forecasts": sarimax_after_all_forecasts,
            "best_df": sarimax_after_all_best_df,
            "label_prefix": "Best RMSE SARIMAX",
            "color": "magenta"
        }
    ],
    main_title="All States: SARIMAX w/ After COVID Dummy Predictions\nHoldout + 2-Year Future Forecast"
)
```


All States: SARIMAX w/ After COVID Dummy Predictions
Holdout + 2-Year Future Forecast
CAICLAIMS: California
Training Data
Actual Future Values
Best RMSE ARIMA: ARIMA(8, 1, 0)
Best RMSE SARIMAX: SARIMAX(2, 1, 2) + Fourier + after-COVID dummy
Train/Test Split
Dataset End (2026-04-11)
Best RMSE ARIMA: ARIMA(8, 1, 0) RMSE = 4,599.13
Best RMSE SARIMAX: SARIMAX(2, 1, 2) + Fourier + after-COVID dummy RMSE = 4,917.40
Number of claims
Weekly Index (Visible Window): 2021-06-05 to 2028-04-08
NYICLAIMS: New York
Best RMSE ARIMA: ARIMA(8, 1, 5)
Best RMSE SARIMAX: SARIMAX(5, 1, 4) + Fourier + after-COVID dummy
Best RMSE ARIMA: ARIMA(8, 1, 5) RMSE = 4,871.00
Best RMSE SARIMAX: SARIMAX(5, 1, 4) + Fourier + after-COVID dummy RMSE = 8,840.23
INICLAIMS: Indiana
Best RMSE ARIMA: ARIMA(1, 1, 0)
Best RMSE SARIMAX: SARIMAX(5, 1, 3) + Fourier + after-COVID dummy
Best RMSE ARIMA: ARIMA(1, 1, 0) RMSE = 912.34
Best RMSE SARIMAX: SARIMAX(5, 1, 3) + Fourier + after-COVID dummy RMSE = 1,520.78
HIICLAIMS: Hawaii
Best RMSE ARIMA: ARIMA(8, 1, 0)
Best RMSE SARIMAX: SARIMAX(0, 1, 5) + Fourier + after-COVID dummy
Best RMSE ARIMA: ARIMA(8, 1, 0) RMSE = 197.62
Best RMSE SARIMAX: SARIMAX(0, 1, 5) + Fourier + after-COVID dummy RMSE = 304.72

4.4 4. Rolling-CV SARIMAX with Fourier Frequencies, During COVID Dummy

4.4.1 Parameter Selection (Grid Search)

Code cell purpose: What: This cell runs rolling-CV SARIMAX models with a during-COVID dummy for all states, selects the best rolling-CV specification, and generates holdout plus future forecasts. Why: Rolling validation better reflects the forecasting task and the during-COVID indicator directly controls for the extraordinary pandemic shock period.

[75]:
```
sarimax_during_cv_all_results = {}
sarimax_during_cv_all_best = []
sarimax_during_cv_all_forecasts = {}

cv_required_cols = {"p", "d", "q", "CV_RMSE", "CV_AIC", "CV_BIC", "n_success",␣
 ↪"n_splits"}

cv_initial_train_size = 1500
cv_horizon = 52
cv_step = 26

for code in datasets:
    df, y, dates, train_end_state = get_state_split(code,␣
 ↪all_state_test_horizon)

    y_train = y.iloc[:train_end_state]
    y_test = y.iloc[train_end_state:]
    test_dates = dates.iloc[train_end_state:]

    X = make_exog_during(dates)
    X_train = X.iloc[:train_end_state]
    X_test = X.iloc[train_end_state:][X_train.columns]


    # Extend dates and exog for future forecasting beyond the holdout period
    dates_extended = extend_dates(dates, all_state_test_horizon)

    X_extended = make_exog_during(dates_extended)
    X_train_extended = X_extended.iloc[:train_end_state]
    X_forecast_extended = X_extended.iloc[
        train_end_state : len(y) + all_state_test_horizon
    ][X_train_extended.columns]

    future_dates = make_future_dates(dates, all_state_test_horizon)
    pred_dates = pd.concat(
        [test_dates.reset_index(drop=True), future_dates],
        ignore_index=True
    )
```

```
    forecast_steps = len(y_test) + all_state_test_horizon


    cache_path = model_cache_dir / f"sarimax_during_cv_results_{code}.csv"
    results_df = load_valid_cache(cache_path, cv_required_cols)

    if results_df.empty:
        print(f"Running rolling-CV SARIMAX during-COVID for {code}")
        rows = Parallel(n_jobs=-1, verbose=10)(
            delayed(rolling_cv_sarimax)(y_train, X_train, order,␣
↪cv_initial_train_size, cv_horizon, cv_step)
            for order in all_state_param_grid
        )
        results_df = pd.DataFrame(rows)
        results_df.to_csv(cache_path, index=False)

    valid_results = results_df.dropna(subset=["CV_RMSE"]).
↪sort_values("CV_RMSE").reset_index(drop=True)
    best_row = valid_results.iloc[0]
    best_order = (int(best_row["p"]), int(best_row["d"]), int(best_row["q"]))

    best_model = SARIMAX(
        y_train,
        exog=X_train_extended,
        order=best_order,
        enforce_stationarity=False,
        enforce_invertibility=False
    ).fit(disp=False)


    # Forecasting through the end of the holdout plus the future forecast␣
↪horizon
    best_pred = best_model.get_forecast(
        steps=forecast_steps,
        exog=X_forecast_extended
    ).predicted_mean

    best_pred.index = np.arange(train_end_state, train_end_state +␣
↪forecast_steps)

    sarimax_during_cv_all_results[code] = results_df
    sarimax_during_cv_all_forecasts[code] = forecast_frame_extended(pred_dates,␣
↪y_test, best_pred)

    sarimax_during_cv_all_best.append({
        "StateCode": code,
```

```
        "State": state_names[code],
        "Model": f"SARIMAX{best_order} + Fourier + during-COVID dummy",
        "RMSE": calc_rmse(y_test, best_pred.iloc[:len(y_test)]),
        "MAE": calc_mae(y_test, best_pred.iloc[:len(y_test)]),
        "Selection": "Best rolling-CV RMSE among SARIMAX grid",
        "Plot_Label": f"SARIMAX{best_order} + Fourier\n+ during-COVID dummy"
    })
```

4.4.2 Best Models by Two-Year Holdout RMSE

Code cell purpose: What: This cell creates and displays the best all-state rolling-CV SARIMAX results table. Why: The table records the selected rolling-validation specifications and errors before those forecasts are compared against other model families.

[76]:
```
sarimax_during_cv_all_best_df = pd.DataFrame(sarimax_during_cv_all_best)
display(sarimax_during_cv_all_best_df.iloc[:, :-1].sort_values(["State",␣
  ↪"RMSE"]).round(3))
```

```
   StateCode       State                                              Model  \
0  CAICLAIMS  California  SARIMAX(0, 1, 1) + Fourier + during-COVID dummy
3  HIICLAIMS      Hawaii  SARIMAX(5, 1, 3) + Fourier + during-COVID dummy
2  INICLAIMS     Indiana  SARIMAX(2, 1, 1) + Fourier + during-COVID dummy
1  NYICLAIMS    New York  SARIMAX(4, 1, 3) + Fourier + during-COVID dummy

      RMSE      MAE                                Selection
0 5,712.83 4,378.27  Best rolling-CV RMSE among SARIMAX grid
3   736.02   648.89  Best rolling-CV RMSE among SARIMAX grid
2 1,537.35 1,188.45  Best rolling-CV RMSE among SARIMAX grid
1 8,986.61 8,328.73  Best rolling-CV RMSE among SARIMAX grid
```

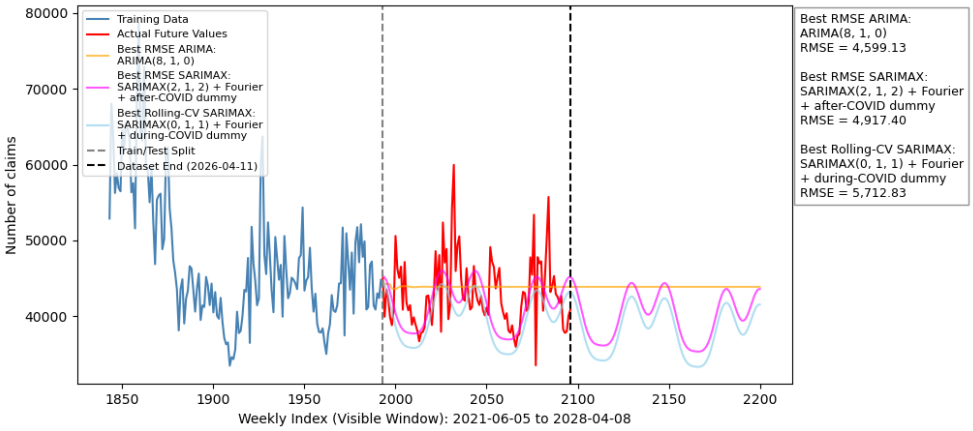
4.4.3 Plotted SARIMAX Predictions

Code cell purpose: What: This cell plots ARIMA, SARIMAX with an after-COVID dummy, and rolling-CV SARIMAX with a during-COVID dummy for all states. Why: The combined plot shows how each added modeling choice changes the forecast path relative to both the baseline and the actual holdout data.

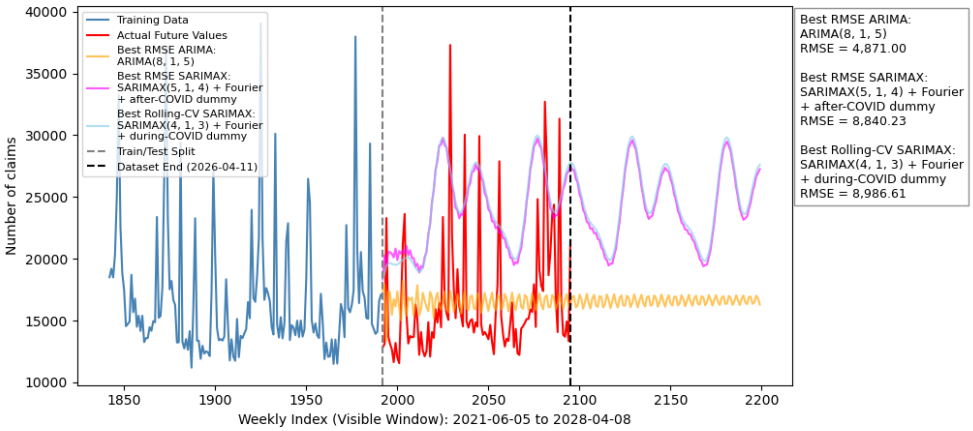
[77]:
```
# Plot with helper function
plot_all_state_predictions(
    model_sources=[
        {
            "forecasts": arima_all_forecasts,
            "best_df": arima_all_best_df,
            "label_prefix": "Best RMSE ARIMA",
            "color": "orange"
        },
        {
            "forecasts": sarimax_after_all_forecasts,
            "best_df": sarimax_after_all_best_df,
            "label_prefix": "Best RMSE SARIMAX",
            "color": "magenta"
        },
        {
            "forecasts": sarimax_during_cv_all_forecasts,
            "best_df": sarimax_during_cv_all_best_df,
            "label_prefix": "Best Rolling-CV SARIMAX",
            "color": "skyblue"
        }
    ],
    main_title="All States: Rolling-CV SARIMAX w/ During COVID Dummy␣
 ↪Predictions\nHoldout + 2-Year Future Forecast"
)
```

All States: Rolling-CV SARIMAX w/ During COVID Dummy Predictions
Holdout + 2-Year Future Forecast

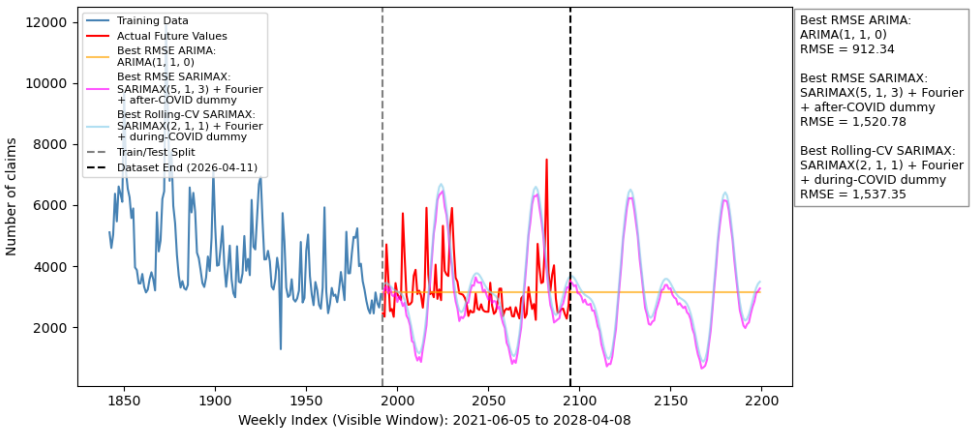
CAICLAIMS: California



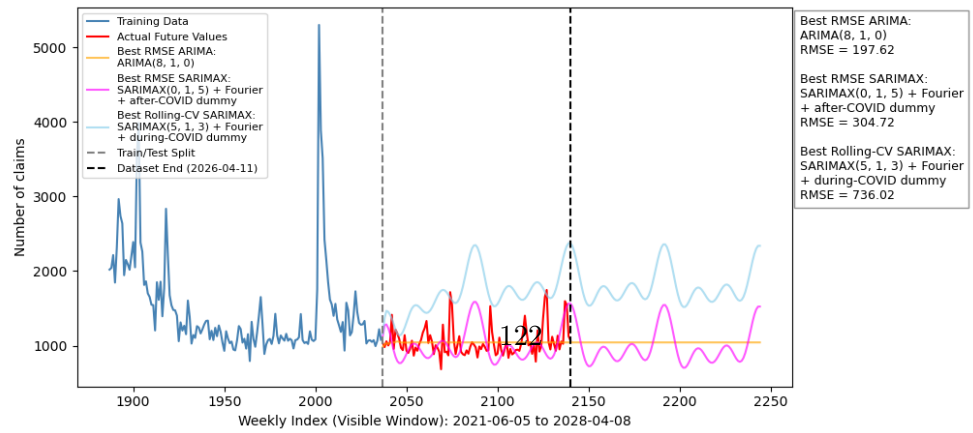
NYICLAIMS: New York



INICLAIMS: Indiana



HIICLAIMS: Hawaii



4.5 5. Rolling-CV SARIMAX with Fourier Frequencies, During COVID Dummy, Google Trends Data

Unlike previous models, we do not make future predictions past the holdout set since that would depend on future Google Trends data. If we were to forecast past the end of the dataset, we would have to forecast the Google Trends time series as well.

4.5.1 Pull and Process Google Trends Data

Code cell purpose: What: This cell pulls or loads cached Google Trends data for all four states using the same unemployment-related keyword set. Why: State-specific Trends features create a scalable external-data augmentation that can be used consistently in the all-state SARIMAX models.

[78]:
```
trend_geos = {
    "CAICLAIMS": "US-CA",
    "NYICLAIMS": "US-NY",
    "INICLAIMS": "US-IN",
    "HIICLAIMS": "US-HI"
}

trend_state_abbrev = {
    "CAICLAIMS": "CA",
    "NYICLAIMS": "NY",
    "INICLAIMS": "IN",
    "HIICLAIMS": "HI"
}

## Already defined in CAICLAIMS Model Testing section
# trend_keywords = [
#     "unemployment",
#     "unemployment benefits",
#     "file for unemployment",
#     "unemployment office",
#     "apply for unemployment",
# ]

## Pull Google Trends data for all states
pytrends = TrendReq(hl="en-US", tz=360)

for code, geo in trend_geos.items():
    abbrev = trend_state_abbrev[code]
    trends_path = data_dir / f"google_trends_{abbrev}.csv"

    if trends_path.exists():
        gt_pull = pd.read_csv(trends_path, index_col=0, parse_dates=True)
        print(f"Loaded cached Google Trends data for {code}")
```

```
    else:
        attempts = 0
        success = False

        while not success and attempts < 5:
            try:
                pytrends.build_payload(
                    kw_list=trend_keywords,
                    timeframe="2004-01-01 2026-04-01",
                    geo=geo
                )

                gt_pull = pytrends.interest_over_time()
                gt_pull = gt_pull.drop(columns=["isPartial"], errors="ignore")
                gt_pull.index = pd.to_datetime(gt_pull.index)
                gt_pull = gt_pull.resample("W-SAT").mean()

                gt_pull.to_csv(trends_path)

                print(f"Saved Google Trends data for {code}")
                success = True

                time.sleep(30)

            except TooManyRequestsError:
                attempts += 1
                wait_time = 60 * attempts
                print(f"Too many requests for {code}. Waiting {wait_time}␣
↪seconds before retrying...")
                time.sleep(wait_time)

        if not success:
            print(f"Skipping {code}: Google Trends request failed after 5␣
↪attempts.")
```

```
Loaded cached Google Trends data for CAICLAIMS
Loaded cached Google Trends data for NYICLAIMS
Loaded cached Google Trends data for INICLAIMS
Loaded cached Google Trends data for HIICLAIMS
```

Code cell purpose: What: This cell defines helper functions for combining during-COVID SARIMAX features with Google Trends features and for creating state-specific Google Trends modeling samples. Why: These helpers make the Trends-augmented SARIMAX workflow reusable across all four states without duplicating data-preparation logic.

[79]:
```
def make_exog_during_gt(dates, gt_aligned):
    X_base = make_exog_during(dates)
    X_gt = rename_google_trends_features(gt_aligned)
```

```
    return pd.concat([X_base, X_gt.reset_index(drop=True)], axis=1)

## Google Trends state split helpers
def get_google_trends_weekly(code):
    abbrev = trend_state_abbrev[code]
    gt_raw = read_google_trends_state(abbrev)
    gt_clean = clean_google_trends_cols(gt_raw)
    return convert_google_dates_to_claims_week(gt_clean)

def get_gt_start_idx(dates, gt_weekly):
    start_date = gt_weekly["Date"].min()
    return int(np.where(pd.Series(dates) >= start_date)[0][0])

def get_state_split_gt(code):
    df = claims_dfs[code].sort_values("Date").reset_index(drop=True)
    gt_weekly = get_google_trends_weekly(code)
    start_idx = get_gt_start_idx(df["Date"], gt_weekly)
    return df.iloc[start_idx:].reset_index(drop=True), gt_weekly
```

4.5.2 Parameter Selection (Grid Search)

Code cell purpose: What: This cell runs rolling-CV SARIMAX + Google Trends models for all states and stores the selected specifications, errors, and forecasts. Why: This tests whether adding search-interest data improves out-of-sample forecasting beyond the claims-only and COVID-control specifications.

[80]:
```
## Run rolling-CV SARIMAX + Google Trends for all states
sarimax_gt_cv_all_results = {}
sarimax_gt_cv_all_best = []
sarimax_gt_cv_all_forecasts = {}

gt_cv_initial_train_size = 600
gt_cv_horizon = 52
gt_cv_step = 26

for code in datasets:
    df_gt, gt_weekly = get_state_split_gt(code)

    y = df_gt["Claims"].reset_index(drop=True)
    dates = df_gt["Date"].reset_index(drop=True)

    train_end_state = len(y) - all_state_test_horizon

    y_train = y.iloc[:train_end_state]
    y_test = y.iloc[train_end_state:]
    test_dates = dates.iloc[train_end_state:]

    gt_aligned = align_google_trends(dates, gt_weekly)
    X = make_exog_during_gt(dates, gt_aligned)

    X_train = X.iloc[:train_end_state]
    X_test = X.iloc[train_end_state:][X_train.columns]

    cache_path = model_cache_dir / f"sarimax_gt_cv_results_{code}.csv"
    results_df = load_valid_cache(cache_path, cv_required_cols)

    if results_df.empty:
        print(f"Running rolling-CV SARIMAX + Google Trends for {code}")
        rows = Parallel(n_jobs=-1, verbose=10)(
            delayed(rolling_cv_sarimax)(y_train, X_train, order,␣
↪gt_cv_initial_train_size, gt_cv_horizon, gt_cv_step)
            for order in all_state_param_grid
        )
        results_df = pd.DataFrame(rows)
        results_df.to_csv(cache_path, index=False)
```

```
    valid_results = results_df.dropna(subset=["CV_RMSE"]).
↪sort_values("CV_RMSE").reset_index(drop=True)
    best_row = valid_results.iloc[0]
    best_order = (int(best_row["p"]), int(best_row["d"]), int(best_row["q"]))

    best_model = SARIMAX(y_train, exog=X_train, order=best_order,
                         enforce_stationarity=False,␣
↪enforce_invertibility=False).fit(disp=False)
    best_pred = best_model.get_forecast(steps=len(y_test), exog=X_test).
↪predicted_mean
    best_pred.index = y_test.index

    sarimax_gt_cv_all_results[code] = results_df
    sarimax_gt_cv_all_forecasts[code] = forecast_frame(test_dates, y_test,␣
↪best_pred)

    sarimax_gt_cv_all_best.append({
        "StateCode": code,
        "State": state_names[code],
        "Model": f"SARIMAX{best_order} + Fourier + during-COVID dummy + Google␣
↪Trends",
        "RMSE": calc_rmse(y_test, best_pred),
        "MAE": calc_mae(y_test, best_pred),
        "Selection": "Best rolling-CV RMSE among SARIMAX + Google Trends grid;␣
↪2004+ sample",
        "Plot_Label": f"SARIMAX{best_order} + Fourier\n+ during-COVID dummy\n+␣
↪Google Trends"
    })
```

4.5.3 Best Models by Two-Year Holdout RMSE

Code cell purpose: What: This cell creates and displays the best all-state SARIMAX + Google Trends results table. Why: The table records the selected Trends-augmented model specifications and error metrics needed for the final model comparison.

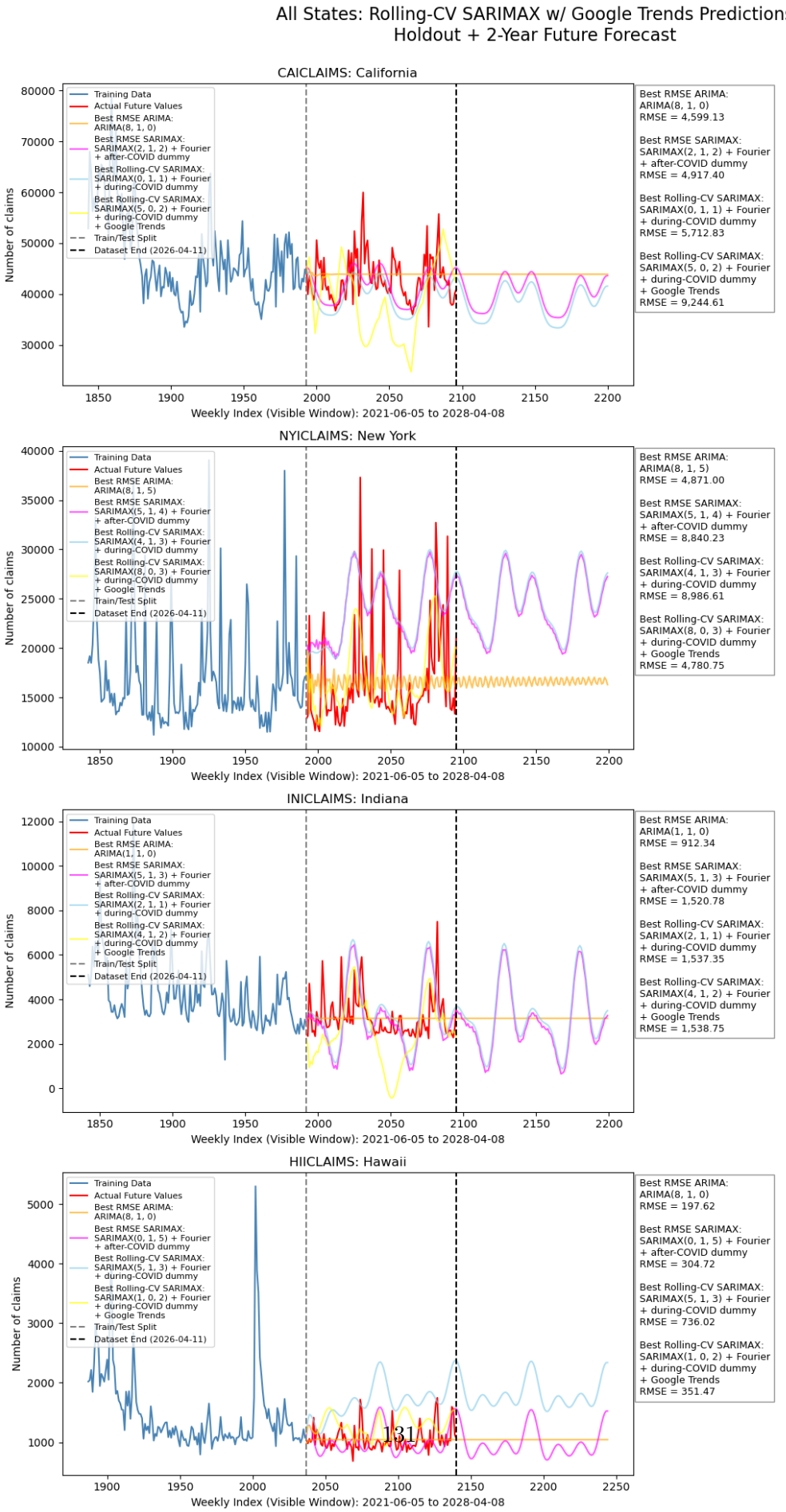
[81]:
```
sarimax_gt_cv_all_best_df = pd.DataFrame(sarimax_gt_cv_all_best)
display(sarimax_gt_cv_all_best_df.iloc[:, :-1].sort_values(["State", "RMSE"]).
  ↪round(3))
```

```
   StateCode       State                                                Model  \
0  CAICLAIMS  California  SARIMAX(5, 0, 2) + Fourier + during-COVID dumm…
3  HIICLAIMS      Hawaii  SARIMAX(1, 0, 2) + Fourier + during-COVID dumm…
2  INICLAIMS     Indiana  SARIMAX(4, 1, 2) + Fourier + during-COVID dumm…
1  NYICLAIMS    New York  SARIMAX(8, 0, 3) + Fourier + during-COVID dumm…

       RMSE      MAE                                          Selection
0 9,244.61 7,537.05  Best rolling-CV RMSE among SARIMAX + Google Tr…
3   351.47   298.14  Best rolling-CV RMSE among SARIMAX + Google Tr…
2 1,538.75 1,185.39  Best rolling-CV RMSE among SARIMAX + Google Tr…
1 4,780.75 3,320.83  Best rolling-CV RMSE among SARIMAX + Google Tr…
```

4.5.4 Plotted SARIMAX Predictions

Code cell purpose: What: This cell plots the ARIMA, SARIMAX, rolling-CV SARIMAX, and rolling-CV SARIMAX + Google Trends forecasts for all states. Why: The plot allows the Google Trends augmentation to be evaluated visually against the earlier econometric forecasting models.

[82]:
```
# Plot with helper function
plot_all_state_predictions(
    model_sources=[
        {
            "forecasts": arima_all_forecasts,
            "best_df": arima_all_best_df,
            "label_prefix": "Best RMSE ARIMA",
            "color": "orange"
        },
        {
            "forecasts": sarimax_after_all_forecasts,
            "best_df": sarimax_after_all_best_df,
            "label_prefix": "Best RMSE SARIMAX",
            "color": "magenta"
        },
        {
            "forecasts": sarimax_during_cv_all_forecasts,
            "best_df": sarimax_during_cv_all_best_df,
            "label_prefix": "Best Rolling-CV SARIMAX",
            "color": "skyblue"
        },
        {
            "forecasts": sarimax_gt_cv_all_forecasts,
            "best_df": sarimax_gt_cv_all_best_df,
            "label_prefix": "Best Rolling-CV SARIMAX",
            "color": "yellow"
        }
    ],
    main_title="All States: Rolling-CV SARIMAX w/ Google Trends␣
 ↪Predictions\nHoldout + 2-Year Future Forecast"
)
```


All States: Rolling-CV SARIMAX w/ Google Trends Predictions
Holdout + 2-Year Future Forecast
CAICLAIMS: California
Training Data
Actual Future Values
Best RMSE ARIMA: ARIMA(8, 1, 0)
Best RMSE SARIMAX: SARIMAX(2, 1, 2) + Fourier + after-COVID dummy
Best Rolling-CV SARIMAX: SARIMAX(0, 1, 1) + Fourier + during-COVID dummy
Best Rolling-CV SARIMAX: SARIMAX(5, 0, 2) + Fourier + during-COVID dummy + Google Trends
Train/Test Split
Dataset End (2026-04-11)
Best RMSE ARIMA: ARIMA(8, 1, 0) RMSE = 4,599.13
Best RMSE SARIMAX: SARIMAX(2, 1, 2) + Fourier + after-COVID dummy RMSE = 4,917.40
Best Rolling-CV SARIMAX: SARIMAX(0, 1, 1) + Fourier + during-COVID dummy RMSE = 5,712.83
Best Rolling-CV SARIMAX: SARIMAX(5, 0, 2) + Fourier + during-COVID dummy + Google Trends RMSE = 9,244.61
Number of claims
Weekly Index (Visible Window): 2021-06-05 to 2028-04-08
NYICLAIMS: New York
Best RMSE ARIMA: ARIMA(8, 1, 5)
Best RMSE SARIMAX: SARIMAX(5, 1, 4) + Fourier + after-COVID dummy
Best Rolling-CV SARIMAX: SARIMAX(4, 1, 3) + Fourier + during-COVID dummy
Best Rolling-CV SARIMAX: SARIMAX(8, 0, 3) + Fourier + during-COVID dummy + Google Trends
Best RMSE ARIMA: ARIMA(8, 1, 5) RMSE = 4,871.00
Best RMSE SARIMAX: SARIMAX(5, 1, 4) + Fourier + after-COVID dummy RMSE = 8,840.23
Best Rolling-CV SARIMAX: SARIMAX(4, 1, 3) + Fourier + during-COVID dummy RMSE = 8,986.61
Best Rolling-CV SARIMAX: SARIMAX(8, 0, 3) + Fourier + during-COVID dummy + Google Trends RMSE = 4,780.75
INICLAIMS: Indiana
Best RMSE ARIMA: ARIMA(1, 1, 0)
Best RMSE SARIMAX: SARIMAX(5, 1, 3) + Fourier + after-COVID dummy
Best Rolling-CV SARIMAX: SARIMAX(2, 1, 1) + Fourier + during-COVID dummy
Best Rolling-CV SARIMAX: SARIMAX(4, 1, 2) + Fourier + during-COVID dummy + Google Trends
Best RMSE ARIMA: ARIMA(1, 1, 0) RMSE = 912.34
Best RMSE SARIMAX: SARIMAX(5, 1, 3) + Fourier + after-COVID dummy RMSE = 1,520.78
Best Rolling-CV SARIMAX: SARIMAX(2, 1, 1) + Fourier + during-COVID dummy RMSE = 1,537.35
Best Rolling-CV SARIMAX: SARIMAX(4, 1, 2) + Fourier + during-COVID dummy + Google Trends RMSE = 1,538.75
HIICLAIMS: Hawaii
Best RMSE ARIMA: ARIMA(8, 1, 0)
Best RMSE SARIMAX: SARIMAX(0, 1, 5) + Fourier + after-COVID dummy
Best Rolling-CV SARIMAX: SARIMAX(5, 1, 3) + Fourier + during-COVID dummy
Best Rolling-CV SARIMAX: SARIMAX(1, 0, 2) + Fourier + during-COVID dummy + Google Trends
Best RMSE ARIMA: ARIMA(8, 1, 0) RMSE = 197.62
Best RMSE SARIMAX: SARIMAX(0, 1, 5) + Fourier + after-COVID dummy RMSE = 304.72
Best Rolling-CV SARIMAX: SARIMAX(5, 1, 3) + Fourier + during-COVID dummy RMSE = 736.02
Best Rolling-CV SARIMAX: SARIMAX(1, 0, 2) + Fourier + during-COVID dummy + Google Trends RMSE = 351.47

4.6 6. LightGBM ML Comparison Model

In this case, with the use of of LightGBM, we enter the realm of a supervised-learning problem, where we understand what we are attemtping to predict and have actual data with which we can compare to our predictions (so an x and a y, essentially). Here, the predictors include lagged claims, rolling averages, week-of-year seasonality, trend, and the same COVID indicators used previously in SARIMAX. This model is forecast recursively, which means that, after predicting the first holdout week, that prediction is added to the history and used to build lag features for the next week. In doing so, we avoid giving the ML model actual future holdout claims as lag inputs, saving a lot of hassle/potential errors.

Code cell purpose: What: This cell imports LightGBM and installs it only if the current environment does not already have it. Why: LightGBM is the machine-learning comparison model, so this step ensures the notebook can run in a fresh environment without manual package setup.

[83]:
```
## Importing LightGBM, installing if needed
try:
    from lightgbm import LGBMRegressor
except ImportError:
    import sys
    import subprocess
    subprocess.check_call([sys.executable, "-m", "pip", "install", "lightgbm"])
    from lightgbm import LGBMRegressor
```

4.6.1 Helper Functions and Setup

Code cell purpose: Cell sets up a couple helpers for LightGBM usage, as aforementioned for calendar features, lag features, rolling averages, cached-parameter cleaning, and recursive forecasting. This matters because machine learning needs a supervised feature table while still respecting the time order of the forecasting problem. Without having the clear setup, our model will be quite poor.

[84]:
```
## Lags and rolling windows used for supervised ML features
lgbm_lags = [1, 2, 4, 13, 26, 52]
lgbm_windows = [4, 13, 26, 52]

## Adds trend, annual seasonality, and COVID indicators
def add_ml_date_features(frame):
    frame["week"] = frame["Date"].dt.isocalendar().week.astype(int)
    frame["trend"] = np.arange(len(frame))
    frame["week_sin"] = np.sin(2 * np.pi * frame["week"] / 52)
    frame["week_cos"] = np.cos(2 * np.pi * frame["week"] / 52)
    frame["covid_during"] = frame["Date"].between(covid_break_start,␣
 ↪covid_break_end).astype(int)
    frame["covid_after"] = (frame["Date"] > covid_break_end).astype(int)
    return frame

## Adds lagged claims values
def add_lag_features(frame, lags):
    for lag in lags:
        frame[f"lag_{lag}"] = frame["Claims"].shift(lag)
    return frame

## Adds rolling means using only past claims
def add_roll_features(frame, windows):
    for window in windows:
        frame[f"roll_mean_{window}"] = frame["Claims"].shift(1).rolling(window).
 ↪mean()
    return frame

## Builds the ML training table
def make_lgbm_frame(df, lags, windows):
    frame = df[["Date", "Claims"]].copy()
    frame = add_ml_date_features(frame)
    frame = add_lag_features(frame, lags)
    frame = add_roll_features(frame, windows)
    return frame.dropna().reset_index(drop=True)

## Returns model feature columns
def ml_feature_columns(frame):
    return [col for col in frame.columns if col not in ["Date", "Claims"]]
```

```
## Cleans cached LightGBM parameters so integer parameters remain integers
def clean_lgbm_params(row):
    params = row.drop("Valid_RMSE").to_dict()
    for key in ["n_estimators", "num_leaves", "min_child_samples"]:
        params[key] = int(params[key])
    params["learning_rate"] = float(params["learning_rate"])
    return params

## Calendar values for one recursive prediction row
def ml_calendar_row(date, t):
    week = int(pd.Timestamp(date).isocalendar().week)
    row = {"week": week, "trend": t}
    row["week_sin"] = np.sin(2 * np.pi * week / 52)
    row["week_cos"] = np.cos(2 * np.pi * week / 52)
    row["covid_during"] = int(covid_break_start <= pd.Timestamp(date) <=␣
 ↪covid_break_end)
    row["covid_after"] = int(pd.Timestamp(date) > covid_break_end)
    return row

## Builds one recursive forecast row from the available history
def history_feature_row(history, date, t, lags, windows):
    row = ml_calendar_row(date, t)
    row.update({f"lag_{lag}": history[-lag] for lag in lags})
    row.update({f"roll_mean_{window}": np.mean(history[-window:]) for window in␣
 ↪windows})
    return pd.DataFrame([row])

## Forecasts the test period without using actual test claims as lag inputs
def recursive_lgbm_forecast(model, history, dates, start_t, features):
    preds = []
    hist = list(history)
    for i, date in enumerate(dates):
        row = history_feature_row(hist, date, start_t + i, lgbm_lags,␣
 ↪lgbm_windows)[features]
        pred = max(0, float(model.predict(row)[0]))
        preds.append(pred)
        hist.append(pred)
    return pd.Series(preds)
```

4.6.2 Hyperparameter Selection

Code cell purpose: Cell below sets up the compact LightGBM tuning grid, also defines a helper which evaluates one candidate model using a time-ordered validation split within the training period. Matters because the ML model should tune the hyperparameters w/o looking at the final two-year holdout window used for the ARIMA/SARIMAX comparison… if not, everything falters.

[85]:
```
## Compact LightGBM candidate grid
lgbm_param_grid = [
    {"n_estimators": 250, "learning_rate": 0.03, "num_leaves": 15,␣
 ↪"min_child_samples": 20},
    {"n_estimators": 350, "learning_rate": 0.03, "num_leaves": 31,␣
 ↪"min_child_samples": 20},
    {"n_estimators": 250, "learning_rate": 0.05, "num_leaves": 15,␣
 ↪"min_child_samples": 30},
    {"n_estimators": 350, "learning_rate": 0.05, "num_leaves": 31,␣
 ↪"min_child_samples": 30}
]

## Fits one LightGBM candidate and returns validation RMSE
def lgbm_valid_row(params, X_fit, y_fit, X_valid, y_valid):
    model = LGBMRegressor(objective="regression", random_state=148,␣
 ↪verbosity=-1, **params)
    model.fit(X_fit, y_fit)
    pred = model.predict(X_valid)
    return {**params, "Valid_RMSE": calc_rmse(y_valid, pred)}
```

4.6.3 Fitting the Models for All States

Code cell purpose: This cell fits LightGBM for every state, caches validation-grid results, produces recursive holdout forecasts, and records RMSE/MAE for comparison. This matters as it means we directly test ML benchmark while keeping evaluation window identical to prior ARIMA and SARIMAX sections.

[86]:
```
## Storage objects for all-state LightGBM results
lgbm_all_results = {}
lgbm_all_best = []
lgbm_all_forecasts = {}
lgbm_required_cols = {"n_estimators", "learning_rate", "num_leaves",␣
↪"min_child_samples", "Valid_RMSE"}

## Looping through all states
for code in datasets:
    df, y, dates, train_end_state = get_state_split(code,␣
↪all_state_test_horizon)
    test_start_date = dates.iloc[train_end_state]
    y_test = y.iloc[train_end_state:].reset_index(drop=True)
    test_dates = dates.iloc[train_end_state:].reset_index(drop=True)
    ml_frame = make_lgbm_frame(df, lgbm_lags, lgbm_windows)
    train_frame = ml_frame.loc[ml_frame["Date"] < test_start_date].
↪reset_index(drop=True)
    features = ml_feature_columns(train_frame)

    valid_size = min(52, max(26, len(train_frame) // 10))
    fit_frame = train_frame.iloc[:-valid_size].reset_index(drop=True)
    valid_frame = train_frame.iloc[-valid_size:].reset_index(drop=True)
    cache_path = model_cache_dir / f"all_state_lightgbm_grid_results_{code}.csv"

    results_df = load_valid_cache(cache_path, lgbm_required_cols)
    if results_df.empty:
        print(f"Running LightGBM validation grid for {code}")
        rows = [lgbm_valid_row(params, fit_frame[features],␣
↪fit_frame["Claims"], valid_frame[features], valid_frame["Claims"]) for␣
↪params in lgbm_param_grid]
        results_df = pd.DataFrame(rows)
        results_df.to_csv(cache_path, index=False)

    valid_results = results_df.dropna(subset=["Valid_RMSE"]).
↪sort_values("Valid_RMSE").reset_index(drop=True)
    if valid_results.empty:
        raise ValueError(f"No LightGBM candidates successfully fit for {code}")

    best_params = clean_lgbm_params(valid_results.iloc[0])
    best_model = LGBMRegressor(objective="regression", random_state=148,␣
↪verbosity=-1, **best_params)
```

```
    best_model.fit(train_frame[features], train_frame["Claims"])


    # Forecasting through the end of the holdout plus the future forecast␣
 ↪horizon
    future_dates = make_future_dates(dates, all_state_test_horizon)

    pred_dates = pd.concat(
        [test_dates.reset_index(drop=True), future_dates],
        ignore_index=True
    )

    forecast_steps = len(y_test) + all_state_test_horizon

    y_pred = recursive_lgbm_forecast(
        best_model,
        y.iloc[:train_end_state],
        pred_dates,
        train_end_state,
        features
    )

    forecast_df = forecast_frame_extended(pred_dates, y_test, y_pred)


    forecast_df.to_csv(model_cache_dir /␣
 ↪f"all_state_lightgbm_holdout_predictions_{code}.csv", index=False)
    lgbm_all_results[code] = results_df
    lgbm_all_forecasts[code] = forecast_df
    lgbm_all_best.append({
        "StateCode": code,
        "State": state_names[code],
        "Model": "LightGBM recursive lag model",
        "RMSE": calc_rmse(y_test, y_pred.iloc[:len(y_test)]),
        "MAE": calc_mae(y_test, y_pred.iloc[:len(y_test)]),
        "Selection": "Best time-ordered validation RMSE in compact LightGBM␣
 ↪grid",
        "Plot_Label": "LightGBM recursive lag model"
    })

## Displaying
lgbm_all_best_df = pd.DataFrame(lgbm_all_best)
display(lgbm_all_best_df.sort_values(["State", "RMSE"]).round(3))
```

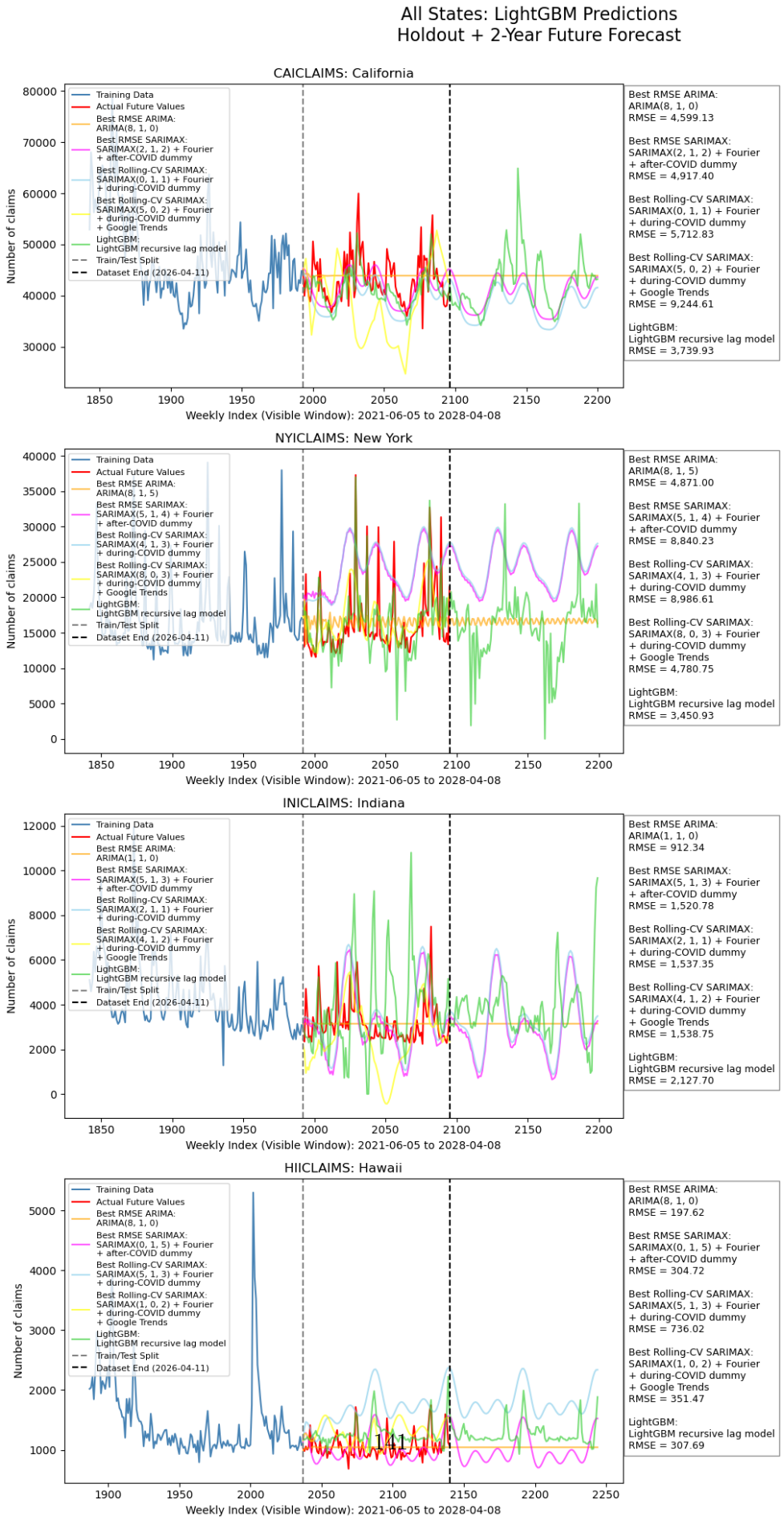
	StateCode	State	Model	RMSE	MAE	\
0	CAICLAIMS	California	LightGBM recursive lag model	3,739.93	2,860.31	
3	HIICLAIMS	Hawaii	LightGBM recursive lag model	307.69	243.65	

```
2  INICLAIMS     Indiana  LightGBM recursive lag model 2,127.70 1,526.30
1  NYICLAIMS    New York  LightGBM recursive lag model 3,450.93 2,515.21

                                          Selection  \
0  Best time-ordered validation RMSE in compact L…
3  Best time-ordered validation RMSE in compact L…
2  Best time-ordered validation RMSE in compact L…
1  Best time-ordered validation RMSE in compact L…

                    Plot_Label
0  LightGBM recursive lag model
3  LightGBM recursive lag model
2  LightGBM recursive lag model
1  LightGBM recursive lag model
```

4.6.4 Plotted LightGBM Predictions

Code cell purpose: What: This cell plots all-state forecasts from ARIMA, the SARIMAX variants, the Google Trends model, and LightGBM. Why: Viewing the ML forecast alongside the econometric forecasts makes it easier to compare whether the flexible model captures holdout dynamics differently.

[87]:
```
# Plot with helper function
plot_all_state_predictions(
    model_sources=[
        {
            "forecasts": arima_all_forecasts,
            "best_df": arima_all_best_df,
            "label_prefix": "Best RMSE ARIMA",
            "color": "orange"
        },
        {
            "forecasts": sarimax_after_all_forecasts,
            "best_df": sarimax_after_all_best_df,
            "label_prefix": "Best RMSE SARIMAX",
            "color": "magenta"
        },
        {
            "forecasts": sarimax_during_cv_all_forecasts,
            "best_df": sarimax_during_cv_all_best_df,
            "label_prefix": "Best Rolling-CV SARIMAX",
            "color": "skyblue"
        },
        {
            "forecasts": sarimax_gt_cv_all_forecasts,
            "best_df": sarimax_gt_cv_all_best_df,
            "label_prefix": "Best Rolling-CV SARIMAX",
            "color": "yellow"
        },
        {
            "forecasts": lgbm_all_forecasts,
            "best_df": lgbm_all_best_df,
            "label_prefix": "LightGBM",
            "color": "limegreen"
        }
    ],
    main_title="All States: LightGBM Predictions\nHoldout + 2-Year Future␣
 ↪Forecast"
)
```


All States: LightGBM Predictions
Holdout + 2-Year Future Forecast
CAICLAIMS: California
Training Data
Actual Future Values
Best RMSE ARIMA:
ARIMA(8, 1, 0)
Best RMSE SARIMAX:
SARIMAX(2, 1, 2) + Fourier
+ after-COVID dummy
Best Rolling-CV SARIMAX:
SARIMAX(0, 1, 1) + Fourier
+ during-COVID dummy
Best Rolling-CV SARIMAX:
SARIMAX(5, 0, 2) + Fourier
+ during-COVID dummy
+ Google Trends
LightGBM:
LightGBM recursive lag model
Train/Test Split
Dataset End (2026-04-11)
Best RMSE ARIMA:
ARIMA(8, 1, 0)
RMSE = 4,599.13
Best RMSE SARIMAX:
SARIMAX(2, 1, 2) + Fourier
+ after-COVID dummy
RMSE = 4,917.40
Best Rolling-CV SARIMAX:
SARIMAX(0, 1, 1) + Fourier
+ during-COVID dummy
RMSE = 5,712.83
Best Rolling-CV SARIMAX:
SARIMAX(5, 0, 2) + Fourier
+ during-COVID dummy
+ Google Trends
RMSE = 9,244.61
LightGBM:
LightGBM recursive lag model
RMSE = 3,739.93
Number of claims
Weekly Index (Visible Window): 2021-06-05 to 2028-04-08
NYICLAIMS: New York
Best RMSE ARIMA:
ARIMA(8, 1, 5)
Best RMSE SARIMAX:
SARIMAX(5, 1, 4) + Fourier
+ after-COVID dummy
Best Rolling-CV SARIMAX:
SARIMAX(4, 1, 3) + Fourier
+ during-COVID dummy
Best Rolling-CV SARIMAX:
SARIMAX(8, 0, 3) + Fourier
+ during-COVID dummy
+ Google Trends
LightGBM:
LightGBM recursive lag model
Train/Test Split
Dataset End (2026-04-11)
RMSE = 4,871.00
RMSE = 8,840.23
RMSE = 8,986.61
RMSE = 4,780.75
RMSE = 3,450.93
Number of claims
Weekly Index (Visible Window): 2021-06-05 to 2028-04-08
INICLAIMS: Indiana
Best RMSE ARIMA:
ARIMA(1, 1, 0)
Best RMSE SARIMAX:
SARIMAX(5, 1, 3) + Fourier
+ after-COVID dummy
Best Rolling-CV SARIMAX:
SARIMAX(2, 1, 1) + Fourier
+ during-COVID dummy
Best Rolling-CV SARIMAX:
SARIMAX(4, 1, 2) + Fourier
+ during-COVID dummy
+ Google Trends
LightGBM:
LightGBM recursive lag model
Train/Test Split
Dataset End (2026-04-11)
RMSE = 912.34
RMSE = 1,520.78
RMSE = 1,537.35
RMSE = 1,538.75
RMSE = 2,127.70
Number of claims
Weekly Index (Visible Window): 2021-06-05 to 2028-04-08
HIICLAIMS: Hawaii
Best RMSE ARIMA:
ARIMA(8, 1, 0)
Best RMSE SARIMAX:
SARIMAX(0, 1, 5) + Fourier
+ after-COVID dummy
Best Rolling-CV SARIMAX:
SARIMAX(5, 1, 3) + Fourier
+ during-COVID dummy
Best Rolling-CV SARIMAX:
SARIMAX(1, 0, 2) + Fourier
+ during-COVID dummy
+ Google Trends
LightGBM:
LightGBM recursive lag model
Train/Test Split
Dataset End (2026-04-11)
RMSE = 197.62
RMSE = 304.72
RMSE = 736.02
RMSE = 351.47
RMSE = 307.69
Number of claims
Weekly Index (Visible Window): 2021-06-05 to 2028-04-08

5 Results and Discussion

5.1 1. Model Comparison for All States

The table below gives the final model comparison. Each pair of state and model is evaluated on the same final 104 weeks/two-year period for that state, with RMSE as the main common metric. MAE is included because it is easier to interpret as an "average absolute miss" in claims.

Code cell purpose: Cell below stacks the ARIMA, SARIMAX, and LightGBM results for all states into one comparison table, helps identify best model for each state by holdout RMSE. Matters because it gives us one clean table for comparing econometric and ML performance across states.

[88]:
```
## Final all-state comparison across the five model classes
all_state_model_comparison = pd.concat([
    arima_all_best_df.assign(Model_Class="ARIMA"),
    sarimax_after_all_best_df.assign(Model_Class="Simple SARIMAX"),
    sarimax_during_cv_all_best_df.assign(Model_Class="Rolling-CV SARIMAX"),
    sarimax_gt_cv_all_best_df.assign(Model_Class="Rolling-CV SARIMAX + Google␣
 ↪Trends"),
    lgbm_all_best_df.assign(Model_Class="LightGBM")
], ignore_index=True)

all_state_model_comparison = all_state_model_comparison.sort_values(
    ["State", "RMSE"]
).reset_index(drop=True)

best_model_by_state = (
    all_state_model_comparison
    .sort_values(["State", "RMSE"])
    .groupby("State", as_index=False)
    .first()
)

display(all_state_model_comparison.round(3))
display(best_model_by_state.round(3))
```

	StateCode	State	Model	\
0	CAICLAIMS	California	LightGBM recursive lag model	
1	CAICLAIMS	California	ARIMA(8, 1, 0)	
2	CAICLAIMS	California	SARIMAX(2, 1, 2) + Fourier + after-COVID dummy	
3	CAICLAIMS	California	SARIMAX(0, 1, 1) + Fourier + during-COVID dummy	
4	CAICLAIMS	California	SARIMAX(5, 0, 2) + Fourier + during-COVID dumm…	
5	HIICLAIMS	Hawaii	ARIMA(8, 1, 0)	
6	HIICLAIMS	Hawaii	SARIMAX(0, 1, 5) + Fourier + after-COVID dummy	
7	HIICLAIMS	Hawaii	LightGBM recursive lag model	
8	HIICLAIMS	Hawaii	SARIMAX(1, 0, 2) + Fourier + during-COVID dumm…	
9	HIICLAIMS	Hawaii	SARIMAX(5, 1, 3) + Fourier + during-COVID dummy	
10	INICLAIMS	Indiana	ARIMA(1, 1, 0)	

```
11  INICLAIMS     Indiana     SARIMAX(5, 1, 3) + Fourier + after-COVID dummy
12  INICLAIMS     Indiana    SARIMAX(2, 1, 1) + Fourier + during-COVID dummy
13  INICLAIMS     Indiana  SARIMAX(4, 1, 2) + Fourier + during-COVID dumm…
14  INICLAIMS     Indiana                       LightGBM recursive lag model
15  NYICLAIMS    New York                       LightGBM recursive lag model
16  NYICLAIMS    New York  SARIMAX(8, 0, 3) + Fourier + during-COVID dumm…
17  NYICLAIMS    New York                                    ARIMA(8, 1, 5)
18  NYICLAIMS    New York     SARIMAX(5, 1, 4) + Fourier + after-COVID dummy
19  NYICLAIMS    New York    SARIMAX(4, 1, 3) + Fourier + during-COVID dummy

       RMSE      MAE                                        Selection  \
0  3,739.93 2,860.31  Best time-ordered validation RMSE in compact L…
1  4,599.13 3,754.13                Best holdout RMSE among ARIMA grid
2  4,917.40 3,708.22       Best holdout RMSE among simple SARIMAX grid
3  5,712.83 4,378.27           Best rolling-CV RMSE among SARIMAX grid
4  9,244.61 7,537.05  Best rolling-CV RMSE among SARIMAX + Google Tr…
5    197.62   136.54                Best holdout RMSE among ARIMA grid
6    304.72   232.38       Best holdout RMSE among simple SARIMAX grid
7    307.69   243.65  Best time-ordered validation RMSE in compact L…
8    351.47   298.14  Best rolling-CV RMSE among SARIMAX + Google Tr…
9    736.02   648.89           Best rolling-CV RMSE among SARIMAX grid
10   912.34   641.19                Best holdout RMSE among ARIMA grid
11 1,520.78 1,192.38       Best holdout RMSE among simple SARIMAX grid
12 1,537.35 1,188.45           Best rolling-CV RMSE among SARIMAX grid
13 1,538.75 1,185.39  Best rolling-CV RMSE among SARIMAX + Google Tr…
14 2,127.70 1,526.30  Best time-ordered validation RMSE in compact L…
15 3,450.93 2,515.21  Best time-ordered validation RMSE in compact L…
16 4,780.75 3,320.83  Best rolling-CV RMSE among SARIMAX + Google Tr…
17 4,871.00 3,418.95                Best holdout RMSE among ARIMA grid
18 8,840.23 8,234.57       Best holdout RMSE among simple SARIMAX grid
19 8,986.61 8,328.73           Best rolling-CV RMSE among SARIMAX grid

                                      Plot_Label  \
0                       LightGBM recursive lag model
1                                    ARIMA(8, 1, 0)
2     SARIMAX(2, 1, 2) + Fourier\n+ after-COVID dummy
3    SARIMAX(0, 1, 1) + Fourier\n+ during-COVID dummy
4   SARIMAX(5, 0, 2) + Fourier\n+ during-COVID dum…
5                                    ARIMA(8, 1, 0)
6     SARIMAX(0, 1, 5) + Fourier\n+ after-COVID dummy
7                       LightGBM recursive lag model
8   SARIMAX(1, 0, 2) + Fourier\n+ during-COVID dum…
9    SARIMAX(5, 1, 3) + Fourier\n+ during-COVID dummy
10                                   ARIMA(1, 1, 0)
11    SARIMAX(5, 1, 3) + Fourier\n+ after-COVID dummy
12   SARIMAX(2, 1, 1) + Fourier\n+ during-COVID dummy
13  SARIMAX(4, 1, 2) + Fourier\n+ during-COVID dum…
14                      LightGBM recursive lag model
```

```
15                       LightGBM recursive lag model
16  SARIMAX(8, 0, 3) + Fourier\n+ during-COVID dum…
17                                    ARIMA(8, 1, 5)
18    SARIMAX(5, 1, 4) + Fourier\n+ after-COVID dummy
19   SARIMAX(4, 1, 3) + Fourier\n+ during-COVID dummy

                          Model_Class
0                            LightGBM
1                               ARIMA
2                       Simple SARIMAX
3                  Rolling-CV SARIMAX
4   Rolling-CV SARIMAX + Google Trends
5                               ARIMA
6                       Simple SARIMAX
7                            LightGBM
8   Rolling-CV SARIMAX + Google Trends
9                  Rolling-CV SARIMAX
10                              ARIMA
11                      Simple SARIMAX
12                 Rolling-CV SARIMAX
13  Rolling-CV SARIMAX + Google Trends
14                           LightGBM
15                           LightGBM
16  Rolling-CV SARIMAX + Google Trends
17                              ARIMA
18                      Simple SARIMAX
19                 Rolling-CV SARIMAX

        State  StateCode                          Model     RMSE      MAE  \
0  California  CAICLAIMS  LightGBM recursive lag model 3,739.93 2,860.31
1      Hawaii  HIICLAIMS                ARIMA(8, 1, 0)   197.62   136.54
2     Indiana  INICLAIMS                ARIMA(1, 1, 0)   912.34   641.19
3    New York  NYICLAIMS  LightGBM recursive lag model 3,450.93 2,515.21

                                      Selection  \
0  Best time-ordered validation RMSE in compact L…
1                Best holdout RMSE among ARIMA grid
2                Best holdout RMSE among ARIMA grid
3  Best time-ordered validation RMSE in compact L…

                    Plot_Label Model_Class
0  LightGBM recursive lag model    LightGBM
1                ARIMA(8, 1, 0)       ARIMA
2                ARIMA(1, 1, 0)       ARIMA
3  LightGBM recursive lag model    LightGBM
```

5.2 2. Plotted Holdout Predictions for All States and Models

The final plots focus on the holdout period because this is where the models are evaluated for efficacy. For each state, the plot compares actual claims against the ARIMA forecast, the SARIMAX forecast, and the LightGBM recursive forecast.

Code cell purpose: What: This cell plots the final 104-week holdout forecasts for all five model families across the four states. Why: The holdout-only comparison focuses the figure on the exact evaluation window used to judge model performance.

[89]:
```
## Plot final 104-week holdout forecasts for all five models
fig, axes = plt.subplots(2, 2, figsize=(14, 9), sharex=False)
axes = axes.flatten()

for ax, code in zip(axes, datasets):
    actual_df = arima_all_forecasts[code].iloc[:all_state_test_horizon]
    arima_df = arima_all_forecasts[code].iloc[:all_state_test_horizon]
    after_df = sarimax_after_all_forecasts[code].iloc[:all_state_test_horizon]
    during_df = sarimax_during_cv_all_forecasts[code].iloc[:
 ↪all_state_test_horizon]
    gt_df = sarimax_gt_cv_all_forecasts[code].iloc[:all_state_test_horizon]
    lgbm_df = lgbm_all_forecasts[code].iloc[:all_state_test_horizon]

    ax.plot(actual_df["Date"], actual_df["Actual"], label="Actual Future␣
 ↪Values", color="steelblue")
    ax.plot(arima_df["Date"], arima_df["Prediction"], label="Best RMSE ARIMA",␣
 ↪color="orange", alpha=0.5)
    ax.plot(after_df["Date"], after_df["Prediction"], label="Best RMSE SARIMAX␣
 ↪+ after COVID dummy", color="magenta", alpha=0.5)
    ax.plot(during_df["Date"], during_df["Prediction"], label="Best Rolling-CV␣
 ↪SARIMAX + during COVID dummy", color="skyblue", alpha=0.5)
    ax.plot(gt_df["Date"], gt_df["Prediction"], label="Best Rolling-CV SARIMAX␣
 ↪+ during COVID dummy + Google Trends", color="yellow", alpha=0.5)
    ax.plot(lgbm_df["Date"], lgbm_df["Prediction"], label="LightGBM␣
 ↪Predictions", color="limegreen", alpha=0.5)

    ax.set_xlim(actual_df["Date"].min(), actual_df["Date"].max())

    ax.set_title(f"{state_names[code]}: Final 104-Week Holdout")
    ax.set_xlabel("Date")
    ax.set_ylabel("Number of claims")
    ax.legend(fontsize=6, loc="upper left")

plt.tight_layout()
plt.show()
```


California: Final 104-Week Holdout
New York: Final 104-Week Holdout
Indiana: Final 104-Week Holdout
Hawaii: Final 104-Week Holdout
Actual Future Values
Best RMSE ARIMA
Best RMSE SARIMAX + after COVID dummy
Best Rolling-CV SARIMAX + during COVID dummy
Best Rolling-CV SARIMAX + during COVID dummy + Google Trends
LightGBM Predictions
Number of claims
Date
2024-07
2024-10
2025-01
2025-04
2025-07
2025-10
2026-01
2026-04

5.3 3. How We Interpreted the COVID Structural Break

In the case of the COVID structural break, this was handled uniquely when it came to the baseline versus ML models.

- The ARIMA model serves as a simple time-series model benchmark, not having any explicit COVID information/behaviors added.
- The simple SARIMAX model uses an after-COVID dummy to allow for a post-pandemic level shift.
- The rolling-CV SARIMAX models use a during-COVID dummy to treat the acute pandemic period as a temporary shock. The Google Trends SARIMAX model uses the same during-COVID structure plus search-interest variables.
- The LightGBM model uses both during- and after-COVID indicators as supervised-learning features.

The LightGBM model also utilizes lagged claims and rolling averages which allows the model to learn nonlinear relationships between recent claims history, seasonal timing, and the COVID-period indicators. By utilizing a recursive forecast design, the ML model is stopped from using actual holdout claims as lag inputs, meaning the comparison remains aligned with the multi-step ARIMA/SARIMAX forecasting setup.

5.4 4. Determining the Best Parameter Choices for the Econometric Forecasting Models

Below, we have a table which touches directly on the selected ARIMA and SARIMAX specifications for each state. Here, our goal was to report both the model which had the lowest error but also interpret what certain parameter choices mean. Specifically, in the case of ARIMA and SARIMAX, the parameters have the same basic meaning: `p` controls how many autoregressive lags are included, `d` controls the degree of differencing used to remove non-stationary movement, and `q` controls the moving-average error correction.

Overall, the main selection rules are:

- For ARIMA and simple SARIMAX, model order is selected by holdout RMSE.
- For the rolling-CV SARIMAX models, model order is selected by rolling-CV RMSE, then the selected model is evaluated on the same final 104-week holdout.

The table reports both the selection RMSE and the final holdout RMSE.

Code cell purpose: This cell extracts the best ARIMA and SARIMAX parameter choices for every state and places them in one table. This matters because the table makes the selected `(p, d, q)` orders explicit and shows whether the more flexible SARIMAX setup improves holdout accuracy relative to the simpler ARIMA baseline.

[90]:
```
## Selected econometric parameters with final holdout RMSE
def best_holdout_row(results):
    return results.dropna(subset=["RMSE"]).sort_values("RMSE").iloc[0]

def best_cv_row(results):
    return results.dropna(subset=["CV_RMSE"]).sort_values("CV_RMSE").iloc[0]

def holdout_result(code, model_class):
    query = "StateCode == @code and Model_Class == @model_class"
    return all_state_model_comparison.query(query).iloc[0]

selected_econometric_params = []

for code in datasets:
    arima_best = best_holdout_row(arima_all_results[code])
    sarimax_after_best = best_holdout_row(sarimax_after_all_results[code])
    sarimax_during_best = best_cv_row(sarimax_during_cv_all_results[code])
    sarimax_gt_best = best_cv_row(sarimax_gt_cv_all_results[code])

    model_specs = [
        ("ARIMA", arima_best, "RMSE", "AIC", "BIC"),
        ("Simple SARIMAX", sarimax_after_best, "RMSE", "AIC", "BIC"),
        ("Rolling-CV SARIMAX", sarimax_during_best, "CV_RMSE", "CV_AIC",␣
 ↪"CV_BIC"),
        ("Rolling-CV SARIMAX + Google Trends", sarimax_gt_best, "CV_RMSE",␣
 ↪"CV_AIC", "CV_BIC")
```

```
    ]

    for model_class, row, select_col, aic_col, bic_col in model_specs:
        holdout = holdout_result(code, model_class)
        selected_econometric_params.append({
            "StateCode": code,
            "State": state_names[code],
            "Model_Class": model_class,
            "Order": f"({int(row['p'])}, {int(row['d'])}, {int(row['q'])})",
            "Selection_RMSE": row[select_col],
            "AIC": row[aic_col],
            "BIC": row[bic_col],
            "Holdout_RMSE": holdout["RMSE"],
            "Holdout_MAE": holdout["MAE"]
        })

selected_econometric_params_df = pd.DataFrame(selected_econometric_params)
selected_econometric_params_df = selected_econometric_params_df.
 ↪sort_values(["State"])
display(selected_econometric_params_df.round(3))
```

	StateCode	State	Model_Class	Order	\
0	CAICLAIMS	California	ARIMA	(8, 1, 0)	
1	CAICLAIMS	California	Simple SARIMAX	(2, 1, 2)	
2	CAICLAIMS	California	Rolling-CV SARIMAX	(0, 1, 1)	
3	CAICLAIMS	California	Rolling-CV SARIMAX + Google Trends	(5, 0, 2)	
12	HIICLAIMS	Hawaii	ARIMA	(8, 1, 0)	
13	HIICLAIMS	Hawaii	Simple SARIMAX	(0, 1, 5)	
14	HIICLAIMS	Hawaii	Rolling-CV SARIMAX	(5, 1, 3)	
15	HIICLAIMS	Hawaii	Rolling-CV SARIMAX + Google Trends	(1, 0, 2)	
8	INICLAIMS	Indiana	ARIMA	(1, 1, 0)	
9	INICLAIMS	Indiana	Simple SARIMAX	(5, 1, 3)	
10	INICLAIMS	Indiana	Rolling-CV SARIMAX	(2, 1, 1)	
11	INICLAIMS	Indiana	Rolling-CV SARIMAX + Google Trends	(4, 1, 2)	
4	NYICLAIMS	New York	ARIMA	(8, 1, 5)	
5	NYICLAIMS	New York	Simple SARIMAX	(5, 1, 4)	
6	NYICLAIMS	New York	Rolling-CV SARIMAX	(4, 1, 3)	
7	NYICLAIMS	New York	Rolling-CV SARIMAX + Google Trends	(8, 0, 3)	

	Selection_RMSE	AIC	BIC	Holdout_RMSE	Holdout_MAE
0	4,599.13	45,577.51	45,627.88	4,599.13	3,754.13
1	4,917.40	45,519.25	45,591.99	4,917.40	3,708.22
2	47,304.29	36,588.53	36,642.92	5,712.83	4,378.27
3	30,633.85	17,271.85	17,369.72	9,244.61	7,537.05
12	197.62	33,832.79	33,883.36	197.62	136.54
13	304.72	33,690.37	33,768.99	304.72	232.38
14	1,834.34	26,127.54	26,220.21	736.02	648.89

15	2,259.03	11,984.06	12,063.33	351.47	298.14
8	912.34	37,884.10	37,895.30	912.34	641.19
9	1,520.78	37,573.23	37,668.33	1,520.78	1,192.38
10	6,940.34	31,272.96	31,338.23	1,537.35	1,188.45
11	7,739.76	14,507.04	14,600.25	1,538.75	1,185.39
4	4,866.19	42,313.41	42,391.76	4,871.00	3,418.95
5	4,950.30	42,231.75	42,332.43	8,840.23	8,234.57
6	23,203.85	34,857.85	34,944.85	8,986.61	8,328.73
7	20,623.61	16,408.62	16,525.03	4,780.75	3,320.83

Table above directly informs us about specific econometric parameter choices. Mainly, a lower value of "d" usually means the model can work with the level of claims after controlling for lag structure, while "d = 1" means the selected model relies on week-to-week changes rather than raw levels. Higher "p" values give the model more direct lag dependence, while higher "q" values let the model correct for recent forecast errors. SARIMAX uses the same order logic, but it also receives trend, seasonal Fourier terms, and COVID-period indicators, so its selected order should be interpreted as the remaining autoregressive structure after those controls are included.

5.5 5. Exploring the Grid-Best ARIMA and SARIMAX Specifications

Even though we have a table showcasing good models, the best model should not simply be deemed that which appears first. To avoid this issue, we have an output below which provides the top candidate ARIMA/SARIMAX specifications per state based on our parameter grids, allowing us to have full robustness when checking the models we built. Mainly, if there is extreme similarities in the RMSE values of top models, then broader modeling choice matters more than direct ordering of the models. However, if there are clearly major differences in RMSE, then order does matter for the models.

Code cell purpose: What: This cell defines helper functions for re-evaluating selected SARIMAX orders on the final holdout and formatting order tuples. Why: These helpers support a robustness check of top candidate specifications using a common final holdout metric.

[91]:
```
## Holdout evaluation helper for selected SARIMAX orders
def eval_sarimax_holdout(y_train, y_test, X_train, X_test, order):
    p, d, q = map(int, order)
    try:
        model = SARIMAX(y_train, exog=X_train, order=(p, d, q),
                        enforce_stationarity=False,␣
 ↪enforce_invertibility=False).fit(disp=False)
        pred = model.get_forecast(steps=len(y_test), exog=X_test).predicted_mean
        pred.index = y_test.index
        return calc_rmse(y_test, pred)
    except Exception:
        return np.nan

def order_tuple(row):
    return (int(row["p"]), int(row["d"]), int(row["q"]))
```

Code cell purpose: What: This cell builds or loads the cached top econometric candidate table across ARIMA and SARIMAX model classes. Why: Comparing top-ranked candidates helps verify that the final model choice is robust rather than driven by a single nearly tied grid-search result.

[92]:
```
## Top candidate models, evaluated using final holdout RMSE where possible
## This cell can be slow because rolling-CV SARIMAX candidates are refit on␣
 ↪holdout
## We cache the top-10 table and then display k = 1 and k = 10 views from the␣
 ↪cache

top_k_build = 10
top_candidates_cache_path = model_cache_dir / "top_k_econometrics_candidates.
 ↪csv"

top_candidates_required_cols = {
    "StateCode", "State", "Model_Class", "Rank", "Order",
    "Selection_RMSE", "Holdout_RMSE", "AIC", "BIC"
}
```

```
top_econometric_candidates_df = load_valid_cache(
    top_candidates_cache_path,
    top_candidates_required_cols
)

cache_has_enough_ranks = (
    not top_econometric_candidates_df.empty
    and top_econometric_candidates_df["Rank"].max() >= top_k_build
)

if cache_has_enough_ranks:
    print(f"Loading cached top-{top_k_build} econometric candidates from␣
 ↪{top_candidates_cache_path}")

else:
    print(f"Building and caching top-{top_k_build} econometric candidates...")

    top_econometric_candidates = []

    for code in datasets:
        df, y, dates, train_end_state = get_state_split(code,␣
 ↪all_state_test_horizon)

        y_train = y.iloc[:train_end_state]
        y_test = y.iloc[train_end_state:]

        X_after = make_exog_after(dates)
        X_after_train = X_after.iloc[:train_end_state]
        X_after_test = X_after.iloc[train_end_state:][X_after_train.columns]

        X_during = make_exog_during(dates)
        X_during_train = X_during.iloc[:train_end_state]
        X_during_test = X_during.iloc[train_end_state:][X_during_train.columns]

        df_gt, gt_weekly = get_state_split_gt(code)
        y_gt = df_gt["Claims"].reset_index(drop=True)
        dates_gt = df_gt["Date"].reset_index(drop=True)

        train_end_gt_state = len(y_gt) - all_state_test_horizon
        y_gt_train = y_gt.iloc[:train_end_gt_state]
        y_gt_test = y_gt.iloc[train_end_gt_state:]

        gt_aligned = align_google_trends(dates_gt, gt_weekly)
        X_gt = make_exog_during_gt(dates_gt, gt_aligned)
        X_gt_train = X_gt.iloc[:train_end_gt_state]
        X_gt_test = X_gt.iloc[train_end_gt_state:][X_gt_train.columns]
```

```
        arima_top = (
            arima_all_results[code]
            .dropna(subset=["RMSE"])
            .sort_values("RMSE")
            .head(top_k_build)
        )

        after_top = (
            sarimax_after_all_results[code]
            .dropna(subset=["RMSE"])
            .sort_values("RMSE")
            .head(top_k_build)
        )

        during_top = (
            sarimax_during_cv_all_results[code]
            .dropna(subset=["CV_RMSE"])
            .sort_values("CV_RMSE")
            .head(top_k_build)
        )

        gt_top = (
            sarimax_gt_cv_all_results[code]
            .dropna(subset=["CV_RMSE"])
            .sort_values("CV_RMSE")
            .head(top_k_build)
        )

        for rank, row in arima_top.reset_index(drop=True).iterrows():
            top_econometric_candidates.append({
                "StateCode": code,
                "State": state_names[code],
                "Model_Class": "ARIMA",
                "Rank": rank + 1,
                "Order": f"({int(row['p'])}, {int(row['d'])}, {int(row['q'])})",
                "Selection_RMSE": row["RMSE"],
                "Holdout_RMSE": row["RMSE"],
                "AIC": row["AIC"],
                "BIC": row["BIC"]
            })

        for rank, row in after_top.reset_index(drop=True).iterrows():
            top_econometric_candidates.append({
                "StateCode": code,
                "State": state_names[code],
                "Model_Class": "Simple SARIMAX",
```

```
                "Rank": rank + 1,
                "Order": f"({int(row['p'])}, {int(row['d'])}, {int(row['q'])})",
                "Selection_RMSE": row["RMSE"],
                "Holdout_RMSE": row["RMSE"],
                "AIC": row["AIC"],
                "BIC": row["BIC"]
            })

        for rank, row in during_top.reset_index(drop=True).iterrows():
            order = order_tuple(row)

            holdout_rmse = eval_sarimax_holdout(
                y_train,
                y_test,
                X_during_train,
                X_during_test,
                order
            )

            top_econometric_candidates.append({
                "StateCode": code,
                "State": state_names[code],
                "Model_Class": "Rolling-CV SARIMAX",
                "Rank": rank + 1,
                "Order": f"({order[0]}, {order[1]}, {order[2]})",
                "Selection_RMSE": row["CV_RMSE"],
                "Holdout_RMSE": holdout_rmse,
                "AIC": row["CV_AIC"],
                "BIC": row["CV_BIC"]
            })

        for rank, row in gt_top.reset_index(drop=True).iterrows():
            order = order_tuple(row)

            holdout_rmse = eval_sarimax_holdout(
                y_gt_train,
                y_gt_test,
                X_gt_train,
                X_gt_test,
                order
            )

            top_econometric_candidates.append({
                "StateCode": code,
                "State": state_names[code],
                "Model_Class": "Rolling-CV SARIMAX + Google Trends",
                "Rank": rank + 1,
```

```
                "Order": f"({order[0]}, {order[1]}, {order[2]})",
                "Selection_RMSE": row["CV_RMSE"],
                "Holdout_RMSE": holdout_rmse,
                "AIC": row["CV_AIC"],
                "BIC": row["CV_BIC"]
            })

    top_econometric_candidates_df = pd.DataFrame(top_econometric_candidates)

    top_econometric_candidates_df = top_econometric_candidates_df.sort_values(
        ["State", "Model_Class", "Rank"]
    ).reset_index(drop=True)

    top_econometric_candidates_df.to_csv(top_candidates_cache_path, index=False)

    print(f"Saved top-{top_k_build} econometric candidates to␣
↪{top_candidates_cache_path}")
```

```
Loading cached top-10 econometric candidates from
Model_Cache/top_k_econometrics_candidates.csv
```

Code cell purpose: What: This cell defines a display helper for the cached top econometric candidate table and displays the top two candidates per state and model class. Why: Showing multiple leading candidates makes the model-selection logic more transparent when top RMSE values are close.

[93]:
```
def display_top_econometric_candidates(k):
    print(f"Top {k} econometric candidates per state and model class:")

    out = (
        top_econometric_candidates_df
        .loc[top_econometric_candidates_df["Rank"] <= k]
        .sort_values(["State", "Model_Class", "Rank"])
        .reset_index(drop=True)
    )

    display(out.round(3))

# Change k to display top k candidates per state and model type from cached␣
↪table
display_top_econometric_candidates(2)
```

```
Top 2 econometric candidates per state and model class:
```

	StateCode	State	Model_Class	Rank	\
0	CAICLAIMS	California	ARIMA	1	
1	CAICLAIMS	California	ARIMA	2	
2	CAICLAIMS	California	Rolling-CV SARIMAX	1	
3	CAICLAIMS	California	Rolling-CV SARIMAX	2	

4	CAICLAIMS	California	Rolling-CV SARIMAX + Google Trends	1
5	CAICLAIMS	California	Rolling-CV SARIMAX + Google Trends	2
6	CAICLAIMS	California	Simple SARIMAX	1
7	CAICLAIMS	California	Simple SARIMAX	2
8	HIICLAIMS	Hawaii	ARIMA	1
9	HIICLAIMS	Hawaii	ARIMA	2
10	HIICLAIMS	Hawaii	Rolling-CV SARIMAX	1
11	HIICLAIMS	Hawaii	Rolling-CV SARIMAX	2
12	HIICLAIMS	Hawaii	Rolling-CV SARIMAX + Google Trends	1
13	HIICLAIMS	Hawaii	Rolling-CV SARIMAX + Google Trends	2
14	HIICLAIMS	Hawaii	Simple SARIMAX	1
15	HIICLAIMS	Hawaii	Simple SARIMAX	2
16	INICLAIMS	Indiana	ARIMA	1
17	INICLAIMS	Indiana	ARIMA	2
18	INICLAIMS	Indiana	Rolling-CV SARIMAX	1
19	INICLAIMS	Indiana	Rolling-CV SARIMAX	2
20	INICLAIMS	Indiana	Rolling-CV SARIMAX + Google Trends	1
21	INICLAIMS	Indiana	Rolling-CV SARIMAX + Google Trends	2
22	INICLAIMS	Indiana	Simple SARIMAX	1
23	INICLAIMS	Indiana	Simple SARIMAX	2
24	NYICLAIMS	New York	ARIMA	1
25	NYICLAIMS	New York	ARIMA	2
26	NYICLAIMS	New York	Rolling-CV SARIMAX	1
27	NYICLAIMS	New York	Rolling-CV SARIMAX	2
28	NYICLAIMS	New York	Rolling-CV SARIMAX + Google Trends	1
29	NYICLAIMS	New York	Rolling-CV SARIMAX + Google Trends	2
30	NYICLAIMS	New York	Simple SARIMAX	1
31	NYICLAIMS	New York	Simple SARIMAX	2

	Order	Selection_RMSE	Holdout_RMSE	AIC	BIC
0	(8, 1, 0)	4,599.13	4,599.13	45,577.51	45,627.88
1	(4, 1, 0)	4,602.98	4,602.98	45,662.47	45,690.45
2	(0, 1, 1)	47,304.29	5,712.83	36,588.53	36,642.92
3	(1, 1, 0)	47,526.39	5,723.09	36,835.71	36,890.10
4	(5, 0, 2)	30,633.85	9,244.61	17,271.85	17,369.72
5	(1, 0, 3)	30,707.57	9,679.99	17,297.20	17,381.11
6	(2, 1, 2)	4,917.40	4,917.40	45,519.25	45,591.99
7	(2, 1, 3)	4,932.25	4,932.25	45,485.31	45,563.64
8	(8, 1, 0)	197.62	197.62	33,832.79	33,883.36
9	(0, 1, 3)	197.68	197.68	33,855.56	33,878.03
10	(5, 1, 3)	1,834.34	756.54	26,127.54	26,220.21
11	(2, 1, 1)	1,876.90	670.77	26,275.09	26,340.53
12	(1, 0, 2)	2,259.03	351.47	11,984.06	12,063.33
13	(1, 0, 3)	2,317.82	319.28	12,011.60	12,095.51
14	(0, 1, 5)	304.72	304.72	33,690.37	33,768.99
15	(0, 1, 6)	306.66	306.66	33,658.26	33,742.49
16	(1, 1, 0)	912.34	912.34	37,884.10	37,895.30
17	(0, 1, 2)	912.40	912.40	37,884.97	37,901.75

18	(2, 1, 1)	6,940.34	1,537.35	31,272.96	31,338.23
19	(8, 1, 4)	7,036.75	2,345.51	31,134.11	31,248.25
20	(4, 1, 2)	7,739.76	1,538.75	14,507.04	14,600.25
21	(6, 1, 1)	7,789.04	1,507.04	14,468.63	14,566.44
22	(5, 1, 3)	1,520.78	1,520.78	37,573.23	37,668.33
23	(4, 1, 4)	1,520.85	1,520.85	37,574.50	37,669.60
24	(8, 1, 5)	4,866.19	4,866.19	42,313.41	42,391.76
25	(7, 1, 0)	4,938.91	4,938.91	42,384.92	42,429.69
26	(4, 1, 3)	23,203.85	8,973.50	34,857.85	34,944.85
27	(2, 1, 2)	26,226.62	9,580.08	34,880.97	34,951.67
28	(8, 0, 3)	20,623.61	4,773.84	16,408.62	16,525.03
29	(1, 0, 6)	20,710.24	4,763.94	16,475.14	16,572.95
30	(5, 1, 4)	4,950.30	4,950.30	42,231.75	42,332.43
31	(1, 1, 2)	4,975.78	4,975.78	42,379.85	42,446.98

5.6 6. LightGBM Parameter Choices and Validation Logic

LightGBM is selected differently from ARIMA/SARIMAX. Rather than choosing `(p, d, q)` orders, the model chooses hyperparameters which control the complexity of boosted decision trees. In our notebook, `n_estimators` controls the number of boosting rounds, `learning_rate` controls how aggressively each tree updates the prediction, `num_leaves` controls how flexible each tree can be, and `min_child_samples` regularizes the tree by requiring enough observations in each leaf.

The LightGBM grid is chosen using a time-ordered validation split inside the training period, NOT the final test/holdout period. This preserves the final 104 weeks/two-year period as a fair test set shared across ARIMA, SARIMAX, and LightGBM.

Code cell purpose: This cell reports the selected LightGBM hyperparameters for every state and connects the validation score to the final holdout score. This matters because the ML model is tuned on a training-only validation window but evaluated on the same final holdout window as the econometric models. Ensuring consistency and effective testing.

[94]:
```
## Storage for selected LightGBM hyperparameter rows
selected_lgbm_params = []

## Pulling selected LightGBM parameters and final holdout results
for code in datasets:
    best_valid = lgbm_all_results[code].dropna(subset=["Valid_RMSE"]).
 ↪sort_values("Valid_RMSE").iloc[0]
    holdout_row = all_state_model_comparison.query("StateCode == @code and␣
 ↪Model_Class == 'LightGBM'").iloc[0]

    selected_lgbm_params.append({
        "StateCode": code,
        "State": state_names[code],
        "n_estimators": int(best_valid["n_estimators"]),
        "learning_rate": float(best_valid["learning_rate"]),
        "num_leaves": int(best_valid["num_leaves"]),
        "min_child_samples": int(best_valid["min_child_samples"]),
        "Validation_RMSE": best_valid["Valid_RMSE"],
        "Holdout_RMSE": holdout_row["RMSE"],
        "Holdout_MAE": holdout_row["MAE"]
    })

## Final LightGBM parameter table
selected_lgbm_params_df = pd.DataFrame(selected_lgbm_params)
selected_lgbm_params_df = selected_lgbm_params_df.sort_values("State")
display(selected_lgbm_params_df.round(3))
```

	StateCode	State	n_estimators	learning_rate	num_leaves	\
0	CAICLAIMS	California	250	0.03	15	
3	HIICLAIMS	Hawaii	350	0.03	31	
2	INICLAIMS	Indiana	350	0.03	31	
1	NYICLAIMS	New York	350	0.05	31	

	min_child_samples	Validation_RMSE	Holdout_RMSE	Holdout_MAE
0	20	3,640.69	3,739.93	2,860.31
3	20	693.10	307.69	243.65
2	20	1,361.76	2,127.70	1,526.30
1	30	2,880.33	3,450.93	2,515.21

In terms of interpretation here, LightGBM is considered a more flexible model as compared to the main structural model. This is because of the fact that, while ARIMA and SARIMAX impose explicit time-series structure, the LightGBM model instead learns nonlinear relationships from engineered predictors like lags, rolling means, seasonality, trend, and COVID indicators. In turn, if we see that LightGBM performs best, the interpretation is that nonlinear lag and seasonal patterns helped. But, if SARIMAX performs best, the interpretation is that a more structured time-series model with explicit COVID and seasonal controls generalized better (as compared to the baseline boring ARIMA model).

5.7 7. LightGBM Feature Importance Across States

We wanted to increase the interpretability of our LightGBM model, hence the following chunk of code. Mainly, feature importance is not something like a causal estimate, but instead helps identify whether the model is relying mostly on recent lagged claims, longer-run rolling averages, seasonal timing, trend, or COVID-period indicators. This is useful for comparing the ML process against the econometric process, as the econometric models specify lag and seasonal structure directly, while the LightGBM ML model learns which engineered predictors are most useful.

Code cell purpose: Cell below refits each state's selected LightGBM model on the training period and reports the most important predictors. This matters because here we delve into model logic and variable selection, and feature importance gives us a more-concrete way to explain what the ML model used most heavily.

[95]:
```
## Storage for feature-importance rows
lgbm_feature_importance = []

## Refit selected LightGBM models to inspect feature importance
for code in datasets:
    df, y, dates, train_end_state = get_state_split(code,␣
 ↪all_state_test_horizon)
    test_start_date = dates.iloc[train_end_state]
    ml_frame = make_lgbm_frame(df, lgbm_lags, lgbm_windows)
    train_frame = ml_frame.loc[ml_frame["Date"] < test_start_date].
 ↪reset_index(drop=True)
    features = ml_feature_columns(train_frame)
    best_valid = lgbm_all_results[code].dropna(subset=["Valid_RMSE"]).
 ↪sort_values("Valid_RMSE").iloc[0]
    best_params = clean_lgbm_params(best_valid)
    model = LGBMRegressor(objective="regression", random_state=148,␣
 ↪verbosity=-1, **best_params)
    model.fit(train_frame[features], train_frame["Claims"])

    for feature, importance in zip(features, model.feature_importances_):
        lgbm_feature_importance.append({"StateCode": code, "State":␣
 ↪state_names[code], "Feature": feature, "Importance": importance})

## Displaying the top predictors for each state
top_lgbm_features_df = pd.DataFrame(lgbm_feature_importance)
top_lgbm_features_df = top_lgbm_features_df.sort_values(["State",␣
 ↪"Importance"], ascending=[True, False])
top_lgbm_features_df = top_lgbm_features_df.groupby("State", as_index=False).
 ↪head(8)
display(top_lgbm_features_df.reset_index(drop=True))
```

	StateCode	State	Feature	Importance
0	CAICLAIMS	California	lag_1	501
1	CAICLAIMS	California	week	387

2	CAICLAIMS	California	lag_52	374
3	CAICLAIMS	California	lag_26	360
4	CAICLAIMS	California	roll_mean_52	256
5	CAICLAIMS	California	roll_mean_4	239
6	CAICLAIMS	California	lag_4	237
7	CAICLAIMS	California	week_cos	234
8	HIICLAIMS	Hawaii	lag_1	1632
9	HIICLAIMS	Hawaii	trend	969
10	HIICLAIMS	Hawaii	lag_52	960
11	HIICLAIMS	Hawaii	lag_2	835
12	HIICLAIMS	Hawaii	roll_mean_52	793
13	HIICLAIMS	Hawaii	roll_mean_4	766
14	HIICLAIMS	Hawaii	lag_26	691
15	HIICLAIMS	Hawaii	week_sin	660
16	INICLAIMS	Indiana	lag_1	1339
17	INICLAIMS	Indiana	lag_52	1222
18	INICLAIMS	Indiana	lag_2	928
19	INICLAIMS	Indiana	lag_26	894
20	INICLAIMS	Indiana	roll_mean_4	779
21	INICLAIMS	Indiana	lag_4	742
22	INICLAIMS	Indiana	roll_mean_52	688
23	INICLAIMS	Indiana	week	685
24	NYICLAIMS	New York	lag_52	1585
25	NYICLAIMS	New York	lag_1	1226
26	NYICLAIMS	New York	lag_26	1056
27	NYICLAIMS	New York	roll_mean_52	809
28	NYICLAIMS	New York	trend	698
29	NYICLAIMS	New York	lag_2	679
30	NYICLAIMS	New York	lag_13	628
31	NYICLAIMS	New York	roll_mean_4	616

5.8 8. Final Model Choice and Modeling Logic Comparison

For the final recommendation, we choose models on a state-by-state basis because the four states have different labor-market scales, seasonal patterns, and shock dynamics. Although ARIMA is retained as an important benchmark, we do not allow it to determine the final recommended model. The reason is that the ARIMA forecasts often achieve low holdout RMSE by producing relatively stable paths, but these paths do not always capture the variation and model structure that we want from a forecasting tool. Thus, the final recommendation is selected only from the SARIMAX and LightGBM families.

Within that restricted candidate set, we still use final 104-week holdout RMSE as the primary selection metric, with MAE included as a secondary measure because it is easier to interpret as the average absolute miss in weekly claims. This lets us preserve the same out-of-sample evaluation framework while making the final choice among models that include either explicit structural controls, as in SARIMAX, or richer lag/rolling/calendar features, as in LightGBM.

5.8.1 Candidate Model Comparison Table

Code cell purpose: Cell below creates one final recommendation table that combines the best non-ARIMA models for each state with the selected parameter or hyperparameter details. This matters because it gives us a concise table for stating which model should be emphasized and why.

[96]:
```
# Final model choice excluding ARIMA from recommendation set
# ARIMA remains a benchmark, but is not considered for final recommended model.

recommendation_candidate_classes = [
    "Simple SARIMAX",
    "Rolling-CV SARIMAX",
    "Rolling-CV SARIMAX + Google Trends",
    "LightGBM"
]

final_candidate_model_comparison = (
    all_state_model_comparison
    .loc[all_state_model_comparison["Model_Class"].
 ↪isin(recommendation_candidate_classes)]
    .sort_values(["State", "RMSE"])
    .reset_index(drop=True)
)

best_non_arima_model_by_state = (
    final_candidate_model_comparison
    .sort_values(["State", "RMSE"])
    .groupby("State", as_index=False)
    .first()
)

display(final_candidate_model_comparison.loc[:,␣
 ↪final_candidate_model_comparison.columns != "Plot_Label"].round(3))
```

```
display(best_non_arima_model_by_state.loc[:, best_non_arima_model_by_state.
  ↪columns != "Plot_Label"].round(3))
```

```
    StateCode       State                                                Model  \
0   CAICLAIMS  California                        LightGBM recursive lag model
1   CAICLAIMS  California     SARIMAX(2, 1, 2) + Fourier + after-COVID dummy
2   CAICLAIMS  California    SARIMAX(0, 1, 1) + Fourier + during-COVID dummy
3   CAICLAIMS  California  SARIMAX(5, 0, 2) + Fourier + during-COVID dumm…
4   HIICLAIMS      Hawaii     SARIMAX(0, 1, 5) + Fourier + after-COVID dummy
5   HIICLAIMS      Hawaii                        LightGBM recursive lag model
6   HIICLAIMS      Hawaii  SARIMAX(1, 0, 2) + Fourier + during-COVID dumm…
7   HIICLAIMS      Hawaii    SARIMAX(5, 1, 3) + Fourier + during-COVID dummy
8   INICLAIMS     Indiana     SARIMAX(5, 1, 3) + Fourier + after-COVID dummy
9   INICLAIMS     Indiana    SARIMAX(2, 1, 1) + Fourier + during-COVID dummy
10  INICLAIMS     Indiana  SARIMAX(4, 1, 2) + Fourier + during-COVID dumm…
11  INICLAIMS     Indiana                        LightGBM recursive lag model
12  NYICLAIMS    New York                        LightGBM recursive lag model
13  NYICLAIMS    New York  SARIMAX(8, 0, 3) + Fourier + during-COVID dumm…
14  NYICLAIMS    New York     SARIMAX(5, 1, 4) + Fourier + after-COVID dummy
15  NYICLAIMS    New York    SARIMAX(4, 1, 3) + Fourier + during-COVID dummy

       RMSE      MAE                                          Selection  \
0  3,739.93 2,860.31  Best time-ordered validation RMSE in compact L…
1  4,917.40 3,708.22        Best holdout RMSE among simple SARIMAX grid
2  5,712.83 4,378.27            Best rolling-CV RMSE among SARIMAX grid
3  9,244.61 7,537.05  Best rolling-CV RMSE among SARIMAX + Google Tr…
4    304.72   232.38        Best holdout RMSE among simple SARIMAX grid
5    307.69   243.65  Best time-ordered validation RMSE in compact L…
6    351.47   298.14  Best rolling-CV RMSE among SARIMAX + Google Tr…
7    736.02   648.89            Best rolling-CV RMSE among SARIMAX grid
8  1,520.78 1,192.38        Best holdout RMSE among simple SARIMAX grid
9  1,537.35 1,188.45            Best rolling-CV RMSE among SARIMAX grid
10 1,538.75 1,185.39  Best rolling-CV RMSE among SARIMAX + Google Tr…
11 2,127.70 1,526.30  Best time-ordered validation RMSE in compact L…
12 3,450.93 2,515.21  Best time-ordered validation RMSE in compact L…
13 4,780.75 3,320.83  Best rolling-CV RMSE among SARIMAX + Google Tr…
14 8,840.23 8,234.57        Best holdout RMSE among simple SARIMAX grid
15 8,986.61 8,328.73            Best rolling-CV RMSE among SARIMAX grid

                         Model_Class
0                           LightGBM
1                       Simple SARIMAX
2                    Rolling-CV SARIMAX
3   Rolling-CV SARIMAX + Google Trends
4                       Simple SARIMAX
5                           LightGBM
6   Rolling-CV SARIMAX + Google Trends
```

```
7                   Rolling-CV SARIMAX
8                       Simple SARIMAX
9                   Rolling-CV SARIMAX
10  Rolling-CV SARIMAX + Google Trends
11                            LightGBM
12                            LightGBM
13  Rolling-CV SARIMAX + Google Trends
14                      Simple SARIMAX
15                  Rolling-CV SARIMAX
```

```
        State  StateCode                                          Model  \
0  California  CAICLAIMS                     LightGBM recursive lag model
1      Hawaii  HIICLAIMS  SARIMAX(0, 1, 5) + Fourier + after-COVID dummy
2     Indiana  INICLAIMS  SARIMAX(5, 1, 3) + Fourier + after-COVID dummy
3    New York  NYICLAIMS                     LightGBM recursive lag model

      RMSE      MAE                                        Selection  \
0 3,739.93 2,860.31  Best time-ordered validation RMSE in compact L…
1   304.72   232.38        Best holdout RMSE among simple SARIMAX grid
2 1,520.78 1,192.38        Best holdout RMSE among simple SARIMAX grid
3 3,450.93 2,515.21  Best time-ordered validation RMSE in compact L…

      Model_Class
0        LightGBM
1  Simple SARIMAX
2  Simple SARIMAX
3        LightGBM
```

5.8.2 Final Model Choices by State

Code cell purpose: What: This cell assembles the final recommended model table with selected details, RMSE, MAE, and the model-selection reason for each state. Why: This gives a concise state-by-state summary of the preferred forecasting model after comparing the econometric and ML candidates.

[97]:
```
# Storage for final recommendation rows
final_recommendation_rows = []

# Combining best non-ARIMA model choices with relevant parameter details
for _, row in best_non_arima_model_by_state.iterrows():
    code = row["StateCode"]
    model_class = row["Model_Class"]
    detail = row["Model"]

    if model_class == "LightGBM":
        params = selected_lgbm_params_df.query("StateCode == @code").iloc[0]
        detail = (
            f"LightGBM: leaves={params['num_leaves']}, "
            f"lr={params['learning_rate']}, "
            f"trees={params['n_estimators']}"
        )

    final_recommendation_rows.append({
        "StateCode": code,
        "State": row["State"],
        "Recommended_Model": model_class,
        "Selected_Details": detail,
        "Holdout_RMSE": row["RMSE"],
        "Holdout_MAE": row["MAE"],
        "Reason": "Lowest final 104-week holdout RMSE among SARIMAX and␣
 ↪LightGBM candidates"
    })

# Displaying final recommendations
final_model_recommendations_df = pd.DataFrame(final_recommendation_rows)
final_model_recommendations_df = final_model_recommendations_df.
 ↪sort_values("State")

display(final_model_recommendations_df.round(3))
```

```
   StateCode       State Recommended_Model  \
0  CAICLAIMS  California          LightGBM
1  HIICLAIMS      Hawaii    Simple SARIMAX
2  INICLAIMS     Indiana    Simple SARIMAX
3  NYICLAIMS    New York          LightGBM
```

	Selected_Details	Holdout_RMSE	Holdout_MAE	\
0	LightGBM: leaves=15, lr=0.03, trees=250	3,739.93	2,860.31	
1	SARIMAX(0, 1, 5) + Fourier + after-COVID dummy	304.72	232.38	
2	SARIMAX(5, 1, 3) + Fourier + after-COVID dummy	1,520.78	1,192.38	
3	LightGBM: leaves=31, lr=0.05, trees=350	3,450.93	2,515.21	

	Reason
0	Lowest final 104-week holdout RMSE among SARIM…
1	Lowest final 104-week holdout RMSE among SARIM…
2	Lowest final 104-week holdout RMSE among SARIM…
3	Lowest final 104-week holdout RMSE among SARIM…

5.8.3 Final Model Plotted Predictions

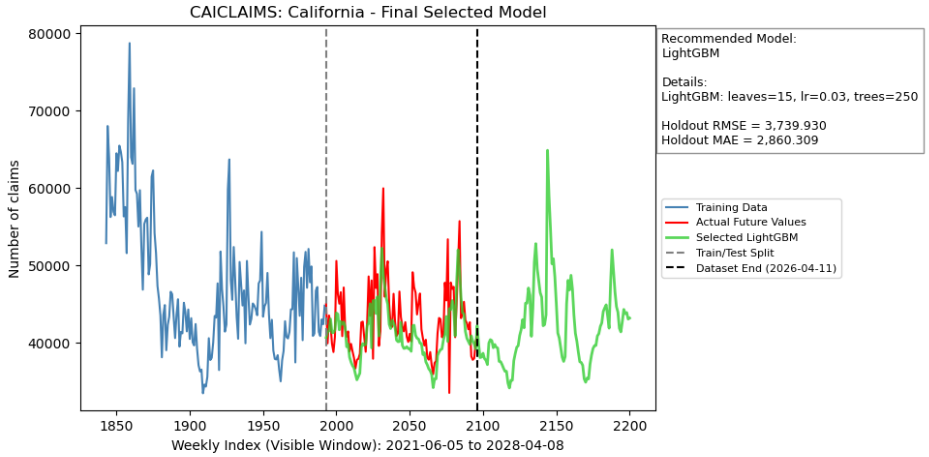
Code cell purpose: What: This cell plots the final selected model's holdout and future forecast for each state. Why: The figure translates the final recommendations into forecast paths, making it clear how the chosen model behaves relative to recent training data and actual holdout values.

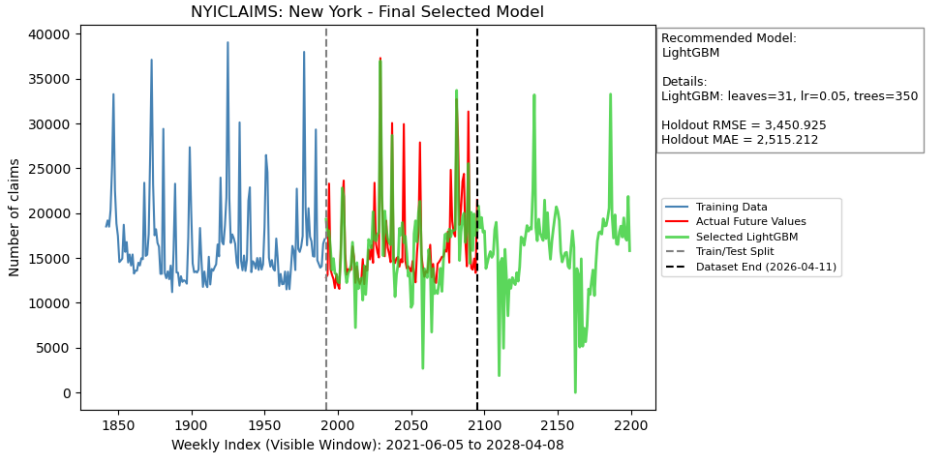
[98]:
```
## Final selected models: plotted predictions for holdout and future forecasts

selected_forecast_sources = {
    "Simple SARIMAX": sarimax_after_all_forecasts,
    "Rolling-CV SARIMAX": sarimax_during_cv_all_forecasts,
    "Rolling-CV SARIMAX + Google Trends": sarimax_gt_cv_all_forecasts,
    "LightGBM": lgbm_all_forecasts
}

selected_colors = {
    "Simple SARIMAX": "magenta",
    "Rolling-CV SARIMAX": "skyblue",
    "Rolling-CV SARIMAX + Google Trends": "gold",
    "LightGBM": "limegreen"
}

selected_label_names = {
    "Simple SARIMAX": "Selected Simple SARIMAX",
    "Rolling-CV SARIMAX": "Selected Rolling-CV SARIMAX",
    "Rolling-CV SARIMAX + Google Trends": "Selected Rolling-CV SARIMAX + GT",
    "LightGBM": "Selected LightGBM"
}

fig, axes = plt.subplots(4, 1, figsize=(14, 20), sharex=False)

if not isinstance(axes, np.ndarray):
    axes = np.array([axes])

for ax, code in zip(axes, datasets):
    df, y, dates, train_end_state = get_state_split(code,␣
 ↪all_state_test_horizon)

    y_train = y.iloc[:train_end_state]
    y_test = y.iloc[train_end_state:]

    ttrain_state = np.arange(train_end_state)
    ttest_state = np.arange(train_end_state, len(y))

    rec_row = final_model_recommendations_df.query("StateCode == @code").iloc[0]
    selected_class = rec_row["Recommended_Model"]
    selected_detail = rec_row["Selected_Details"]
    selected_rmse = rec_row["Holdout_RMSE"]
```

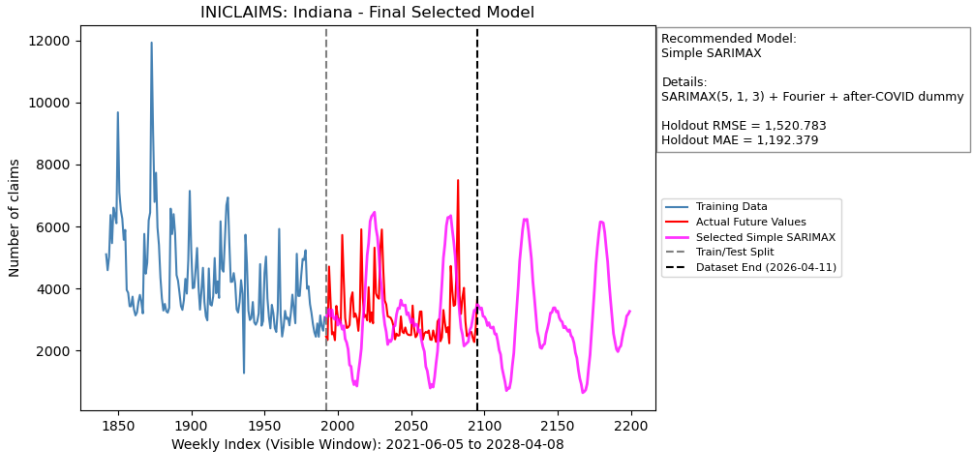
```
    selected_mae = rec_row["Holdout_MAE"]

    forecast_df = selected_forecast_sources[selected_class][code].copy()

    # Forecast x-axis: holdout start through future extension
    forecast_x = np.arange(train_end_state, train_end_state + len(forecast_df))

    # Plot last 150 training weeks
    ax.plot(ttrain_state[-150:], y_train.iloc[-150:], label="Training Data",␣
↪color="steelblue")

    # Plot actual holdout values
    ax.plot(ttest_state, y_test, label="Actual Future Values", color="red")

    # Plot selected model forecast (holdout + future)
    ax.plot(
        forecast_x,
        forecast_df["Prediction"],
        label=selected_label_names[selected_class],
        color=selected_colors[selected_class],
        alpha=0.8,
        linewidth=2
    )

    # Train/test split marker
    ax.axvline(x=train_end_state, color="gray", linestyle="--", label="Train/
↪Test Split")

    # Dataset end date marker
    ax.axvline(x=len(y) - 1, color="black", linestyle="--", label=f"Dataset End␣
↪({dates.iloc[-1].date()})")

    visible_start_date = dates.iloc[max(0, train_end_state - 150)].date()

    forecast_end_date = dates.iloc[-1].date()
    if "Date" in forecast_df.columns:
        forecast_end_date = max(
            forecast_end_date,
            pd.to_datetime(forecast_df["Date"]).max().date()
        )

    ax.set_title(f"{code}: {state_names[code]} - Final Selected Model")
    ax.set_xlabel(f"Weekly Index (Visible Window): {visible_start_date} to␣
↪{forecast_end_date}")
    ax.set_ylabel("Number of claims")

    model_info_text = (
```

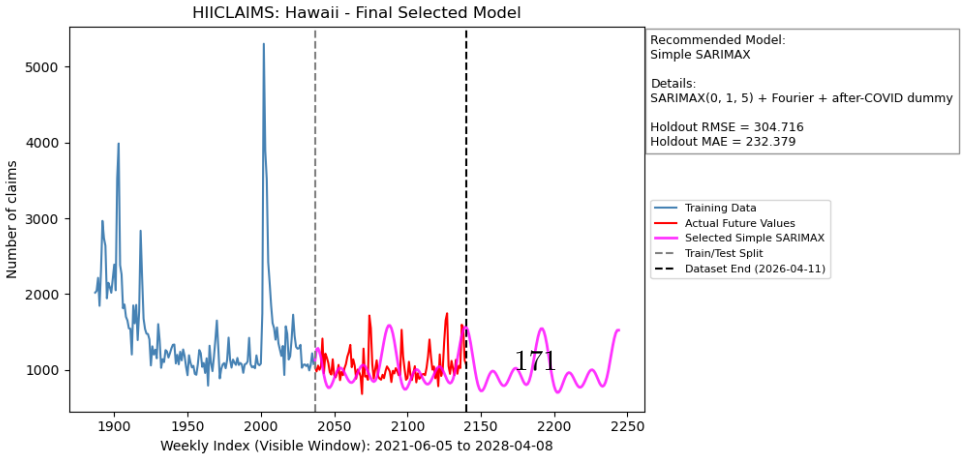
```
        f"Recommended Model:\n"
        f"{selected_class}\n\n"
        f"Details:\n{selected_detail}\n\n"
        f"Holdout RMSE = {selected_rmse:,.3f}\n"
        f"Holdout MAE = {selected_mae:,.3f}"
    )

    ax.text(
        1.01,
        0.98,
        model_info_text,
        transform=ax.transAxes,
        va="top",
        ha="left",
        fontsize=9,
        bbox=dict(facecolor="white", alpha=0.85, edgecolor="gray")
    )

    ax.legend(
        fontsize=8,
        loc="upper left",
        bbox_to_anchor=(1.01, 0.55),
        borderaxespad=0
    )

fig.suptitle(
    "Final Selected State-Level Forecast Models: Holdout + 2-Year Future␣
 ↪Forecast",
    fontsize=16
)

plt.tight_layout(rect=[0, 0, 0.73, 0.98])
plt.show()
```

Final Selected State-Level Forecast Models: Holdout + 2-Year Future Forecast









5.9 9. Conclusion

Overall, the modeling exercise shows that no single non-ARIMA model is uniformly best across all states. The preferred model differs by state, which is consistent with the fact that California, New York, Indiana, and Hawaii have very different claim levels, volatility patterns, and post-COVID behavior.

ARIMA remains useful as a baseline because it shows how far we can get using only the internal time-series structure of claims. However, we do not use ARIMA for the final recommendation because its relatively flat forecasts are less aligned with our goal of producing forecasts that reflect meaningful seasonal, structural, and lag-based variation. The final model choice instead comes from the SARIMAX and LightGBM candidates. SARIMAX is useful because its forecasts benefit from explicitly defined time-series structure, Fourier seasonality, and COVID-period indicators, all of which are interpretable in an economic context. LightGBM is useful when lagged claims, rolling averages, trend, and calendar features provide stronger predictive information, albeit at the cost of interpretability.

The final recommendation for forecasting state-level unemployment claims is therefore state-specific rather than based on a universal choice of model class. For some states, the more structured SARIMAX specification is preferred, while for others, LightGBM performs better on the holdout data. This supports the broader conclusion that time-series forecasting benefits from comparing interpretable econometric models with flexible machine-learning models, then using both statistical performance and economic reasoning to arrive at a final recommendation.

5.10 Appendix: AI Usage Disclosure

AI tools were used as a support tool after the main project pipelines were manually developed. Specifically, AI was used to help generate optimized helper functions once manual versions of the data-loading, cleaning, forecasting, and plotting pipelines had already been coded, which made repeated processes more streamlined and scalable across the four states. AI was also used to improve the structure of the baseline time-series and machine-learning modeling sections, especially by helping organize reusable functions, model-comparison tables, and plotting workflows. For the Google Trends augmentation, AI supported the process of aligning Trends data with weekly FRED claims dates and checking that the external features could be incorporated consistently into SARIMAX models. AI was additionally used to clean up selected code chunks, improve documentation, and double-check that the notebook structure followed the project code standards.

Overall, AI was used more as a review, organization, and optimization tool than as a replacement for manual project work. Group members remained responsible for the research question, data choices, modeling decisions, interpretation of results, and final verification. Verification involved reviewing the AI-assisted code against the intended outputs, checking that helper functions preserved the manually developed logic, confirming that cached data and model outputs were loaded from clear file paths, and ensuring that the notebook remained runnable from start to finish without manual intervention. No separate `agent.md` or `skill.md` files were used for this project.